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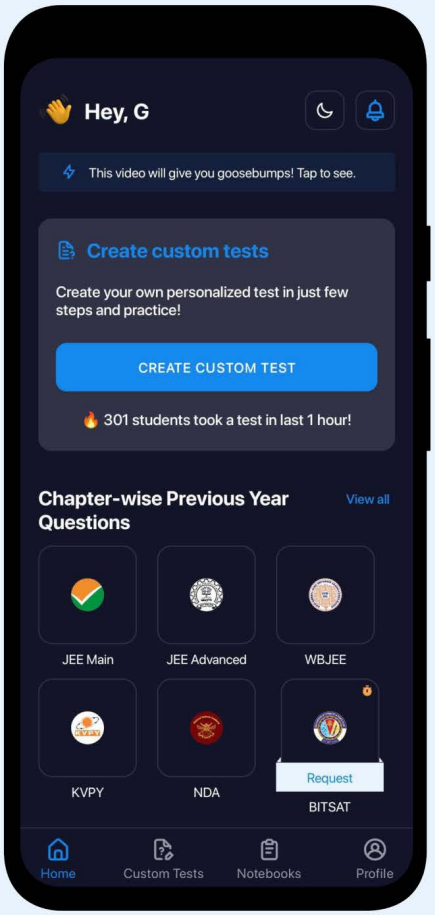

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ELECTROSTATICS

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ELECTROSTATICS

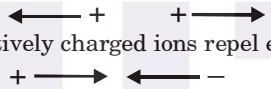
1. INTRODUCTION

When we comb our hair on a dry day and bring the comb close to tiny pieces of paper, we note that they are swiftly attracted by the comb. Similar phenomena occur if we rub a glass rod or an amber rod with a cloth or with a piece of fur. Why does this happens? What really happens in an electric circuit? How do electric motors and generators work?

The answers to all these questions come from a branch of physics known as **electromagnetism,** the study of electric and magnetic interactions. These interactions involve particles that have a property called electric charge, an inherent property of matter that is as fundamental as mass.

We begin our study of electromagnetism in this chapter by the electric charge. We will see that it is quantized and obey a conservation principle. Then we will study the interactions of electric charges that are at rest, called **electrostatic** interactions. These interactions are governed by a simple relationship known as **Coulomb's law.** This law is more conveniently described by using the concept of **electric field.**

2. ELECTRIC CHARGE



Two positively charged ions repel each other

A positively charged ion and electron repel each other

Atoms are composed of three particles with an electric charge : electrons, protons and uncharged neutrons. Electrons have a negative charge, and protons have a positive charge. The negative charge of the electron has the same magnitude, or amount of charge, as the positive charge of the proton.

Electrical charge is quantized, with the smallest unit of electrical charge found in a single electron or proton. If we double the number of electrons or protons, we double the amount of negative or positive electric charge. If we triple the electrons or protons, we triple the electric charge. By increasing the amount of single unit electric charges, we have a net charge. A net charge is the summed value of single electrical charges.

We don't typically refer to single units of charge. We typically refer to net charges. Atoms are composed of protons and electrons and generally have a net neutral electrical charge. The magnitude of total positive and total negative charges inside the neutral atom is equal. If we were to remove an electron (called a Valence electron) from the atom, the net charge of the atom would be positive. There is now more positive charge in the atom. This atom is called a positive ion, and is quantized with a net positive charge.

What happens when we keep removing Valence electrons from a group of neutral atoms? We end up with a group of positive ions and a group of free Valence electrons. Each group has a net positive or net negative electrical charge. We can call each charged group a body of charge.

We can't really say what an electrical charge is, but only describe its behavour. Simply started two single protons or two single electrons repel each other. A single proton and a single electron attract each other. The same can be said for net charges - a net negative charge will attract a net positive charge.

Points to Remember

- there are two kinds of charge, positive and negative.
- like charges repel, unlike charges attract
- positive charge comes from having more protons than electrons; negative charge comes from having more electrons than protons
- charge is quantized, meaning that charge comes in integer multiples of the elementary charge e
- charge is conserved

Probably everyone is familiar with the first three concepts, but what does it mean for charge to be quantized? Charge comes in multiples of an indivisible unit of charge, represented by the letter e. In other words, charge comes in multiples of the charge on the electron or the proton. These things have the same size charge, but the sign is different. A proton has a charge of + e, while an electron has a charge of – e.

Electrons and protons are not the only things that carry charge. Other particles (positrons, for example) also carry charge in multiples of the electronic charge. Those are not going to be discussed, for the most part, in this course, however.

Putting "charge is quantized" in terms of an equation, we say :

$q = n\ e$

q is the symbol used to represent charge, while n is a positive or negative integer, and e is the electronic charge, $1.60 \quad 10^{-19}$ Coulombs.

2.1 The Law of Conservation of Charge

The Law of conservation of charges states that **the net charge of an isolated system remains constant.**

If a system starts out with an equal number of positive and negative charges, there's nothing we can do to create an excess of one kind of charge in that system unless we bring in charge from outside the system (or remove some charge from the system). Likewise, if something starts out with a certain net charge, say + 100 e, it will always have + 100 e unless it is allowed to interact with something external to it.

Charge can be created and destroyed, but only in positive-negative pairs.

Table of elementary particle masses and charges :

Particle	**mass**	**charge**
electron	$9.11 \quad 10^{-31}$ kg	$-1.60 \quad 10^{-10}$ C (– e)
proton	$1.672 \quad 10^{-27}$ kg	$+1.60 \quad 10^{-10}$ C (+ e)
neutron	$1.674 \quad 10^{-27}$ kg	0

3. CONDUCTORS AND INSULATORS

For the purpose of electrostatic theory all substances can be divided into two main groups– conductors and insulators. In conductors electric charges are free to move from one place to another, whereas in insulators they are tightly bound to their respective atoms. In an uncharged body there are equal number of positive and negative charges.

The examples of conductors of electricity are the metals, human body and the earth and that of insulators are glass, hard rubber and plastics. In metals, the free charges are free electrons known as cnduction electrons.

Semiconductors are a third class of materials, and their electrical properties are somewhere between those of insulators and conductors. Silicon and germanium are well known example of semiconductors.

4. CHARGING OF A BODY

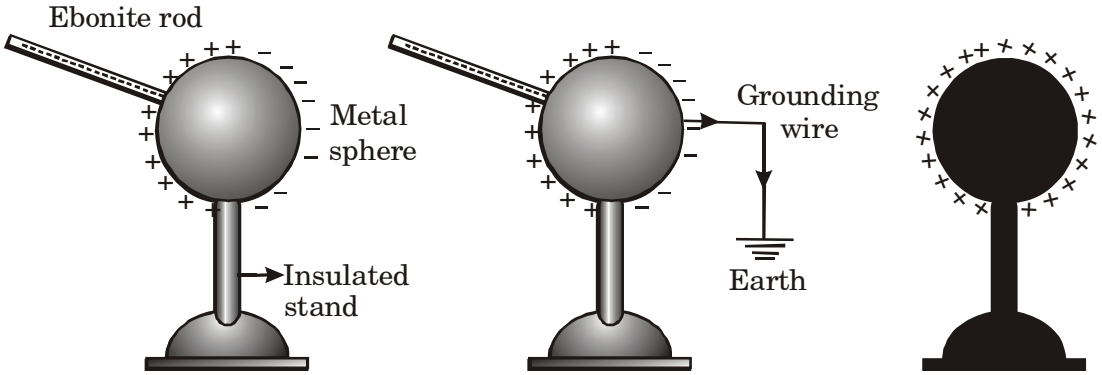
Mainly there are following there methods of charging a body :

(i) Charging by rubbing

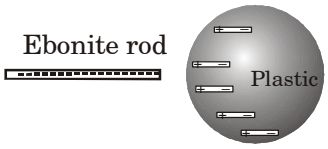
The simplest way to experience electric charges is to rub certain bodies against each other. When a glass rod is rubbed with a silk cloth the glass rod acquires some positive charge and the silk cloth acquires negative charge by the same amount. The explanation of appearance of electric charge on rubbing is simple. All material bodies contain large number of electrons and equal number of protons in their normal state. When rubbed against each other, some electrons from one body pass onto the other body. The body that donates the electron becomes positively charged while that which receives the electrons becomes negatively charged. For example when glass rod is rubbed with silk cloth, glass rod becomes positvely charged because it donate the electrons while the silk cloth becomes negatively charged because it receives electrons. Electricity so obtained by rubbing two objects is also known as **frictional electricity.** The other places where the frictional electricity can be observed are when amber is rubbed with wool or a comb is passed through a dry hair. Clouds also become charged by friction.

(ii) Charging by induction

It is also possible to charge a conductor in a way that does not involve contact.



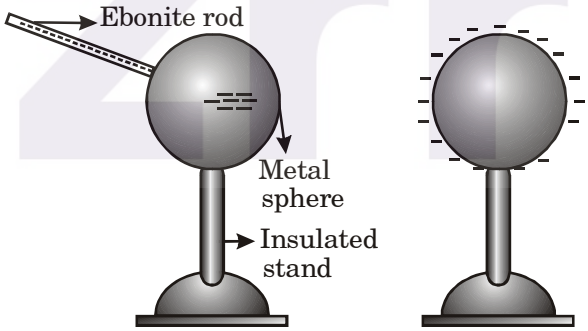
(a) (b) (c)



In figure (a) a negatively charged rod brought close to (but does not touch) a metal sphere. In the sphere, the free electrons close to the rod move to the other side (by repulsion). As a result, the part of the sphere nearer to the rod becomes positively charged and the part farthest from the rod negatively charged. This phenomenon is called **induction.** Now if the rod is removed, the free electrons return to their original places and the charged regions disappear. Under most conditions the earth is a good electric conductor. So when a metal wire is attached between the sphere and the ground as in figure (b) some of the free electrons leave the sphere and distribute themselves on the much larger earth. If the grounding wire is then removed, followed by the ebonite rod, the sphere is left with a net positive charge. The process of giving one object a net electric charge without touching the object to a second charged object is called **charging by induction.** The process could also be used to give the sphere a net negative charge, if a positively charged rod were used. Then, electrons would be drawn up from the ground through the grounding wire and onto the sphere.

If the sphere were made from an insulating material like plastic, instead of metal, the method of producing a net charge by induction would not work, because very little charge would flow through the insulating material and down the grounding wire. However, the electric force of the charged rod would have some effect as shown in figure. The electric force would cause the positive and negative charges in the molecules of the insulating material to separate slightly, with the negative charges being pushed away from the negative rod. The surface of the plastic sphere does acquire a slight induced positive charge, although no net charge is created.

(iii) Charging by contact



When a negatively charged ebonite rod is rubbed on a metal object, such as a sphere, some of the excess electrons from the rod are transferred to the sphere. Once the electrons are on the metal sphere, where they can move readily, they repel one another and spread out over the sphere's surface. The insulated stand prevents them from flowing to the earth. When the rod is removed the sphere is left with a negative charge distributed over its surface. In a similar manner the sphere will be left with a positive charge after being rubbed with a positively charged rod. In this case, electrons from the sphere would be transferred to the rod. The process of giving one object a net electric charge by placing it in contact with another object that is already charged is known as **charging by contact.**

Example 1

How many electrons are there in one coulomb of negative charge?

Solution :

The negative charge is due to the presence of excess electrons, since they carry negative charge. Because an electron has a charge whose magnitude is $e = 1.6 \quad 10^{-19}$ C, the number of electrons is equal to the charge q divided by the charge e on each electron. Therefore, the number n of electrons is

$$n = \frac{q}{e} = \frac{1.0}{1.6 \times 10^{-19}}$$

$$= 6.25 \quad 10^{18}$$ **Ans.**

Exmple 2

If we comb our hair on a dry day and bring the comb near small pieces of paper, the comb attracts the pieces, why?

Solution :

This is an example of frictional electricity and induction. When we comb our hair it gets positively charged by rubbing. When the comb is brought near the pieces of paper some of the electrons accumulate at the edge of the paper piece which is closer to the comb. At the farther end of the piece there is deficiency of electrons and hence, positive charge appears there. Such a redistribution of charge in a material, due to presence of a nearby charged body is called induction. The comb exerts larger attraction on the negative charges on the paper piece as compared to the repulsion on the positive charge. This is because the negative charges are closer to the comb. Hence, there is a net attraction between the comb and the paper piece.

Example 3

Does the attraction between the comb and the piece of papers last for longer period of time?

Solution :

No, because the comb loses its net charge after some time. The excess charge of the comb transfers to earth through our body after some time.

Example 4

Can two similarly charged bodies attract each other?

Solution :

Yes, when the charge on one body (q_1) is much greater than that on the other (q_2) and they are close enough to each other so that force of attraction between q_1 and induced charge on the other exceeds the force of repulsion between q_1 and q_2. However two similar point charges can never attract each other because no induction will take place here.

Example 5

Does the mass of a body change during charging ?

Solution :

Yes, as charging a body means addition or removal of electrons and electron has a mass.

Example 6

Why a third hole in a socket provided for grounding?

Solution :

All electric appliances may end with some charge due to faulty connections. In such a situation charge will be accumulated on the appliance. When the user touches the appliance he may get a shock. By providing the third hole for grounding all accumulated charge is discharged to the ground and the appliance is safe.

5. COULOMB'S LAW

The interaction between charged objects is a non-contact force which acts over some distance of separation. Every electrical interaction involves a force which highlights the importance of these three variables. Whether it is a plastic golf tube attracting paper bits, two like-charged balloons repelling or a charged Styrofoam plate interacting with electrons in a piece of aluminium, there is always two charges and a distance between them as the three critical variables which influence the strength of the interaction.

5.1 Force as a Vector Quantity

The electrical force, like all forces, is typically expressed in units of Newtons. Being a force, the strength of the electrical interaction is a vector quantity which has both magnitude and direction. The direction of the electrical force is dependent upon whether the charged objects are charged with like charge or positive charge and upon their spatial orientation. By knowing the type of charge on the two objects, the direction of the force on either one of them can be determined with a little reasoning. In the diagram below, objects A and B have like charge causing them to repel each other. Thus, the force on object A is directed leftward (away from B) and the force on object B is directed rightward (away from A). On the other hand, objects C and D have opposite charge causing them to attract each other. Thus, the force on object C is directed rightward (toward object D) and the force on object D is directed leftward (toward object C). When it comes to the electrical force vector, perhaps the best way to determine the direction of it is to apply the fundamental rules of charge interaction (opposites attract and likes repel) using a little reasoning.

Determining the Direction of the Electrical Force Vector

Electrical force also has a magnitude or strength. Like most types of forces, there are a variety of factors which influence the magnitude of the electrical force. Two like-charged balloons will repel each other and the strength of their repulsive force can be altered by changing three variables. First, the

quantity of charge on one of the balloons will effect the strength of the repulsive force. The more charged a balloon is, the greater the repulsive force. Second, the quantity of charge on the second balloon will affect the strength of the repulsive force. Gently rub two balloons with animal fur and they repel a little. Rub the two balloons vigorously to impart more charge to both of them, and they repel a lot. Finally, the distance between the two balloons will have a significant and noticeable affect upon the repulsive force. The electrical force is strongest when the balloons are closest together. Decreasing the separation distance increases the force. The magnitude of the force and the distance between the two balloons is said to be inversely related.

5.2 Coulomb's Law Equation

The quantitative expression for the affect of these three variables on electric force is known as Coulomb's law. Coulomb's laws states that the electrical force between two charged objects is directly prportional to the product of the quantity of charge on the objects and inversely proportional to the square of the separation distance between the two objects. In equation form, Coulomb's law can be stated as

$$\boxed{F = \frac{K Q_1 Q_2}{d^2}}$$

where $\mathbf{Q_1}$ represents the quantity of charge on object 1 (in Coulombs), $\mathbf{Q_2}$ represents the quantity of charge on object 2 (in Coulombs), and d represents the distance of separation between the two objects (in meters). The symbol k is a proportionality constant known as the Coulomb's law constant. The value of this constant is dependent upon the medium that the charged objects are immersed in. In the case of air, the value is approximately 9 10^9 Nm^2/C^2.

This constant k is often written as $\frac{1}{4\pi \in_0}$, where $\in_0$ ("epsilon-nought") is another constant. This appears to complicate matters, but it actually simplifies many formulae that we will encounter in later chapters. Thus, eq. can be written as,

$$\boxed{F_e = \frac{1}{4\pi \in_0} \frac{|q_1 q_2|}{r^2}}$$

Here,

$$\boxed{\frac{1}{4\pi \in_0} = 9 \times 10^9 \ N - m^2 / C^2}$$

Since Coulomb's law applies to point charges, the distance d in the equation is the distance between the centers of charge for both objects (not the distance between their nearest surfaces).

The symbols $\mathbf{Q_1}$ and $\mathbf{Q_2}$ in the Coulomb's law equation represent the quantities of charge on the two interacting objects. Since an object can be charged positively or negatively, these quantities are

often expressed as "+" or "–" values. The sign on the charge is simply representative of whether the object has an excess of electrons (a negatively charged object) or a shortage of electrons (a positively charged object). When using the "+" and "–" signs in the calculation of force, the result will be that a "–" value for force is a sign of an attractive force and a "+" value for force signifies a repulsive force. Mathematically, the force value would be found to be positive when $\mathbf{Q_1}$ and $\mathbf{Q_2}$ are like charge-either both "+" or both "–". And the force value would be found to be negative when $\mathbf{Q_1}$ and $\mathbf{Q_2}$ are of opposite charge - one is "+" and the other is "–". This is consistent with the concept that oppositely charged objects have an attractive interactions and like charged objects have a repulsive interaction. In the end, if you're thinking conceptually (and not merely mathematically), you would be very able to determine the nature of the force-attractive or repulsive-without the use of "+" and "–" signs in the equation.

Regarding Coulomb's law following points are worth noting :

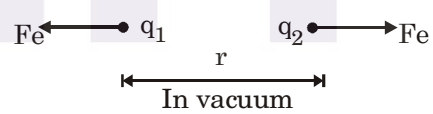
(1) Coulomb's law stated above describes the interaction of two point charges. When two charges exert forces simultaneously on a third charge, the total force acting on that charge is the vector sum of the forces that the two charges would exert individually. This important property, called the **principle of superposition of forces,** holds for any number of charges. Thus,

$$\vec{F}_{net} = \vec{F}_1 + \vec{F}_2 + ... + \vec{F}_n$$

(2) The electric force is an action reaction pair, i.e., the two charges exert equal and opposite forces on each other.

(3) The electric force is conservative in nature.

(4) Coulomb's law as we have stated above can be used for point charges in **vacuum.** If some dielectric (insulator) is present in the space between the charges, the net force acting on each charge is altered because charges are induced in the molecules of the intervening medium. We will describe this effect later. Here at this moment it is enough to say that the force decreases K times if the medium extends till infinity. Here K is a dimensionless constant which depends on the medium and called dielectric constant of the medium. Thus,



$$F_e = \frac{1}{4\pi \in_0} \cdot \frac{q_1 q_2}{r_2} \qquad \text{(in vacuum)}$$

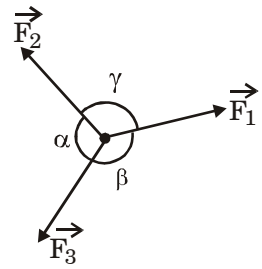
$$F_e' = \frac{F_e}{K} = \frac{1}{4\pi \in_0 K} \cdot \frac{q_1 q_2}{r^2} = \frac{1}{4\pi \in} \cdot \frac{q_1 q_2}{r^2} \qquad \text{(in medium)}$$

Here, $\in = \in_0 K$ is called **permitivity** of the medium.

Points to Remember

1. In few problems of electrostatics **Lami's theorem** is very useful.

According to this theorem, "if three concurrent forces $\vec{F}_1$ $\vec{F}_2$ and $\vec{F}_3$ shown in figure are in equilibrium or if $\vec{F}_1 + \vec{F}_2 + \vec{F}_3 = 0$, then



$$\frac{F_1}{\sin\alpha} = \frac{F_2}{\sin\beta} = \frac{F_3}{\sin\gamma}$$

2. Suppose two charges q_1 and q_2 are placed in vacuum at a distance r_0 and the electric force between them is

$$F_0 = \frac{1}{4\pi \in_0} \cdot \frac{q_1 q_2}{r_0^2}$$

Now the same charges are placed in a dielectric constant K at a distance r (< r_0) such that the electric force between them

$$F = \frac{1}{4\pi \in_0 K} \cdot \frac{q_1 q_2}{r^2}$$

remains the same of $F = F_0$

Then
$$\frac{1}{4\pi \in_0} \cdot \frac{q_1 q_2}{r^2} = \frac{1}{4\pi \in_0} \cdot \frac{q_1 q_2}{r_0^2}$$

or
$$r_0 = \sqrt{K}\, r$$

Thus, we see that, for the electric force between two charges a distance r in a dielectric medium is equivalent to a distance $r_0 = \sqrt{K}\, r$ in vacuum.

Example 7

Suppose that two point charges, each with a charge of + 1.00 Coulomb are separated by a distance of 1.00 meter. Determine the magnitude of the electrical force of repulsion between them.

Solution : **Given :**

Q_1 = 1.00 C
Q_2 = 1.00 C
d = 1.00 m

Find :

F_{elect} = ???

$$F_{elect} = \frac{k \cdot Q_1 \cdot Q_2}{d^2}$$

F_{elect} = (9.0 10^9 N•m²/C²) • (1.00 C) • (1.00 C) / (1.00 m)²

$$\mathbf{F_{elect} = 9.0 \quad 10^9\ N}$$

The force of repulsion of two + 1.00 Coulomb charges held 1.00 meter apart is 9 billion Newtons.

This problem was chosen primarily for its conceptual message. Objects simply do not acquire charges of the order of 1.00 Coulomb. In fact, more likely Q values are on the order of 10^9 or possibly 10^{-6} Coulombs. For this reason, a Greek prefix is often used in front of the Coulomb as a unit of charge. Charge is often expressed in units of micro Coulomb (μC) and nano Coulomb (nC). If a problem states the charge in these units, it is advisable to first convert to Coulombs prior to substitution into the Coulomb's law equation. The following unit equialencies will assist in such conversions.

$$1 \text{ Coulomb} = 10^6 \text{ micro Coulomb}$$

$$1 \text{ Coulomb} = 10^9 \text{ nano Coulomb}$$

Example 8

Two balloons with charges of + 3.37 μC and – 8.21 μC attract each other with a force of 0.626 Newtons. Determine the separation distance between the two balloons.

Given

$Q_1 = + 3.37\ \mu C = + 3.37 \quad 10^{-6}\ C$

$Q_2 = - 8.21\ \mu C = - 8.21 \quad 10^{-6}\ C$

Felect = – 0.0626 N (use a - force value since it is repulsive)

Find :

d = ???

$$F_{elect} = \frac{k \cdot Q_1 \cdot Q_2}{d^2}$$

$$\Rightarrow \quad d^2 = \frac{k \cdot Q_1 \cdot Q_2}{F_{elect}}$$

$$d = \sqrt{\frac{kQ_1\ Q_2}{F_{elect}}}$$

$$d = \sqrt{\frac{\left[\left(9.0 \times 10^9\ N \cdot m^2/C^2\right) \cdot \left(-8.21 \times 10^{-6}\ C\right) \cdot \left(+3.37 \times 10^{-6}\ C\right)\right]}{(-0.062\,N)}}$$

$$d = + 1.99\ m$$

Example 9

What is the smallest electric force between two charges placed at a distance of 1.0 m.

Solution :

$$F_e = \frac{1}{4\pi \in_0} \cdot \frac{q_1 q_2}{r^2}$$

For F_e to be minimum $q_1 q_2$ should be minimum. We know that

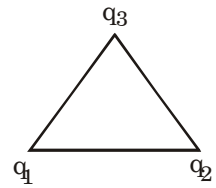
$$(q_1)_{min} = (q_2)_{min} = e = 1.6 \quad 10^{-19}\ C$$

$$\text{we have } (F_e)_{min} = \frac{\left(9.0 \times 10^9\right)}{(1.0)^2}\left(1.6 \times 10^{-19}\right)$$

$$= 2.304 \quad 10^{-28}\ N$$ **Ans.**

Example 10

Three charges q_1 = 1μC, q_2 = – 2μC and q_3 = 3μC are placed on the vertices of an equilateral triangle of side 1.0 m. Find the net electric force acting on charge q_1.



How to Proceed : Charge q_2 will attract charge q_1 (along the line joining them) and charge q_3 will repel charge q_1. Therefore, two forces will act on q_1, one due to q_2 and another due to q_3. Since, the force is a vector quantity both of these forces $\left(\text{say } \vec{F}_1 \text{ and } \vec{F}_2\right)$ will be added by vector method. Following are two methods of their addition.

Solution :

Method 1 : In the figure,

$$\left|\vec{F}_1\right| = F_1 = \frac{1}{4\pi \in_0} \cdot \frac{q_1 q_2}{r^2}$$

= magnitude of force between q_1 and q_2

$$= \frac{\left(9.0 \times 10^9\right)\left(1.0 \times 10^{-6}\right)\left(2.0 \times 10^{-6}\right)}{(1.0)^2}$$

$$= 1.8 \quad 10^{-2} \text{ N}$$

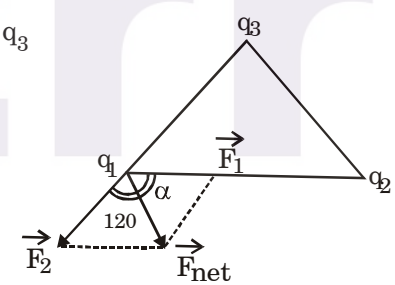
Similarly,

$$\left|\vec{F}_2\right| = F_2 = \frac{1}{4\pi \in_0} \cdot \frac{q_1 q_3}{r^2}$$

= magnitude of force between q_1 and q_3

$$= \frac{\left(9.0 \times 10^9\right)\left(1.0 \times 10^{-6}\right)\left(3.0 \times 10^{-6}\right)}{(1.0)^2}$$

$$= 2.7 \quad 10^{-2} \text{ N}$$



Now,

$$|F_{net}| = \sqrt{F_1^2 + F_2^2 + 2F_1F_2 \cos 120°}$$

$$= \left(\sqrt{(1.8)^2 + (2.7)^2 + 2(1.8)(2.7)\left(-\frac{1}{2}\right)}\right) \times 10^{-2} \text{ N}$$

$$= 2.38 \quad 10^{-2} \text{ N}$$

and

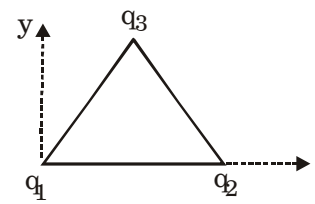
$$\tan\alpha = \frac{F_2 \sin 120°}{F_1 + F_2 \cos 120°}$$

$$= \frac{\left(2.7 \times 10^{-2}\right)(0.87)}{\left(1.8 \times 10^{-2}\right) + \left(2.7 \times 10^{-2}\right)\left(-\frac{1}{2}\right)}$$

or $\alpha = 79.2$.

Thus, the net force on charge q_1 is 2.38 10^{-2} N at an angle α = 79.2 with a line joining q_1 and q_2 as shown in the figure. **Ans.**

Method 2 : In this method let us assume a co-ordinate axes with q_1 at origin as shown in figure.The co-ordinates of q_1, q_2 and q_3 in this co-ordinate system are (0, 0, 0), (1 m, 0, 0) and (0.5 m, 0.87 m, 0) respectively. Now



$\vec{F}_1$ = force on q_1 due to charge q_2

$$= \frac{1}{4\pi \epsilon_0} \cdot \frac{q_1 q_2}{\left|\vec{r}_1 - \vec{r}_2\right|^3}\left(\vec{r}_1 - \vec{r}_2\right)$$

$$= \frac{\left(9.0 \times 10^9\right)\left(1.0 \times 10^{-6}\right)\left(-2.0 \times 10^{-6}\right)}{(1.0)^3}\left[(0-1)\tilde{i} + (0-0)\tilde{j} + (0-0)\tilde{k}\right]$$

$$= \left(1.8 \times 10^{-2}\ \tilde{i}\right) N$$

and $\vec{F}_2$ = force on q_1 due to charge q_3

$$= \frac{1}{4\pi \epsilon_0} \cdot \frac{q_1 q_3}{\left|\vec{r}_1 - \vec{r}_3\right|^3}\left(\vec{r}_1 - \vec{r}_3\right)$$

$$= \frac{\left(9.0 \times 10^9\right)\left(1.0 \times 10^{-6}\right)\left(3.0 \times 10^{-6}\right)}{(1.0)^3}\left[(0-0.5)\tilde{i} + (0-0.87)\tilde{j} + (0-0)\tilde{k}\right]$$

$$= \left(-1.35\tilde{i} - 2.349\ \tilde{j}\right) \times 10^{-2}\ N$$

Therefore, net force on q_1 is $\vec{F} = \vec{F}_1 + \vec{F}_2$

$$= \left(0.45\tilde{i} - 2.349\ \tilde{j}\right) \times 10^{-2}\ N$$ **Ans.**

Note : Once you write a vector in terms of $\tilde{i}$, $\tilde{j}$ and $\tilde{k}$, there is no need of writing the magnitude and direction of vector separately.

Example 11

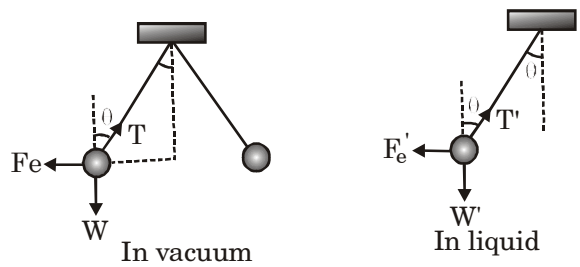
Two identical balls each having a density ρ are suspended from a common point by two insulating strings of equal length. Both the balls have equal mass and charge. In equilibrium each string makes an angle θ with vertical. Now, both the balls are immersed in a liquid. As a result the angle θ does not change. The density of the liquid is σ. Find the dielectric constant of the liquid.

Solution :

Each ball is in equilibrium under the following three forces :

(i) tension, (ii) electric force and (iii) weight

So, Lami's theorem can be applied



In the liquid, $F_e' = \dfrac{F_e}{K}$

where, K = dielectric constant of liquid and W′ = W – upthrust

Applying Lami's theorem in vacuum $\dfrac{W}{\sin(90° + \theta)} = \dfrac{F_e}{\sin(180° - \theta)}$

or $$\frac{W}{\cos\theta} = \frac{F_e}{\sin\theta} \quad ...(i)$$

Similarly in liquid, $$\frac{W'}{\cos\theta} = \frac{F_e'}{\sin\theta} \quad ...(ii)$$

Dividing Eq. (i) by (ii), we get

$$\frac{W}{W'} = \frac{F_e}{F_e'}$$

or $$K = \frac{W}{W - \text{upthrust}} \quad \left(\text{as } \frac{F_e}{F_e'} = K\right)$$

$$= \frac{V\rho g}{V\rho g - V\sigma g} \quad (V = \text{volume of ball})$$

or $$K = \frac{\rho}{\rho - \sigma}$$ **Ans.**

Note : In the liquid F_e and W have been changed. Therefore, T will also change.

6. COMPARING ELECTRICAL AND GRAVITATIONAL FORCES

Electrical force and gravitational force are the two non-contact forces. Coulomb's law equation for electrical force bears a strong resemblance to Newton's equation for universal gravitation.

$$F_{elect} = \frac{k\,Q_1\,Q_2}{d^2} \qquad (k = 9 \times 10^9\ Nm^2/C^2)$$

$$F_{grav} = \frac{G\,m_1\,m_2}{d^2} \qquad \left(G = 6.67 \times 10^{-11}\ Nm^2/kg^2\right)$$

The two equations have a very similar form. Both equations show an inverse square relationship between force and separation distance. And both equations show that the force is proportional to the product of the quantity that causes the force-charge in the case of electrical force and mass in the case of gravitational force. Yet there are some striking differences between these two forces. First, a comparison of the proportionality constants - k versus G - reveals that the Coulomb's law constant (k) is significantly greater than Newton's universal gravitation constant (G). Subsequently a unit of charge will attract a unit of charge with significantly more force than a unit of mass will attract a unit of mass. Second, gravitational forces are only attractive; electrical forces can be either attractive or repulsive.

The inverse square relationship between force and distance which is woven into the equation is common to both non-contact forces. This relationship highlights the importance of separation distance when it comes to the electrical force between charged objects.

7. ELECTRIC FIELD

A charged particle cannot directly interact with another particle kept at a distance. A charge produces something called an electric field in the space found it and this electric field exerts a force on any other charge (except the source charge itself) placed in it.

Thus, the region surrounding a charge or distribution of charge in which its electrical effects can be observed is called the **electric field** of the charge or distribution of charge. Electric field at a point can be defined that all these are functions of positions $\vec{r}\,(x, y, z)$. The field propagates through space with the speed of light, c. Thus, if a charge is suddenly moved, the force it exerts on another charge a distance r away does not change until a time r/c later.

7.1 Electric Field Strength $\left(\vec{E}\right)$

Like its gravitational counterpart, the electric field strength (often called electric field) at a point in an electric field is defined as the electrostatic force $\vec{F}_e$ per unit positive charge. Thus, if the electrostatic force experienced by a small test charge q_0 is $\vec{F}_e$, then field strength at that point is defined as,

$$\boxed{\vec{E} = \lim_{q_0 \to 0} \frac{\vec{F}_e}{q_0}}$$

The electric field is a vector quantity and its direction is the same as the direction of the force $\vec{F}_e$ on a positive test charge. The SI unit of electric field is N/C. Here it should be noted that the test charge

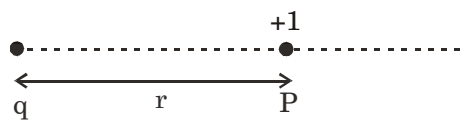
q_0 does not disturb other charges which produces $\vec{E}$. With the concept of electric field, our description of electric interactions has two parts. First, a given charge distribution acts as a source of electric field. Second, the electric field exerts a force on any charge that is present in this field.

7.2 An Electric Field leads to a Force

Suppose there is an electric field strength $\vec{E}$ at some point in an electric field, then the electrostatic force acting on a charge + q is qE in the direction of $\vec{E}$, while on the charge – q it is qE in the opposite direction of $\vec{E}$.

7.3 Electric Field intensity due to a Point Charge

The force experienced by a unit positive charge at a distance r due to point charge 'q' is called electric field intensity at that point due to the charge q.



$$F = \left(\frac{Kq_1q_2}{r^2}\right) \text{ where } K = 9 \quad 10^9 \text{ Nm}^2/\text{C}^2$$

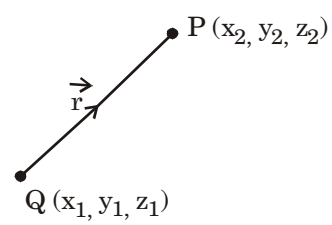
$$\text{and } K = \left(\frac{1}{4\pi \in_0}\right) \text{ where } \varepsilon_0 = 8.85 \quad 10^{-12} \text{ C}^2/\text{Nm}^2$$

We define electric field at a point as

$$E = \left(\frac{KQ.1}{r^2}\right) = \left(\frac{KQ}{r^2}\right)$$

$$\therefore \quad \boxed{E = \frac{KQ}{r^2}} \quad \text{or} \quad \boxed{E = \frac{Q}{4\pi \in_0 r^2}}$$

In vector form



$$\vec{E} = E.\tilde{n}_r$$

$$= \left(\frac{KQ}{r^2}\right).\tilde{n}_r = \left(\frac{KQ}{r^3}\right)\vec{r}$$

$$\therefore \quad \boxed{\vec{E} = \left(\frac{KQ}{r^3}\right)\vec{r}} \qquad \vec{r} = (x_2 - x_1)i + (y_2 - y_1)j + (z_2 - z_1)k$$

$$\boxed{\vec{E} = \left(\frac{Q}{4\pi \in_0 r^3}\right)\vec{r}}$$

- Net field intensity in the space due to a number of point charges at a point is given by principle of superposition.

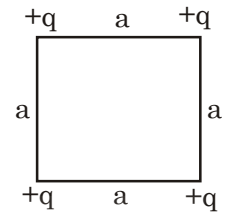
$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + ... + \vec{E}_n$$

[Principle of superposition is applicable only when the effect remains unchanged, due to presence of others (charges in this case).]

ı Net force experienced by a point charge in the electric field

$$\boxed{\vec{F} = q\vec{E}}$$

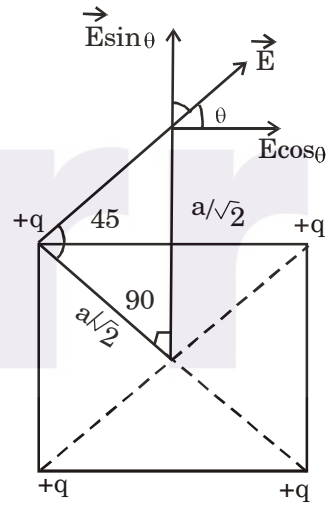
Example 12



Four identical charges are placed at the corners of a square of side a. Find the net field at a point lying on the perpendicular drawn to the plane of square and passing through its centre at a height $\frac{a}{\sqrt{2}}$.

Solution :

Observe that the net field lie in the vertical direction as the opposite charges cancel out each others horizontal component of the electric field.



$$\left|\vec{E}_{net}\right| = 4E\sin\theta$$

$$= 4\left(\frac{Kq}{a^2}\right)\left(\frac{1}{\sqrt{2}}\right)$$

$$\vec{E} = \left(\frac{2\sqrt{2}Kq}{a^2}\right) \text{ (away from centre)}$$

Note : for any symmetric polygon, the field at a perpendicular height on the centre will be vertically directed away from centre.

---

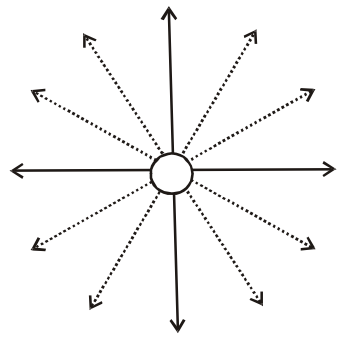
8. ELECTRIC POTENTIAL

"Potential is the potential energy per unit charge." Electric potential at any point in an electric field is defined as **the potential energy per unit charge**, same as field strength is defined as the force per unit charge.

Thus $V = \frac{U}{q_0}$ or $\boxed{U = q_0V}$

The SI unit of potential is Volt (V) which is Joule per Coloumb. So 1 Volt = 1 J/C

If we want a picture of a field, we usually think of many lines lines of "flux" emanating radially away from the charge :



For negative charges, the lines point inward. These flux lines indicate the direction of a positively charged test particle would move due to the electric force between it and the configuration charge. Note that the "density per unit area" of the flux decreases as the distance from the center increases. That is, the lines are much closer together when nearer to their origin. Since we live in three spatial dimensions, and the surface area of a sphere centered on the charge increases with the square of the distance, we have a model for the inverse square force law : the magnitude of the force is proportional to the density of the field lines ("flux density"), which decreases with the square of the distance. We can also see why spherical source is equivalent to a point source at its center : the flux lines emanating from the surface can be extrapolated inwards to the center with no change in the physical forces felt outside the surface.

The electric field can be thought of as the gradient of the electrical potential field. But since the electric field is a force per unit charge, the electric potential must be an energy per unit charge. It is a scalar field, having a magnitude (but no direction) at every point in space.

$$\boxed{V(r) = \frac{q}{4\pi \in_0 r}}$$

also $$U = q_0 V$$

As with the cases of the electric field and force, the electrical potential field superposes lineraly, so that the field at a point due to a collection of configuration charges is equal to the sum of the fields due to the individual charges, at that point. For example, if in a vacuum two protons (q_{c1}) lie in the xy plane at $(x_1, y_1) = (-2, 1)$ Angstroms and an electron (q_{c2}) lies at the origin $(x_2, y_2) = (0, 0)$, the energy of an electron (q_t) at $(x_t, y_t) = (1, 1)$ Angstroms would be

$$U = U_{tc1} + U_{tc2}$$

$$= q_t (V_{c1} + V_{c2}))$$

$$= \frac{(-e)(2e)}{4\pi \in_0 \times 3 \times 10^{-10}} + \frac{(-e)(e)}{4\pi \in_0 \sqrt{2} \times 10^{-10}}$$

$$= 9.33 \quad 10^{-20} \text{ J},$$

where the distances between charges are

$$r = \sqrt{(x_t - x_i)^2 + (y_t - y_i)^2}$$

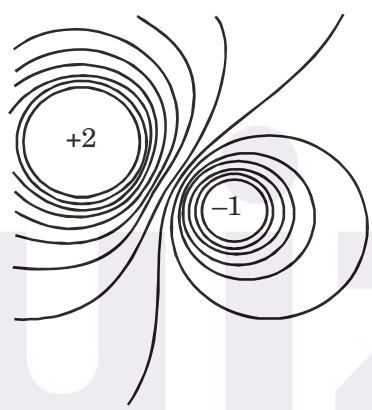
and "i" denotes either of the configuration charges.

8.1 Relation between Electric Potential and Electric Field

In order to compute the direction of the force felt by a test charge in a potential field, one can take the gradient of the potential

$$\boxed{\mathbf{E}_x = -\frac{\delta \mathbf{V}}{\delta x}}$$

If we were to plot the electrical potential field due to a pair of configuration charges, we would see that there are lines, called "equipotentials", on which the electrical potential has the same value everywhere :



The direction of the gradient (and hence the electric field) is perpendicular to the equipotential lines since its component along (tangent to) the equipotentials is zero (V is a constant). We can therefore immediately trace the path of any test particle in the field, since **charges move in the direction of least energy.**

8.1 Absolute Potential at Some Point

Suppose we take the point b at infinity and as a reference point assign the value $V_b = 0$, the equations can be written as

$$V_a - V_b = \frac{(W_{a-b})_{\text{electric force}}}{q_0} = \frac{(W_{b-a})_{\text{external force}}}{q_0}$$

or $$V_a - V_b = \frac{(W_{a-\infty})_{\text{electric force}}}{q_0} = \frac{(W_{\infty-a})_{\text{external force}}}{q_0}$$

Thus, the absolute electric potential at point a in an electric field can be defined as the work done

in displacing a unit positive charge from infinity to a by the external force or the work done per unit positive charge in displacing it from a to infinity.

Note : Following three formulae are very useful in the problems related to work done in electric field.

$$(W_{a-b})_{\text{electric force}} = q_0 (V_a - V_b)$$
$$(W_{a-b})_{\text{external force}} = q_0 (V_b - V_a) = -(W_{a-b})_{\text{electric force}}$$
$$(W_{\infty-a})_{\text{external force}} = q_0 V_a$$

Here, q_0, and V_b are to be substituted with sign.

Example 13

The electric potential at point A is 20 V and at B is – 40 V. Find the work done by an external force and electrostatic force in moving an electron slowly from B to A.

Solution :

Here, the test charge is an electron, i.e.

$$q_0 = -1.6 \quad 10^{-19} \text{ C}$$
$$V_A = 20 \text{ volt}$$

and $$V_B = -40 \text{ volt}$$

Work done by external force

$$(W_{B-A})_{\text{external force}} = q_0 (V_A - V_B)$$
$$= (-1.6 \quad 10^{-19}) [(20) - (-40)]$$
$$= -9.6 \quad 10^{-18} \text{ J}$$ **Ans.**

Work done by electric force

$$(W_{B-A})_{\text{electric force}} = -(W_{B-A})_{\text{external force}}$$
$$= -(9.6 \quad 10^{-18} \text{ J})$$
$$= 9.6 \quad 10^{-18} \text{ J}$$ **Ans.**

Note : Here we can see that the electron (a negative charge) moves from B (lower potential) to A (higher potential) and the work done by electric force is positive. Therefore, we may conclude that whenever a negative charge moves from a lower potential to higher potential work done by the electric force is positive or when a positive charge moves from lower potential to higher potential the work done by the electric force is negative.

Example 14

Find the work done by some external force in moving a charge q = 2 μC from infinity to a point where electric potential is 10^4 V.

Solution :

Using the relation,

$$(W_{\infty-a})_{\text{external force}} = q_0 V_a$$

We have, $$(W_{\infty-a})_{\text{external force}} = (2 \quad 10^{-6}) (10^4)$$
$$= 2 \quad 10^{-2} \text{ J}$$ **Ans.**

8.4 Electric Potential Due to a Point Charge q

Work done against the field in slow moving a unit positive charge from infinity to the point in the field is called electric potential at that point.

From the definition of potential, $V = \frac{U}{q_0} = \frac{\frac{1}{4\pi \epsilon_0} \cdot \frac{q q_0}{r}}{q_0}$

$$V = \frac{1}{4\pi \epsilon_0} \cdot \frac{q}{r}$$

Here, r is the distance from the point charge q to the point at which the potential is evaluated. If q is positive, the potential that it produces is positive at all points; if q is negative, it produces a potential that is negative everywhere. In either case, V is equal to zero at $r = \infty$.

8.5 Electric Potential Due to a System of Charges

Just as the electric field due to a collection of point charges is the vector sum of the fields produced by each charge, the electric potential due to a collection of point charge is the sclar sum of the potentials due to each charge.

$$V = \frac{1}{4\pi \epsilon_0} \sum_i \frac{q_i}{r_i}$$

In this expression, r_i is the distance from the ith charge, q_i, to the point at which V is evaluated. For a continuous distribution of charge along a line, over a surface, or through a volume, we divide the charge into elements dq and the sum in the above equation becomes an integral,

$$V = \frac{1}{4\pi \epsilon_0} \int \frac{dq}{r}$$

Note : In the equation $V = \frac{1}{4\pi \epsilon_0} \sum_i \frac{q_i}{r_i}$ or $V = \frac{1}{4\pi \epsilon_0} \int \frac{dq}{r}$ if the whole charge is at equal distance r_0 from the point where V is to be evaluated, then we can write

$$V = \frac{1}{4\pi \epsilon_0} \cdot \frac{q_{net}}{r_0}$$

Where q_{net} is the algebraic sum of all the charges of which the system is made.

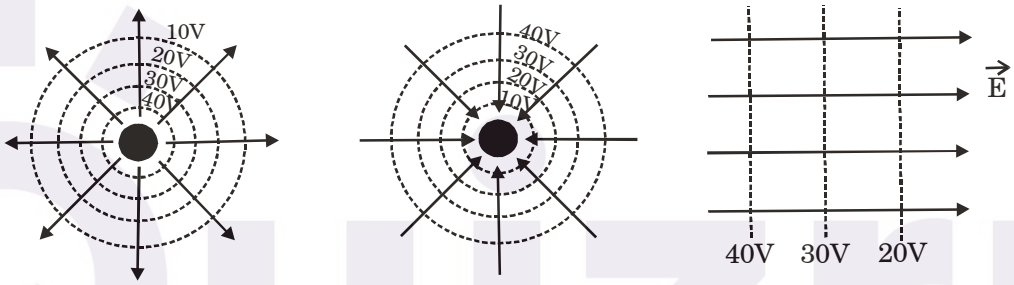
9. EQUIPOTENTIAL SURFACES

The equipotential surfaces in an electric field have the same basic idea as topographic maps used by civil engineers or mountain climbers. On a topographic map, contour lines are drawn passing through the points having the same elevation. The potential energy of a mass m does not change along a contour line as the elevation is same everywhere.

An equipotential surface is a three dimensional surface on which the electrical potential V is the same at every point on it. An equipotential surface has the following characteristics.

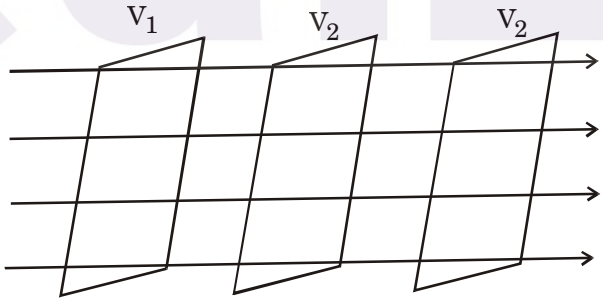
1. Potential difference between any two points in an equipotential surface is zero.
2. If a test charge q_0 is moved from one point to the other on such a surface, the electric potential energy q_0V remains constant.
3. No work is done by the electric force when the test charge is moved along this surface.
4. Two equipotential surfaces can never intersect each other because otherwise the point of intersection will have two potentials which is of course not acceptable.
5. As the work done by electric force is zero when a test charge is moved along the equipotential surface, it follows that $\vec{E}$ must be perpendicular to the surface at every point so that the electric force $q_0\vec{E}$ will always be perpendicular to the displacement of a charge moving on the surface.

Thus, **field lines and equipotential surfaces are always mutually perpendicular.** Some equipotential surfaces are shown below :

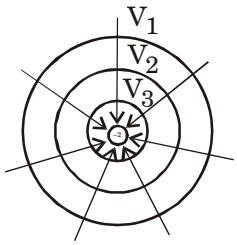


Equipotential surfaces are always perpendicular to field lines.

(i) When field lines are parallel, equipotential surface are planar sheets ($V_1 > V_2 > V_3$)



(ii) For a point charge, equipotential surface are concentric spherical surfaces centred at q. ($V_1 > V_2 > V_3$ when $q < 0$)

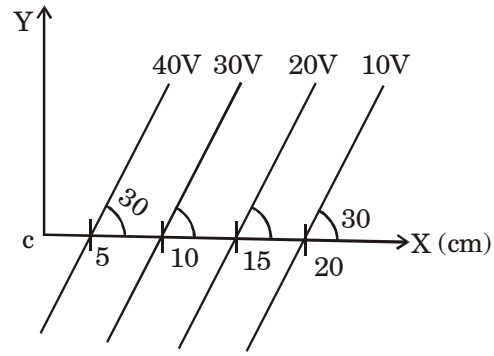


Note : Equipotential surface is locus of points where potential remain constant.

To find the field intensity when Equipotential Surface are given

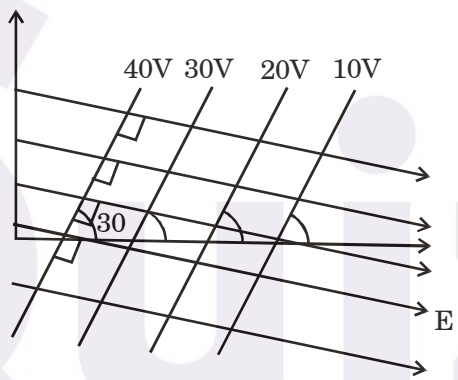
Example 15

The figure shows parallel equipotential surfaces which are parallel planes. Find the magnitude and direction of field intensity.



Solution :

Since the equipotential surface are equidistant and parallel, hence electric field intensity is magnitude an perpendicular to the equipotential surfaces



(Direction of E will be perpendicular to the shown line)

the rate of fall of potential with distance is uniform.

$$\therefore \qquad E_x = \frac{-dV}{dx} = -\left(\frac{\Delta V}{\Delta x}\right) \quad \left[\because \left(\frac{dV}{dx}\right) \text{ is uniform}\right]$$

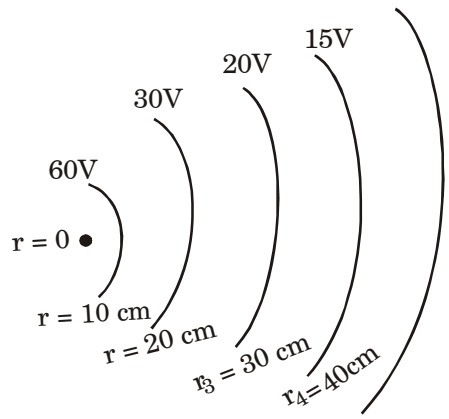
$$E \cos 60 = \frac{(-40 - 30)}{(5-10)\times 10^{-2}} \text{ V/m}$$

$$\frac{E}{2} = \left(\frac{10}{5}\right)\times 10^2 = V/m = 2 \quad 10^2 \text{ V/m}$$

$$\Rightarrow \qquad \boxed{E = 4 \quad 10^2 \text{ V/m}}$$

Example 16

The given figure represents part of concentric spheres which are equipotential surfaces. Find the magnitude of the field.



Solution : From the given diagram, we conclude that,

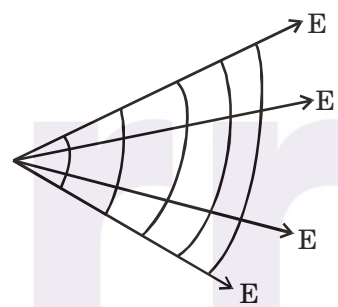
$$(Vr) = \text{const.} = C \text{ (say)}$$

$$\Rightarrow V = (C/r) \text{ where } C = 60 \quad 10^{-1} = 6 \text{ (V/m)}$$

$$\Rightarrow \left(\frac{dV}{dr}\right) = \left(\frac{-c}{r^2}\right)$$

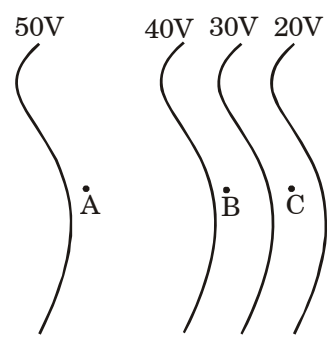
$$Er = -\left(\frac{dV}{dr}\right) = \left(C/_{r^2}\right)$$

$$\boxed{E = \frac{6}{r^2}}$$



Example 17

Find the relation between the field intensity at A, B and C.



Solution : $(V_C > V_B > V_A)$

Reason : $E = -\left(\frac{dV}{dr}\right)$

Since ΔV is same between two equipotential surface.

and $E \propto \frac{1}{r}$ $\therefore$ where Δr is small E will be large.

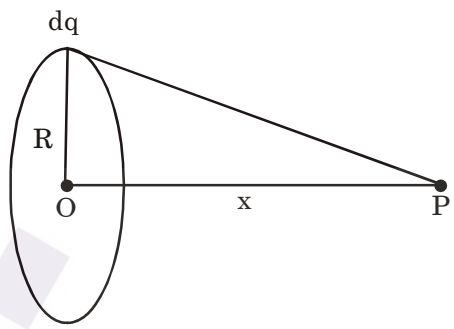
In general rule : To find the field due any type of charge distribution.

use $\vec{dE}$ → dE_x, → dE_y, → dE_z] Integrate separately

9. Electric Potential And field intensity due to Various Charge Distribution

I. Uniformly charged circular ring

(i) Electrical potential



Consider an infinitesmal element of charge dq on the circumference of the ring, then the electric potential due to this charge is

$$dV = \frac{dq}{4\pi\varepsilon_0 \sqrt{x^2 + R^2}}$$

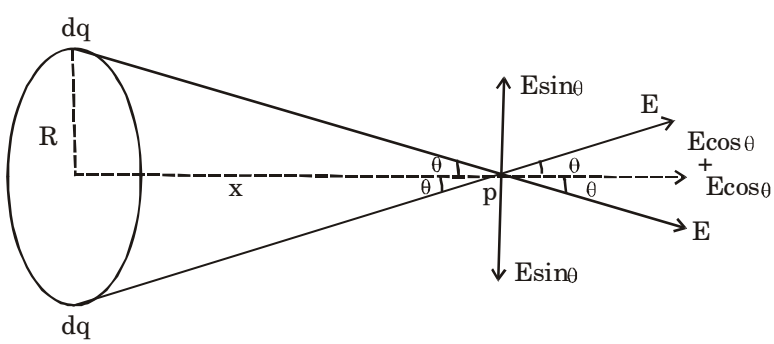
Integrating,

$$\therefore V = \frac{1}{4\pi \in_0 \sqrt{x^2 + R^2}} \int dq$$

$$\Rightarrow \quad \boxed{V = \frac{Q}{4\pi \in_0 \sqrt{x^2 + R^2}}}$$

This result for potential remains same even if the charge distribution is non-uniform.

(ii) **Electric field intensity at an axial point**



Consider small elements of charge dq on the circumference

Note that the net field will be in the axial direction, since the vertical components will cancel out.

$$\therefore \quad dE_x = \int dE \cos\theta$$

$$dE_x = \frac{dq}{(4\pi \epsilon_0)\left(R^2 + x^2\right)} \times \frac{x}{\left(R^2 + x^2\right)^{1/2}} \qquad \left\{\cos\theta = \frac{x}{\sqrt{R^2 + x^2}}\right\}$$

$$\therefore \quad E = \frac{x}{(4\pi \epsilon_0)\left(R^2 + x^2\right)^{3/2}} \int dq = \frac{Qx}{4\pi \epsilon_0 \left(R^2 + x^2\right)^{3/2}}$$

$$\boxed{E = \frac{Q\,x}{4\pi \epsilon_0 \left(R^2 + x^2\right)^{3/2}}}$$

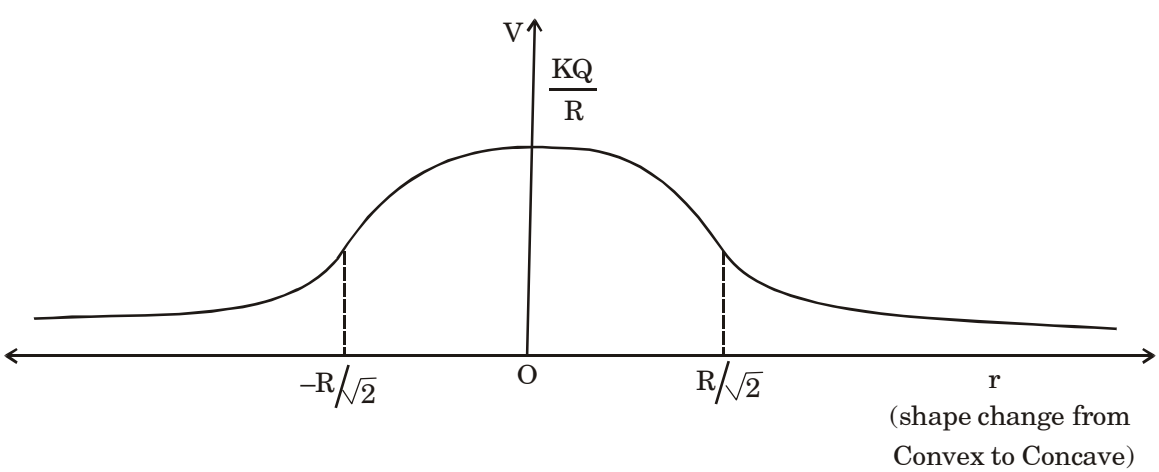
2nd Method : Differentiate the result for the field intensity.

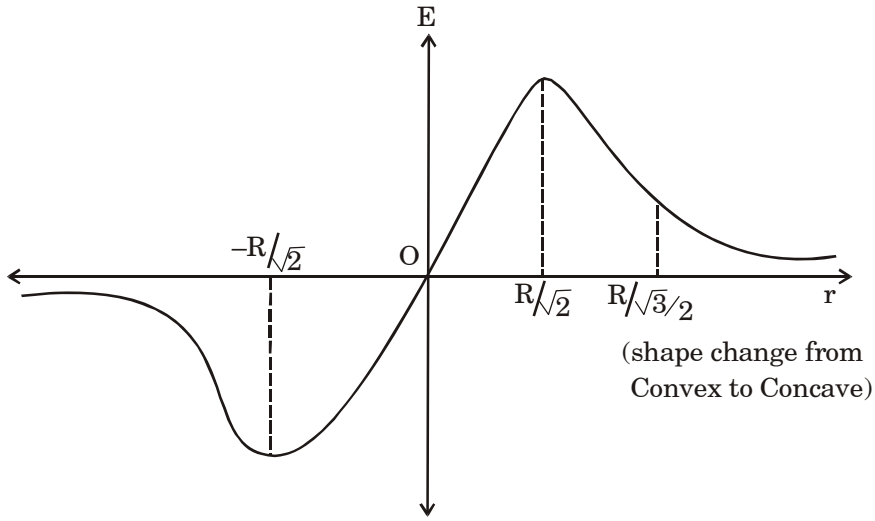
Potential at P due to ring

$$V = \frac{1}{4\pi \epsilon_0} \frac{Q}{\sqrt{R^2 + x^2}}$$

and $$E_r = \frac{-dV}{dx} = \frac{-Q}{4\pi \epsilon_0} \left\{\frac{-2x}{2\left(R^2 + x^2\right)^{3/2}}\right\} = \frac{Q\,x}{4\pi \epsilon_0 \left(x^2 + R^2\right)^{3/2}}$$

(1)





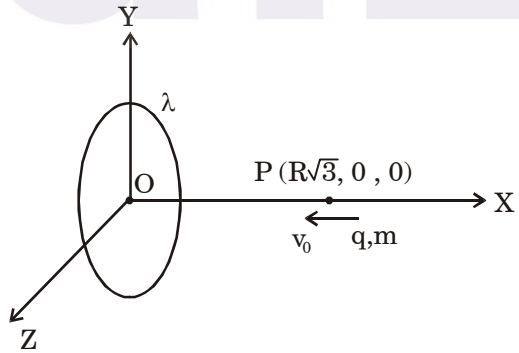
Note : Maximum fied exists at $\left(R/\sqrt{2}\right)$ from the ring.

Example 18

A uniformly charged ring of charge density λ and radius R is located in y-z plane with its centre at origin. Now a point charge q is projected with velocity v_0 from a point P as shown. Find the minimum value of v_0 so that the particle will not return to point P.

Solution :

Let us consider that the ring is placed in the Y-Z plane centred at origin. The point P is then taken to be at P $\left(R\sqrt{3}, 0, 0\right)$ on the X-axis. The point charge is projected towards the centre of the ring along the X-axis.



Concept : v_0 must be sufficient so that the particle just reaches the centre for minimum required velocity.

So, we take $V_P = V_0$ {Given}

and $v_{final} = 0$

Now, applying energy conservation

$$\Delta K + \Delta U = 0$$

$$\Rightarrow \quad \frac{1}{2} m v_P^2 - \frac{1}{2} m v_{final}^2 = q\left(V_0 - V_P\right)$$

also, $$V_0 = \frac{Q}{4\pi\epsilon_0 \sqrt{R^2}} = \frac{Q}{4\pi\epsilon_0 R}$$

$$Q = \lambda\ (2\pi R)$$

$$\Rightarrow \quad V_0 = \frac{\lambda(2\pi R)}{4\pi \epsilon_0 R}$$

$$\Rightarrow \quad V_P = \frac{\lambda\left(2\pi R\right)}{4\pi\epsilon_0 \sqrt{R^2 + \left(R\sqrt{3}\right)^2}} = \frac{\lambda\left(2\pi R\right)}{4\pi \epsilon_0 \left(2R\right)}$$

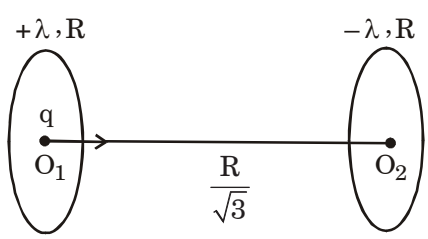
$$\therefore \quad \frac{1}{2} m v_0^2 = q \frac{\lambda\left(2\pi R\right)}{4\pi\epsilon_0}\left(\frac{1}{R} - \frac{1}{2R}\right)$$

$$\Rightarrow \quad \frac{1}{2} m v_0^2 = \frac{q\lambda}{4\epsilon_0}$$

$$\Rightarrow \quad \boxed{v_0 = \sqrt{\frac{\lambda q}{2m\epsilon_0}}}$$

Example 19

Two uniformly charged ring are separated by a distance $R/\sqrt{3}$. A point charge q of mass m is projected with initial velocity v_0 from A to B. Find the velocity when it reaches the centre of negetively charged ring.



Solution :

Let the two rings be centred at O_1 and O_2 having charge densities + λ and – λ rspectively.

A charge q is projected with a velocity v_0 from O_1 toward O_2.

Let v be the velocity when it reaches O_2. Now, potential at centre O_1

$$V_{O_1} = \frac{+\lambda R}{2\epsilon_0 R} - \frac{\lambda R}{2\epsilon_0\left(\sqrt{R^2 + \left(R\sqrt{3}\right)}\right)^2}$$

$$= \frac{\lambda R}{2\epsilon_0 R} - \frac{\lambda R}{4\epsilon_0 R}$$

$$\boxed{V_{O_1} = \frac{\lambda}{4\epsilon_0}}$$

also, potential at O_2

$$V_{O_2} = \frac{-\lambda R}{2\epsilon_0 R} + \frac{\lambda R}{2\epsilon_0\sqrt{R^2 + \left(R\sqrt{3}\right)^2}}$$

$$= -\frac{\lambda}{2\epsilon_0} + \frac{\lambda}{4\epsilon_0}$$

$$\boxed{V_{O_2} = -\frac{\lambda}{4\epsilon_0}}$$

Now, applying energy conservation between O_1 and O_2

$\Delta K + \Delta U = 0$

$$\Rightarrow \left(\frac{1}{2}mv^2 - \frac{1}{2}mv_0^2\right) = q\left(V_{O_1} - V_{O_2}\right)$$

$$\Rightarrow \frac{1}{2}m\left(v^2 - v_0^2\right) = \frac{\lambda q}{2\epsilon_0}$$

$$\Rightarrow \boxed{v = \sqrt{v_0^2 + \frac{\lambda q}{m\epsilon_0}}}$$

Example 20

If a negative charge is slightly displaced from the centre of a uniformly charged ring. Show that it will execute SHM, and find its time period.

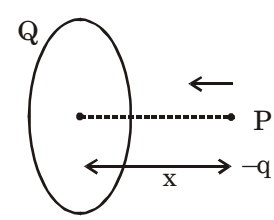
Solution :

When the charge is slightly displaced, it experiences a force towards the centre of the ring as the electric field due to the ring is directed away from it, hence a negatively charged particle will experience a force towards its centre.

Hence, Net force = ma

or $$-qE = m\frac{d^2x}{dt^2}$$

and $$E = \frac{Qx}{4\pi\epsilon_0\left(R^2 + x^2\right)^{3/2}}$$



$$\therefore \qquad m\frac{d^2x}{dt^2} = \frac{-qQx}{4\pi\epsilon_0\left(x^2+R^2\right)^{3/2}}$$

Now, since x << R

$$\therefore \qquad m\frac{d^2x}{dt^2} = -\left(\frac{qQx}{4\pi\epsilon_0 R^3}\right) \quad \text{\{equation for S.H.M.\}}$$

$$\therefore \qquad \boxed{T = 2\pi\sqrt{\frac{4\pi\epsilon_0 m R^3}{Qq}}}$$

Example 21

A small charged particle of mass m and charge q is suspended vertically at one end of ideal spring, and initially lying at the centre of ring in of charge Q in equilibrium. When it is slightly displaced, find its

(a) time period of oscillation

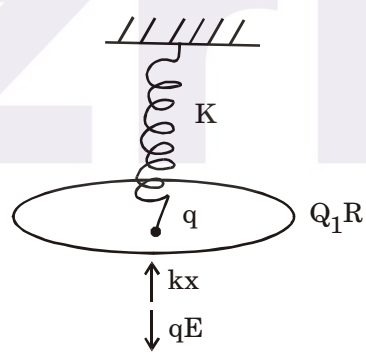
(b) required condition to execute SHM

Solution :

Similar to the example above, the particle experiences a force qE after being displaced. Hence the force balance in this condition is

$$m\left(\frac{d^2x}{dt^2}\right) = -kx + \frac{qQx}{4\pi\epsilon_0\left(x^2+R^2\right)^{3/2}}$$

$$= -\left[k - \frac{qQ}{4\pi\epsilon_0 R^3}\right]x$$



(a) $$T = 2\pi\sqrt{\frac{4\pi\epsilon_0 R^3}{\left(4\pi\epsilon_0 R^3 k - qQ\right)}}$$

(b) Necessary condition, $4\pi\epsilon_0 R^3 k - qQ > 0$

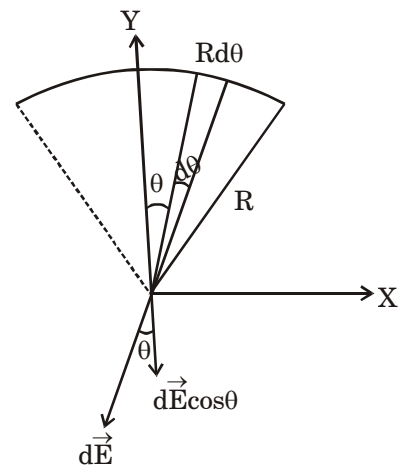
$$\boxed{K > \frac{qQ}{4\pi\epsilon_0 R^3}}$$

Example 22

Find the field intensity at the centre due to a uniformly charged circular arc having an angle θ_0 subtended at the centre.

Solution :

Consider an element of thickness Rdθ making an angle θ with the Y-axis. The electric field $\left(d\vec{E}\right)$ due to this element at the centre is directed at angle θ from the negative Y-axis. The component of this field along the X-axis will cancel out with the compound of electric field from the corresponding element making an angle – θ from the Y-axis. Thus we will have a net electric field in the – Y-axis direction.



$$\therefore \qquad E_{net} = \int dE\cos\theta = \int_{-\frac{\theta_0}{2}}^{\frac{\theta_0}{2}} \frac{\lambda\, R\, d\theta \cos}{4\,\epsilon_0\, R^2}$$

$$= \frac{\lambda}{4\pi\,\epsilon_0\, R}\left[\sin\theta\right]_{-\frac{\theta_0}{2}}^{\frac{\theta_0}{2}} = \frac{\lambda}{4\pi\,\epsilon_0\, R}\left(2\sin\frac{\theta_0}{2}\right)$$

$$\Rightarrow \qquad \boxed{E_{net} = \frac{\lambda}{2\pi\,\epsilon_0\, R}\sin\left(\frac{\theta_0}{2}\right)}$$

Point to Remember

Electric field intensity due to a uniformly charged arc = $\frac{\lambda}{2\pi\,\epsilon_0\, R}\sin\frac{\theta_0}{2} = \frac{Q}{4\pi\,\epsilon_0}\left(\frac{\sin\frac{\theta_0}{2}}{\frac{\theta_0}{2}}\right)$

subtending an ange θ_0 at centre

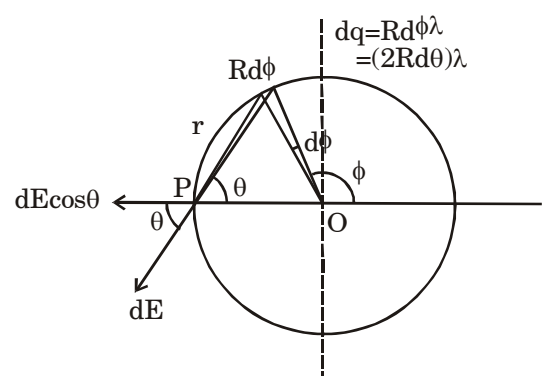
9.1 Field intensity at the circumference of the ring

$\phi = 2\theta$, $(d\phi = 2d\theta)$ {ϕ is the exterior angle}

Also, $r = 2R\cos\theta$ (triangle in semi-circle)

$\therefore \quad dq = (2R\, d\theta)\lambda$

Net field at point P will be along x-axis because of symmetric cancellation of y components.



$$E = \int dE\cos\theta$$

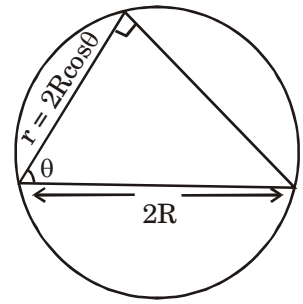
$$= \int\left(\frac{K\,dq}{r^2}\right)\cos\theta$$

$$= \int_{-\pi/2}^{\pi/2} \frac{k(2R\,d\theta)\lambda}{(2R\cos\theta)^2}\cos\theta$$

$$= \int_{-\pi/2}^{\pi/2} \frac{k(2R\,d\theta)\lambda}{(2R\cos\theta)^2}\cos\theta$$

$$- = \left(\frac{k\lambda}{2R}\right)\int_{\pi/2}^{\pi/2} \sec\theta\,d\theta$$

$$E = \text{indeterminate}$$



But note when λ will be some function of θ, E will may become determinate.

Potential energy at any point due to a point charge associated with a charge

$$\vec{F} = q\vec{E}$$

$$U = qV$$

$$\Rightarrow \quad \boxed{U = \frac{Kq_1 q_2}{r_0}}$$

Electric potential at the circumference of the ring

$$dV = \left(\frac{kdq}{r}\right) = \frac{k(2R\,d\theta)\lambda}{(2R\cos\theta)} = \frac{k\lambda\,d\theta}{(\cos\theta)}$$

$$\therefore \quad V = k\lambda\int_{\pi/2}^{\pi/2} \sec\theta d\theta = \text{(indeterminate)}$$

(Note again that though this limit is indeterminate you may calculate it when λ varies with θ)

Example 23

A point charge q is brought to the centre from the circumference of a non-uniformly charged ring of charge density $\lambda = \lambda_0 \cos\theta$. Find the work required by the external agent in doing so.

Solution : Now in this question potential is not indeterminate

$$V = K\lambda_0 \int_{-\pi/2}^{\pi/2} \sec\theta\cos\theta\,d\theta = (K\lambda_0\pi)$$

Now, calculate the potential at the centre

$$\int dV = \int_0^{2\pi} \frac{K(\lambda_0 \cos\theta) R\, d\theta}{R}$$

$$V = K\lambda_0 \left((-\sin\theta)\right)_0^{2\pi} = 0$$

$\therefore$ Work required by external agent

$$\Delta W_{ex} = \Delta U$$
$$= q\,(V_0 - V_p) = q\,(0 - k\lambda_0\pi)$$
$$= -\,kq\lambda_0\pi$$

Example 24

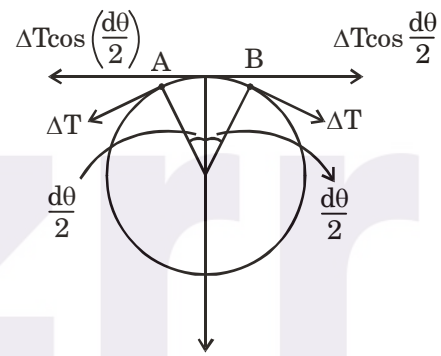
A point charge q is placed at the centre of a uniformly charged (λ) circular ring. Find the increase in the radius of ring if Y = Young's modulus and area of cross-section = S

Solution :

Consider the force on a small element of the ring of charge dQ due to point charge q.

$$F = \frac{K\,q\,dQ}{R^2}$$

As can be seen from the figure, the net force of tension on the element AB comes towards the centre which is



$$2\Delta T \sin\frac{d\theta}{2} \approx \Delta T\, d\theta$$

Hence,

$$F = \Delta T d\theta$$

$$\Rightarrow \quad \frac{K\,q\,dQ}{R^2} = \Delta T d\theta$$

Now, $\quad dQ = \lambda\,(R d\theta)$

$$\therefore \quad \Delta T = \frac{K\lambda\,R\,q}{R^2} = \frac{K\lambda\,q}{R}$$

also, using Young's modulus

$$Y = \frac{\left(\frac{T}{S}\right)}{\left(\frac{\Delta l}{l}\right)}$$

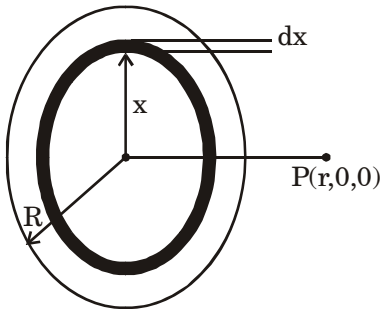
$$\Rightarrow \quad \frac{T}{S} = Y\left(\frac{\Delta R}{R}\right) \qquad \{\therefore\ \Delta l = 2\pi\Delta R\}$$

$$\therefore \quad \Delta R = \left(\frac{K\lambda\,q}{R}\right)\left(\frac{R}{YS}\right) = \boxed{\frac{K\,\lambda\,q}{Y\,S}}$$

II. UNIFORMLY CHARGED DISC

(i) Electric Potential

The disc has a charge density σ (charge per unit area)



For analyzing a charged disc, consider a ring of thickness dx at a distance x from the centre;

$dq = \sigma\,(2\pi x)\,dx$ {charge on the ring of thickness dx}

Now, Potential at P due to the ring

$$dV = \frac{K\,dq}{\sqrt{x^2 + r^2}}$$

On integrating, we get the potential for entire disc.

$$\therefore \quad V = \int_0^R \frac{K\sigma\,(2\pi x)\,dx}{\sqrt{x^2 + r^2}}$$

$$= K\,\sigma\,(2\pi)\int_0^R \frac{x\,dx}{\sqrt{x^2 + r^2}}$$

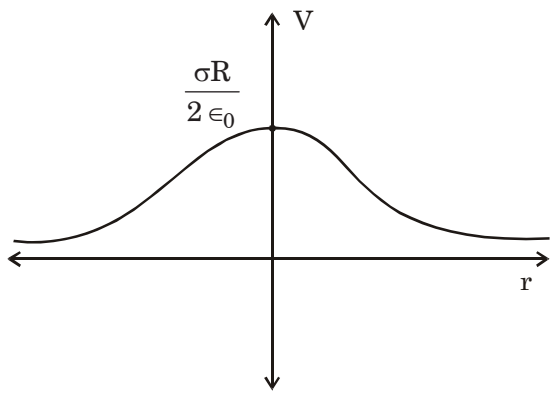
$$\boxed{V = K\,\sigma\,2\pi\left\{\sqrt{R^2 + r^2} - |r|\right\}}$$

(ii) **Electric Field Intensity**

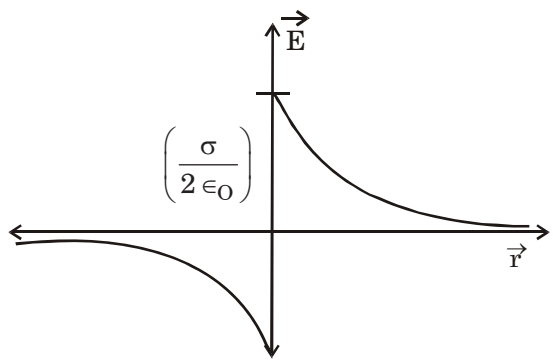
Now, we know that

$$E = \frac{-dV}{dr} = -\frac{\sigma}{2\epsilon_0}\left(\frac{2r}{2\sqrt{R^2 + r^2}} - 1\right]$$

$$\boxed{E = \frac{\sigma}{2\epsilon_0}\left[1 - \frac{r}{\sqrt{R^2 + r^2}}\right]}$$



Plot of potential v/s distance from the disc



Plot of field intensity v/s distance from the disc

If $r \to 0$, $E = \frac{\sigma}{2\epsilon_0}$

Thus, near a uniformly charged surface, E can be taken = $\frac{\sigma}{2\epsilon_0}$

III. UNIFORMLY CHARGED SHELL

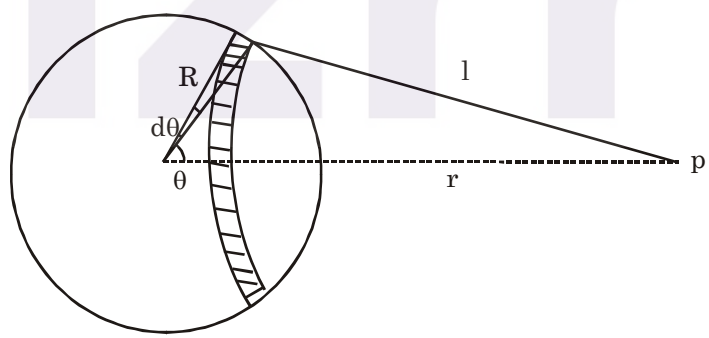
(i) Electric Potential

For analyzing a charged shell, we consider a ring of thickness $Rd\theta$ at an angle of θ from the horizontal.

The shell has a charged density σ and radius R, $\sigma = \frac{Q}{4\pi R^2}$

So, charge on a ring of thickness $Rd\theta$

$$dq = \sigma\,(2\pi R \sin\theta)\,R\,d\theta$$



$$\therefore \quad dV = \frac{dq}{4\pi\epsilon_0\, l}$$

$$dV = \frac{\sigma\left(2\pi R^2 \sin\theta\right)d\theta}{4\pi\epsilon_0\, l}$$

$$= \frac{KQ}{\left(4\pi R^2\right)}\left(2\pi R^2\right)\left(\frac{\sin\theta\, d\theta}{l}\right) = \frac{KQ}{2}\left(\frac{\sin\theta\, d\theta}{l}\right)$$

$$\cos\theta = \frac{r^2 + R^2 - l^2}{2rR}$$

$$2rR\cos\theta = r^2 + R^2 - l^2$$

$$-2r\,R\sin\theta\,d\theta = -2l\,dl$$

$$dl = rR\left(\frac{\sin\theta d\theta}{l}\right)$$

$$V = \int dV = \frac{KQ}{2}\int_{(r-R)}^{r+R}\frac{dl}{rR} = \frac{KQ}{2rR}(r+R-r+R) = \left(\frac{KQ}{r}\right)$$

$$\therefore \quad V = \left(\frac{KQ}{r}\right) \text{ when } r > R$$

for $r < R$

$$V = \int dV = \int_{(R-r)}^{(R+r)}\frac{(KQ)dl}{2rR} = \left(\frac{KQ}{R}\right), r<R$$

Summarizing (Uniformly charged shell)

(i) **For any external part** ($r > R$)

(a) Electric potential

$$= \frac{Q}{4\pi\epsilon_0 r}$$

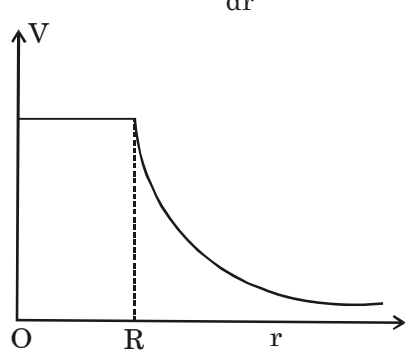
(b) Electric field $= \frac{-dV}{dr} = \frac{Q}{4\pi\epsilon_0 r^2}$

Note : For an external point a shell has similar formulae as a point charge hence this point can be very useful for some objective questions.

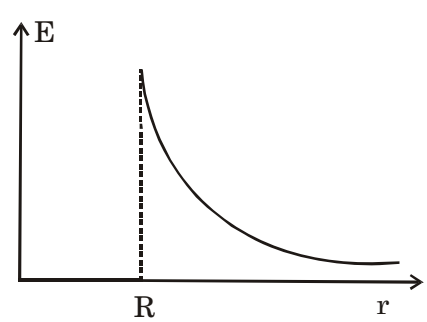
(ii) **For any internal point ($r < R$)**

(a) Electric potential $= \frac{Q}{4\pi\epsilon_0 R}$

(b) Electric field $= \frac{-dV}{dr} = 0$



Plot of potential v/s distance from the centre of shell.



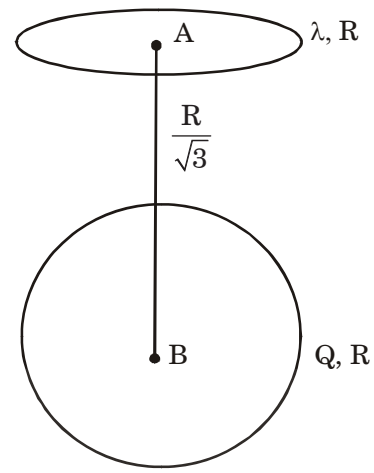
Plot of electric field intensity V/s distance from the centre of shell.

Example 25

A uniformly charged circular ring is placed at $R/\sqrt{3}$ from a uniformly charged shell. Find the force of interaction between the sphere and the ring.

Solution :

Electric field due to the ring at a distance $R\sqrt{3}$ from A



$$E = \frac{Qx}{4\pi \in_0 \left(\sqrt{x^2 + R^2}\right)^3}$$

$$\therefore \quad E = \frac{\lambda(2\pi R)x}{4\pi \in_0 \left(x^2 + R^2\right)^{3/2}}$$

$$E = \frac{\lambda R x}{2\left(x^2 + R^2\right)^{3/2} \in_0} \quad \therefore \quad E_{R/\sqrt{3}} = \left(\frac{\lambda\sqrt{3}}{16 R \in_0}\right)$$

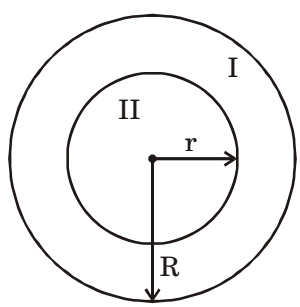
$\therefore$ Since a shell can be treated as a point charge outside its surface.

$$\therefore \quad F = QE = \left(\frac{\lambda Q\sqrt{3}}{16 \in_O R}\right)$$

IV. Uniformly Charged Sphere

(A) Electric Potential

To calculate electric potential at a point inside the sphere, we divide the entire spherical volume into two spherical parts I and II i.e. the region containing charge outside the point and the region inside the point.

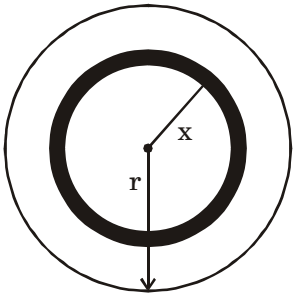


The sphere has a charge density

ρ (charge per unit volume)

$$\rho = \frac{Q}{\frac{4}{3}\pi R^3}$$

Now, **consider region II**



$$dV_1 = K\left(\frac{dq}{r}\right)$$

{Using the formula for potential of external point for concentric shell}

$\therefore$ $$V_1 = \int_0^r K\,\frac{dq}{r}, \qquad \int_0^r dq = \rho\left(\frac{4}{3}\pi r^3\right)$$

$\Rightarrow$ $$V_1 = \left(\frac{K}{r}\right)\left(\rho\,\frac{4}{3}\pi r^3\right)$$

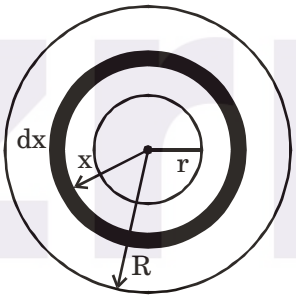
$$\rho = \frac{Q}{\frac{4}{3}\pi R^3}$$

$\Rightarrow$ $$\boxed{V_1 = \frac{K\,Q r^2}{R^3}}$$

Consider Region I

$$dV_2 = K\,\frac{dq}{x},\ dq = \rho\,(4\pi\,x^2)\,dx$$

$\therefore$ $$V_2 = \int_r^R K\,\frac{\left(4\pi x^2\right)dx}{x}\,\rho$$



$\Rightarrow$ $$V_2 = K\,\rho\,(4\pi)\int_r^R x\,dx$$

$$= 4\pi\,\rho K\left(\frac{R^2 - r^2}{2}\right)$$

$$= \frac{4\pi Q K}{\frac{4}{3}\pi R^3}\left(\frac{R^2 - r^2}{2}\right)$$

$$\boxed{V_2 = \frac{3\,KQ}{2\,R^3}\left(R^2 - r^2\right)}$$

Hence, net potential (V_{net})

$$V_{net} = V_1 + V_2 = \frac{KQr^2}{R^3} + \frac{3KQ}{2R^3}\left(R^2 - r^2\right) = \frac{KQ}{2R^3}\left(3R^2 - r^2\right)$$

Summarizing for a uniformly charged sphere :

(i) For an internal point ($r < R$)

Net potential inside the sphere = $\dfrac{Q\left(3R^2 - r^2\right)}{4\pi \in_0 \left(2R^3\right)}$

Net field inside the sphere

$$E = -\frac{dV}{dr} = \frac{2Qr}{4\pi \in_O (2R^3)} = \frac{\rho r}{3 \in_O}$$

This formula is of importance and can be noted in a vector form as well i.e.

$$\boxed{\vec{E} = \frac{\rho \vec{r}}{3 \in_O}}$$

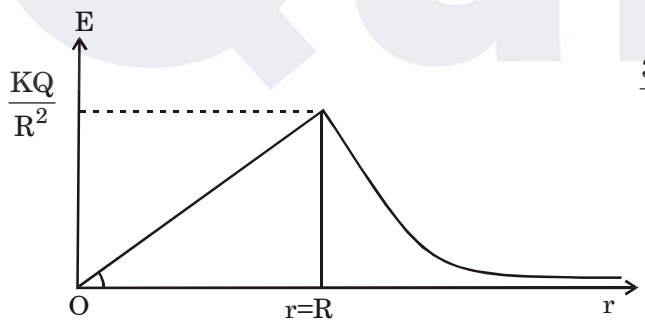
(ii) $r > R$

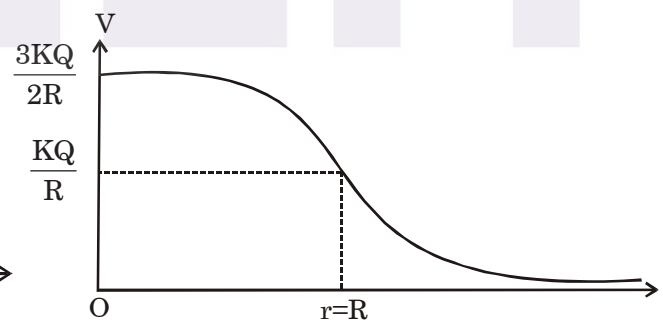
Net potential outside the sphere

$$= \frac{K}{r}\int dq = \left(\frac{KQ}{r}\right) = \left(\frac{Q}{4\pi \in_O r}\right)$$

Net field = $-\left(\dfrac{dV}{dr}\right) = \dfrac{Q}{4\pi \in_O r^2}$

Note : A uniformly charged sphere also behave as a point charge for any external point.



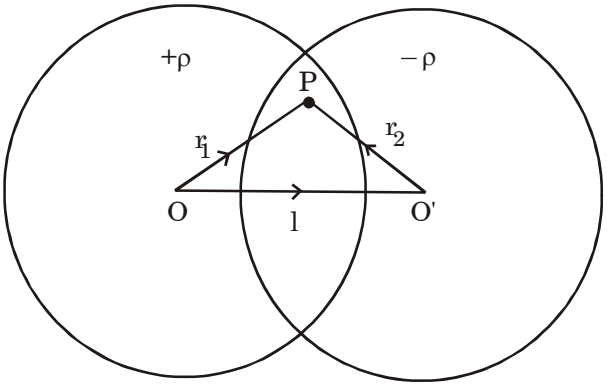


10. ELECTRIC FIELD INTENSITY IN A CAVITY

Example 26

Find the field intensity in the region of intersection of two uniformly charged sphere with volume charge density +ρ and –ρ. The distance between centres is l**.**

Solution :



At the point P, electric field due to sphere O and O´ can be calculated and added vectorially

$$\vec{E}_O = \frac{\rho \vec{r_1}}{3\epsilon_O} \quad , \quad \vec{E}_{O'} = \frac{(-\rho)\vec{r_2}}{3\epsilon_O}$$

$$\vec{E}_{net} = \vec{E} = \vec{E}_O + \vec{E}_{O'}$$

$$\therefore \vec{E} = \left(\frac{\rho\vec{r_1}}{3\epsilon_O}\right) - \left(\frac{\rho\vec{r_2}}{3\epsilon_O}\right) = \frac{\rho}{3\epsilon_O}\left(\vec{r_1} - \vec{r_2}\right) = \frac{\rho}{3\epsilon_O}\vec{OO'} = \left(\frac{\rho\vec{l}}{3\epsilon_O}\right)$$

Remark :

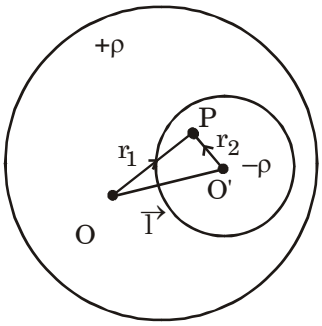
(i) Thus, it can be easily seen that the field intensity comes out to be consistent inside the common area and is directed from positive to negative centres of the sphere.

(ii) Here, ρ and –ρ must be same.

Example 27

CAVITY (it is a special case of previous question).

A sphere with centre O is filled with a charge density + ρ. There exists a cavity inside this sphere which is also a spherical cavity with centre O´ at a distance $\vec{l}$ from the centre O. Find the electric field strength at a point P inside the cavity ?



Solution :

Concept : Fill the cavity with charge densities + ρ and – ρ so that the cavity is filled.

This makes it a special case of previous example where the common volume between the 2 spheres is same as one of the sphere.

Hence, $\vec{E}_0 = \dfrac{\rho \vec{r}_1}{3\epsilon_0}$, $\vec{E}_{0'} = \dfrac{(-\rho)\vec{r}_2}{3\epsilon_0}$

$$\vec{E} = \vec{E}_0 - \vec{E}_{0'} = \frac{\rho\left(\vec{r}_1 - \vec{r}_2\right)}{3\epsilon_0}$$

$$= \frac{\rho \vec{l}}{3\epsilon_0} \quad \left\{\because \vec{r}_1 - \vec{r}_2 = \vec{l}\right\}$$

Example 28

Find the field intensity at the centre of a charged spherical shell whose surface charge density varies as $\sigma = \left(\vec{a}.\vec{r}\right)$ where $\vec{a}$ is a constant vector and $\vec{r}$ is the radius vector.

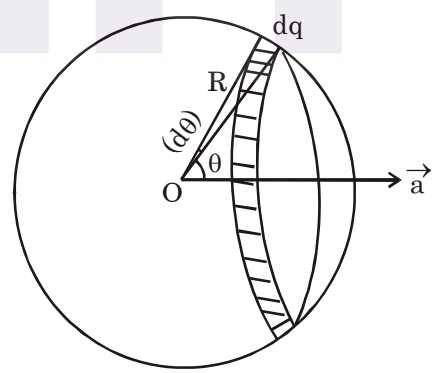
Solution :

For convenience, take the direction of the $\vec{a}$ vector along the diametric line, and the angle θ is measured from the $\vec{a}$ vector direction.

$dq = \sigma(2\pi R\sin\theta)Rd\theta$ {charge of ring at an angle θ}

$= (a\ R\cos\theta)\ (2\pi R^2 \sin\theta\ d\theta)$

$= 2\ \pi\ a\ R^3 \sin\theta \cos\theta\ d\theta$



Now, electric field for a ring is given by

$$\therefore \quad dE = \frac{dqx}{4\pi\epsilon_O R^3} = \frac{(dq)\,R\cos\theta}{4\pi\epsilon_O R^3}$$

$$\therefore \quad E = \frac{2\pi\, aR^3 \sin\theta\cos\theta\, d\theta(R\cos\theta)}{4\pi\epsilon_O R^3}$$

$$= \frac{aR}{2\epsilon_O}\int_0^{\pi}\cos^2\theta\sin\theta\, d\theta$$

$$=$$

$$\therefore \quad \left|\vec{E}\right| = \left(\frac{aR}{3\epsilon_O}\right)$$ and directed opposite to constant vector $\vec{a}$

Example 29

Three point charges q_1 = 1μC, q_2 = – 2μC and q_3 = 3 μC are placed at (1 m, 0, 0), (0, 2m, 0) and (0, 0, 3m) respectively. Find the electric potential at origin.

Solution :

The net electric potential at origin is,

$$V = \frac{1}{4\pi\epsilon_0}\left[\frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3}\right]$$

Substituting the values, we have

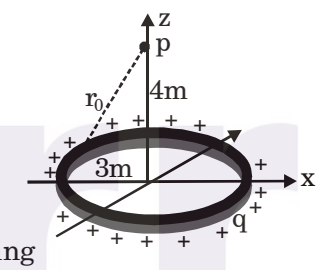
$$V = \left(9.0 \times 10^9\right)\left(\frac{1}{1.0} - \frac{2}{2.0} + \frac{3}{3.0}\right) \times 10^{-6}$$

$$= 9.0 \quad 10^3 \text{ volt.}$$ **Ans.**

Example 30

A charge q = 10 μC is distributed uniformly over the circumference of a ring of radius 3 m placed on x-y plane with its centre at origin. Find the electric potential at a point P (0, 0, 4 m).

Solution :



The electric potential at point P would be,

$$V = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r_0}$$

Here, r_0 = distance of point P from the circumference of ring

$$= \sqrt{(3)^2 + (4)^2} = 5\,\text{m}$$

and $q = 10\ \mu C = 10^{-5}$ C

Substituting the values, we have

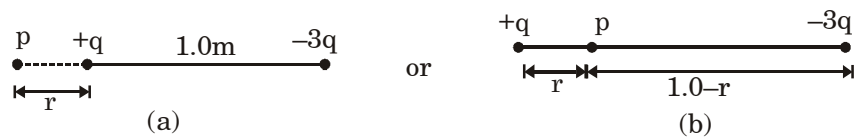
$$V = \frac{\left(9.0 \times 10^9\right)\left(10^{-5}\right)}{(5.0)} = 1.8 \times 10^4 \text{ volt}$$ **Ans.**

Example 31

Find out the points on the line joining two charges + q and – 3q (kept at a distance of 1.0 m) where electric potential is zero.

Solution :

Let P be the point on the axis either to the left or to the right of charge + q at a distance r where potential is zero. Hence,



For case (a) $$V_P = \frac{q}{4\pi\epsilon_O r} - \frac{3q}{4\pi\epsilon_O(1+r)} = 0$$

Solving this, we get r = 0.5 m

Further, for case (b) $$V_P = \frac{q}{4\pi\epsilon_O r} - \frac{3q}{4\pi\epsilon_O(1-r)} = 0$$

which gives r = 0.25 m

Thus, the potential will be zero at point P on the axis which is either 0.5 m to the left or 0.25 m to the right of charge + q.

Example 32

Four point charges + 8 μC, – 1 μC, – 1 μC and + 8 μC are fixed at the points $-\sqrt{27/2}$ m, $-\sqrt{3/2}$ m, $+\sqrt{3/2}$ m and $+\sqrt{27/2}$ m respectively on the Y-axis. A particle of mass 6×10^{-4} kg and charge + 0.1 μC moves along the – X direction. Its speed at x = + ∞ is v_0. Find the least value of v_0 for which the particle will cross the origin. Find also the kinetic energy of the particle at the origin. Assume that space is gravity free. (JEE 2000)

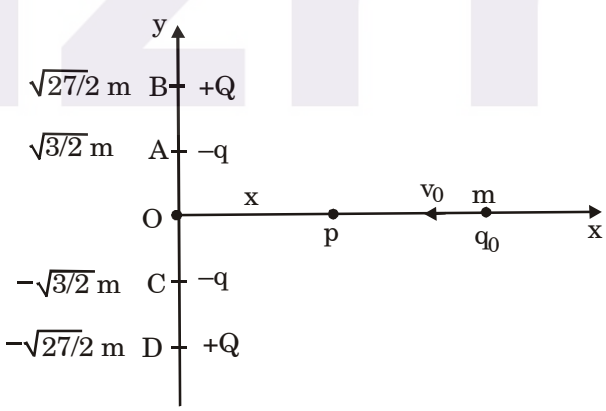
Solution : In the figure,

$q = 1\ \mu C = 10^{-6}$ C

$q_0 = +\ 0.1\ \mu C = 10^{-7}$ C

$m = 6 \quad 10^{-4}$ kg

and $Q = 8\ \mu C = 8 \quad 10^{-6}$ C



Let P be any point at a distance x from origin O. Then

$$AP = CP = \sqrt{\frac{3}{2} + x^2}$$

$$BP = DP = \sqrt{\frac{27}{2} + x^2}$$

Electric potential at point P will be

$$V = \frac{2KQ}{BP} - \frac{2Kq}{AP}$$

where $$K = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9\ Nm^2/C^2$$

$$\therefore \qquad V = 2 \times 9 \times 10^9 \left[\frac{8 \times 10^{-6}}{\sqrt{\frac{27}{2} + x^2}} - \frac{10^{-6}}{\sqrt{\frac{3}{2} + x^2}} \right]$$

$$V = 1.8 \times 10^4 \left[\frac{8}{\sqrt{\frac{27}{2} + x^2}} - \frac{1}{\sqrt{\frac{3}{2} + x^2}} \right] \qquad \text{... (i)}$$

$\therefore$ Electric field at P is

$$E = \frac{-dV}{dx} = -1.8 \times 10^4 \left[(8)\left(-\frac{1}{2}\right)\left(\frac{27}{2} + x^2\right)^{-3/2} - (1)\left(-\frac{1}{2}\right)\left(\frac{3}{2} + x^2\right)^{-3/2} \right](2x)$$

E = 0 on X-axis where

$$\frac{8}{\left(\frac{27}{2} + x^2\right)^{3/2}} = \frac{1}{\left(\frac{3}{2} + x^2\right)^{3/2}}$$

$$\Rightarrow \qquad \frac{(4)^{3/2}}{\left(\frac{27}{2} + x^2\right)^{3/2}} = \frac{1}{\left(\frac{3}{2} + x^2\right)^{3/2}}$$

$$\Rightarrow \qquad \left(\frac{27}{2} + x^2\right) = 4\left(\frac{3}{2} + x^2\right)$$

This equation gives $\quad x = \pm\sqrt{\frac{5}{2}}$ m .... (i)

The least value of kinetic energy of the particle at infinity should be enough to take the particle upto $x = +\sqrt{\frac{5}{2}}$ m because

at $x = +\sqrt{\frac{5}{2}}$ m, E = 0 $\Rightarrow$ Electrostatic force on charge q_0 is zero or $F_e = 0$

for $x > \sqrt{\frac{5}{2}}$ m, E is repulsive (towards positive X-axis)

and for $x < \sqrt{\frac{5}{2}}$ m, E is attractive (towards negative X-axis)

Now, from Eq. (i), potential at $x = \sqrt{\frac{5}{2}}$

$$V = 1.8 \quad 10^4 \left[\frac{8}{\sqrt{\frac{27}{2} + \frac{5}{2}}} - \frac{1}{\sqrt{\frac{3}{2} + \frac{5}{3}}} \right]$$

$V = 2.7 \quad 10^4$ volt

Applying energy conservation at $x = \infty$ and $x = \sqrt{\frac{5}{2}}$

$$\frac{1}{2} m v_0^2 = q_0 V \qquad \text{... (ii)}$$

$\therefore$ $$v_0 = \sqrt{\frac{2 q_0 V}{m}}$$

Substituting the values $$v_0 = \sqrt{\frac{2 \times 10^{-7} \times 2.7 \times 10^4}{6 \times 10^{-4}}}$$

$$v_0 = 3 \text{ m/s}$$ **Ans.**

$\therefore$ Minimum value of v_0 is 3 m/s

potential at origin (x = 0) is

$$V_0 = 1.8 \times 10^4 \left[\frac{8}{\sqrt{\frac{27}{2}}} - \frac{1}{\sqrt{\frac{3}{2}}} \right] = 2.4 \times 10^4 \text{ V}$$

Let T be the kinetic energy of the particle at origin.

Applying energy conservation at x = 0 and at $x = \infty$

$$T + q_0 V_0 = \frac{1}{2} m v_0^2$$

But $$\frac{1}{2} m v_0^2 = q_0 V$$

$\therefore$ $$T = q_0 (V - V_0)$$

$$T = (10^{-7}) (2.7 \quad 10^4 - 2.4 \quad 10^4)$$

$$T = 3 \quad 10^{-4} \text{ J}$$ **Ans.**

Note : E = 0 or F_e on q_0 is zero at x = 0 and $x = \pm\sqrt{\frac{5}{2}}$ m. Of these x = 0 is stable equilibrium position and $x = \pm\sqrt{\frac{5}{2}}$ m is unstable equilibrium position.

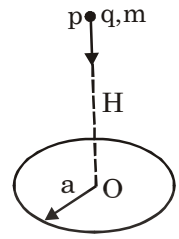
Example 33

A non-conducting disc of radius a **and uniform positive surface charge density σ is placed on the ground with its axis vertical. A particle of mass** m **and positive charge** q **is dropped, along the axis of the disc from a height H with zero initial velocity. The particle has** q/m **= 4ϵ_0 g/σ.**

(a) Find the value of H if the particle just reaches the disc.

(b) Sketch the potential energy of the partical as a function of its height and find its equilibrium position. (JEE 1999)

Solution : As we have derived in the theory



$$V_P = \frac{\sigma}{2\epsilon_0}\left[\sqrt{a^2 + H^2} - H\right]$$

Potential at centre, (O) will be

$$V_0 = \frac{\sigma a}{2\epsilon_0} \qquad (H = 0)$$

(a) Particle is released from P and it just reaches point O. Therefore, from conservation of mechanical energy.

Decrease in gravitational potential energy = increase in electrostatic potential energy

($\Delta KE = 0$ because $K_i = K_f = 0$)

$\therefore \quad mgH = q\,[V_0 - V_P]$

or $$gH = \left(\frac{q}{m}\right)\left(\frac{\sigma}{2\epsilon_0}\right)\left[a - \sqrt{a^2 + H^2} + H\right] \qquad \text{... (i)}$$

$$\frac{q}{m} = \frac{4\epsilon_0 g}{\sigma} \qquad \therefore \qquad \frac{q\sigma}{2\epsilon_0 m} = 2g$$

Substituting in Eq. (i), we get

$$gH = 2g\left[a + H - \sqrt{a^2 + H^2}\right]$$

or $$\frac{H}{2} = (a + H) - \sqrt{a^2 + H^2}$$

or $$\sqrt{a^2 + H^2} = a + \frac{H}{2} \qquad \text{or} \qquad a^2 + H^2 = a^2 + \frac{H^2}{4} + aH$$

or $$\frac{3}{4}H^2 = aH \qquad \text{or} \qquad H = \frac{4}{3}a \text{ and } H = 0$$

$\therefore \quad H = (4/3)\,a$ **Ans.**

(b) Potential energy of the particle at height H = Electrostatic potential energy + gravitational potential energy

$\therefore$ $$U = qV + mgH$$

Here V = Potential at height H

$\therefore$ $$U = \frac{\sigma q}{2\epsilon_0}\left[\sqrt{a^2 + H^2} - H\right] + mgH \quad \text{... (ii)}$$

At equilibrium position, $$F = \frac{-dU}{dH} = 0$$

Differentiating Eq. (ii) w.r.t. H

or $$mg + \frac{\sigma q}{2\epsilon_0}\left[\left(\frac{1}{2}\right)(2H)\frac{1}{\sqrt{a^2 + H^2}} - 1\right] = 0 \qquad \left[\frac{\sigma q}{2\epsilon_0} = 2mg\right]$$

$\therefore$ $$mg + 2mg\left[\frac{H}{\sqrt{a^2 + H^2}} = 1\right] = 0$$

or $$1 + \frac{2H}{\sqrt{a^2 + H^2}} - 2 = 0$$

$$\frac{2H}{\sqrt{a^2 + H^2}} = 1$$

or $$\frac{H^2}{a^2 + H^2} = \frac{1}{4} \text{ or } 3H^2 = a^2$$

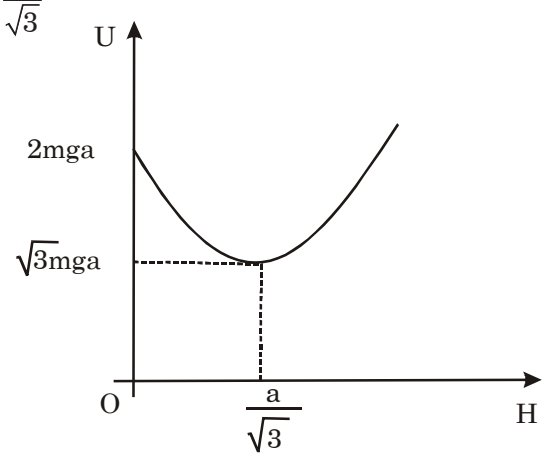
or $$H = \frac{a}{\sqrt{3}}$$ **Ans.**

From Eq. (ii), we can see that,

$$U = 2mga \text{ at } H = 0 \text{ and}$$

$$U = U_{min} = \sqrt{3}\ mga \text{ at } H = \frac{a}{\sqrt{3}}$$

Therefore, U-H graph will be as shown.



Note that at $H = \frac{a}{\sqrt{3}}$, U is minimum.

Therefore, $H = \frac{a}{\sqrt{3}}$ is stable equilibrium position.

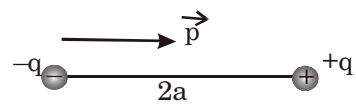
11. ELECTRIC DIPOLE

A pair of equal and opposite point charges ±q, that are separated by a fixed distance is known as **electric dipole.** Electric dipole occurs in nature in a variety of situations The hydrogen fluoride molecule (HF) is typical. When a hydrogen atom combines with a fluorine atom, the single electron of the former is strongly attracted to the latter and spends most of its time near the fluorine atom. As a result, the molecule consists of a strongly negative fluorine ion some (small) distance away from a strongly positive ion, though the molecule is electrically neutral overall.

Every electric dipole is characterized by its **electric dipole moment** which is a vector $\vec{\mathbf{P}}$ directed from the negative to the positive charge.

The magnitude of dipole moment is,

$$\boxed{P = (2a)\ q}$$



Here, 2a is the distance between the two charges.

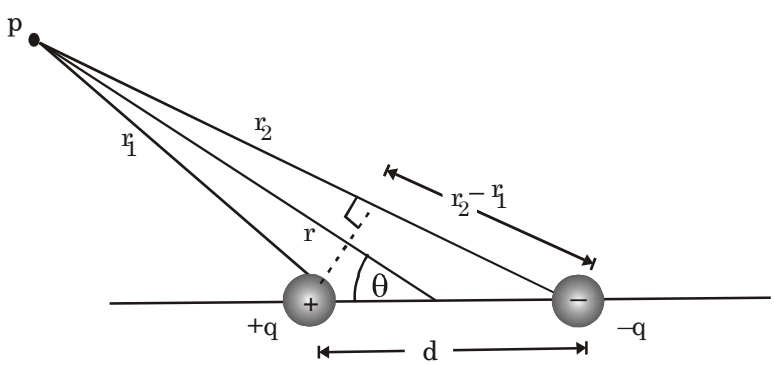
11.1 Electric Dipole Potential

The potential of an electric dipole can be found by superposing the point charge potentials of the two charges.

$$V = Kq\left[\frac{1}{r_1} - \frac{1}{r_2}\right] = Kq\left[\frac{r_2 - r_1}{r_1 r_2}\right]$$

For cases where r >> d, this can be approximated by

$$V = \frac{kp\cos\theta}{r^2}$$



The potential of a dipole is of most interest where r >> d. The standard approximations are

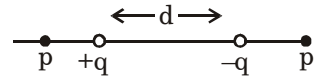
$r_2 - r_1 \approx d \cos\theta$

$r_1 r_2 \approx r^2$

where $\vec{p} = q\vec{d}$ is defined as the dipole moment.

Electric Potential at

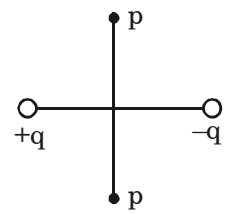
(a) End on position (a point lying on the axis of the dipole)



$\theta = 0$, or π

$$V = \left(\pm\frac{Kp}{r^2}\right)$$

(b) Board side on position (a point lying on the perpendicular bisector



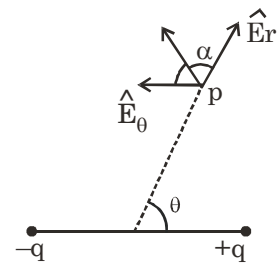
$\theta = \pm\pi/2$

$V = 0$

11.2 Electric field intensity due to an electric dipole

(Note that the electric field intensity must be lying in the same plane containing the point, the axis of the dipole

Find the field along the E_r and E_θ



$$E_r = \frac{-\partial V}{\partial r} = \frac{-\partial}{\partial r}\left(\frac{Kp\cos\theta}{r^2}\right) = \left(\frac{Kp\cos\theta}{r^3}\right)$$

$$E_\theta = \frac{-\partial V}{r\partial\theta} = \frac{-\partial}{\partial\theta}\left(\frac{Kp\cos\theta}{r^2}\right) = \left(\frac{Kp\sin\theta}{r^3}\right)$$

(Note that we put variable of differentiation as $r\partial\theta$ in case of E_θ)

$$\therefore \quad E_{net} = \sqrt{E_r^2 + E_\theta^2}$$

$$= \frac{Kp}{r^3}\sqrt{4\cos^2\theta + \sin^2\theta}$$

$$\boxed{E = \frac{Kp\sqrt{3\cos^2\theta + 1}}{r^3}}$$

To find the direction

$$\tan\alpha = \left(\frac{E_\theta}{E_r}\right) = \frac{\tan\theta}{2}$$

$\therefore \quad \alpha = \tan^{-1}\left(\frac{1}{2}\tan\theta\right)$ where α is the angle with E_r.

(a) **Electric field at end on position** (axial position)

$\theta = 0$ or π,

$$E_r = \left(\frac{2Kp}{r^3}\right) \qquad \left(E_{net} \uparrow\uparrow \vec{p}\right)$$

At end on position, net electric field is parallel to dipole moment direction.

(b) **Electric field at board side on position** (equatorial position)

$\theta = \pm \pi/2$

$$E_r = 0,\ E_\theta = \left(\frac{Kp}{r^3}\right) \quad \therefore \quad E_{net} = E_\theta = \frac{Kp}{r^3}$$

$$\left(\vec{E} \uparrow\downarrow \vec{p}\right)$$

Net electric field is anti-parallel at broad side on position.

11.3 Torque experienced by an electric dipole placed is an uniform electric field

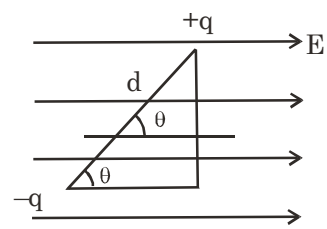
$F_{net} = 0$ [Equal and opposite forces]

These forces which are equal and opposite form couple.

$\therefore \quad \tau = qE\ (d \sin\theta)$

$= P(\sin\theta)E$

$\therefore \quad \boxed{\vec{\tau} = \vec{P} \times \vec{E}}$



Remarks about couple and characteristics

(i) It remains same about whichever point you calculate the torque

(ii) Couple is said to be formed when equal and opposite forces are acting on the body separated by same distance

(iii) Couple is a free vector

Example 34

An electric dipole having dipole moment $\vec{P}$ is placed parallel to electric field. Now it is slightly rotated and released. Find the time period of oscillation.

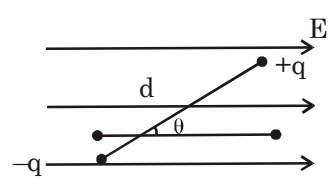
Solution :

When it is kept parallel to electric fied, $\tau = 0$ as $\sin\theta = 0$

Now, when is is rotated by an angle θ, it experiences a torque

$$\tau = pE \sin\theta$$

$$\tau = I\alpha = I\left(\frac{d^2\theta}{dt^2}\right)$$



$\{\sin\theta \approx \theta$ as it is slightly displaced $\therefore\ \theta \approx 0\}$

$$I\left(\frac{d^2\theta}{dt^2}\right) = pE(\theta)$$

$$\left(\frac{d^2\theta}{dt^2}\right) = \left(\frac{pE}{I}\right)\theta$$

$$\therefore \qquad w = \sqrt{\frac{pE}{I}}$$

time period of oscillation $= 2\pi\sqrt{\frac{I}{pE}}$

$$\boxed{\text{where } I = 2 \quad \left(\frac{md^2}{4}\right) = \left(\frac{md^2}{2}\right)}$$

In this question if the two charges are connected at the ends of a rod, then take the moment of inertia of rod also into consideration.

11.4 Potential Energy of an electric dipole in uniform field

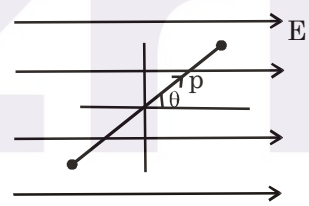
Work done against the field in rotating slowly the dipole from $\theta = 90$ to θ.

Note : U = 0 (assumed) at an angle $\pi/2$ from $\vec{E}$

$$dU = -\tau |d\theta|$$

$$\Delta U = +pE \int_{\pi/2}^{\theta} \sin\theta \, d\theta$$

$$= -pE \left|\cos\theta\right|_{\pi/2}^{\theta}$$



$$\Delta U = -pE\cos\theta$$

$$U_\theta - U_{\pi/2} = pE\cos\theta$$

$$U_{\pi/2} = 0$$

$$\therefore \qquad \boxed{U_\theta = -pE\cos\theta = -p.E}$$

Example 35

Find the word required is rotating a dipole from $\theta = 30$ to $\theta = 60$

Solution : $\Delta W_{ex} = \Delta U$

$$= U_{60} - U_{30}$$

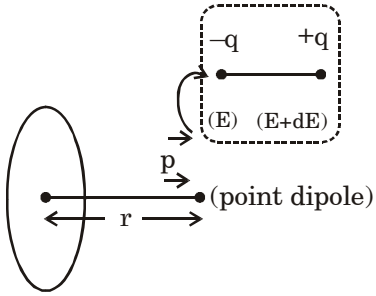
$$= -pE(\cos 60 - \cos 30) = -pE\left(\frac{1-\sqrt{3}}{2}\right)$$

$$\Delta W_{ex} = pE\left(\frac{\sqrt{3}-1}{2}\right)$$

Example 36

A point electric dipole is placed at a distance r from the centre of a uniformly charged ring and has dipole moment equal to $\vec{p}$. Find the force and torque experienced by the dipole and the work required to rotate it by $\pi/_3$.

Solution :



Now calculate the force on this dipole

$$\vec{F} = -q\vec{E} + q\left(\vec{E} + d\vec{E}\right)$$

$$\therefore \qquad \boxed{\vec{F} = qd\vec{E}}$$

Now since the dipole is placed along r, we can change the above vector equation into scalar equation.

$$F = q\ dE$$

$$\Rightarrow \qquad F = q(dr)\left(\frac{dE}{dr}\right)$$

$$\boxed{\mathbf{F = P}\left(\frac{d\mathbf{E}}{dr}\right)}$$ where p = q dr (length of dipole is very small)

Important Note :

This above equation to be used directly to calculate the field when :

(i) dipole is directed along $\vec{r}$

(ii) length of dipole is very small.

But $\vec{\mathbf{F}} = qd\vec{\mathbf{E}}$ can be used in general

Now, $E = \dfrac{KQr}{\left(R^2 + r^2\right)^{3/2}}$ {Field due to Charged ring}

$$\left(\frac{dE}{dr}\right) = KQ\left[\frac{\left(R^2 + r^2\right)^{3/2} - 3r\left(R^2 + r^2\right)^{1/2}}{\left(R^2 + r^2\right)^3}\right]$$

$$= KQ\left[\frac{\left(R^2 + r^2\right)^{3/2} - 3r^2\left(R^2 + r^2\right)^{1/2}}{\left(R^2 + r^2\right)^3}\right]$$

$$= KQ\left(R^2 + r^2\right)^{1/2}\left[\frac{R^2 + r^2 - 3r^2}{\left(R^2 + r^2\right)^3}\right]$$

$$= \frac{KQ\left(R^2 - 2r^2\right)}{\left(R^2 + r^2\right)^{3/2}}$$

$$\therefore \quad F = \frac{pKQ\left(R^2 - 2r^2\right)}{\left(R^2 + r^2\right)^{3/2}}$$

Net force will be towards ring if $r > \dfrac{R}{\sqrt{2}}$ {attractive}

Net force will be away from the ring if $r < \dfrac{R}{\sqrt{2}}$ (repulsive)

(b) $\tau = \left(\vec{P} \times \vec{E}\right) = 0 \quad \left(\vec{P} \uparrow\uparrow \vec{E}\right)$

(c) ΔW_{ex} in rotating from $\theta = 0$, to $\theta = \pi/_3$

$= \Delta U = U_{\pi/3} - U_0$

$= -\ pE_{\pi/3} - (-\ pE_0)$

$= -\ pE\left(\dfrac{1}{2} - 1\right) = \dfrac{pE}{2}$

Example 37

A point electric dipole is placed at a distance r from a uniformly charged long thread (+ λ). Find the force and the torque experienced by dipole if

(a) (i) dipole is placed along r

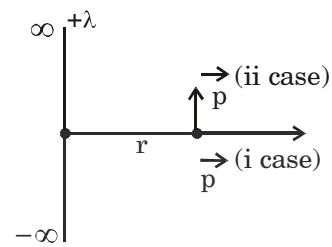
(ii) dipole is placed perpendicular to r and parallel to the thread

(iii) dipole is placed perpendicular to the plane containing the thread and $\vec{r}$.

(b) Also find the work done required to rotate the dipole by $\pi/_3$, $\pi/_2$, π in each case.

Solution : (a) (i) Just use

$$F = P\left(\frac{dE}{dr}\right)$$



$$E = \frac{\lambda}{2\pi \in_0 r}, \quad \frac{dE}{dr} = \frac{-\lambda}{2\pi \in_0 r^2}$$

$$\therefore \quad F = \left(\frac{-p\lambda}{2\pi \in_0 r^2}\right)$$

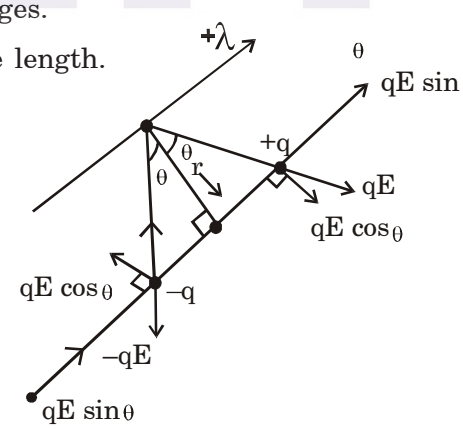
Here force will be attractive between the dipole and the thread.

(ii) F = 0

$\therefore \quad d\vec{E} = 0$ [∴ field remain same at a same perpendicular distance]

(iii) Now there are two method to evaluate the force

Ist method : Just consider force on the two indivisual Charges.



F_{net} obviously directed along the direction of the dipole length.

$F_{net} = 2qE\ (\sin\theta) \approx 2qE(\theta) \quad \{\because \theta \approx \sin\theta)$

$F_{net} = 2qE\theta$

(Now length of dipole = dr = (2rθ)

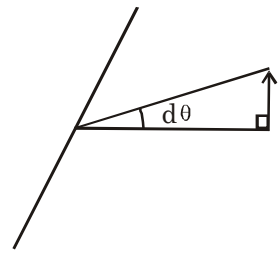
$$\therefore \quad F_{net} = q(2\theta r)\frac{E}{r} = \left(\frac{pE}{r}\right)$$

$\therefore$ Magnitude of the force = $\left(\frac{pE}{r}\right)$, directed along $\vec{p}$.

2nd method : (More effective)

For very small dipole length the field can be assumed to be constant in magnitude. Now use the concept that when a vector is constant in magnitude then it changes direction.

$d\vec{E} = (Ed\theta)(\tilde{u}_\theta)$ (where $\tilde{u}_\theta$ is a unit vector parallel to the length of dipole).



$$\therefore \quad \vec{F} = q(\vec{dE})$$

$$= q\,Ed\theta\,(\tilde{u}_\theta)$$

$$= q\frac{E(rd\theta)}{r}.\tilde{u}_\theta$$

$$\vec{F} = \left(\frac{pE}{r}\right).\tilde{u}_\theta$$

$\therefore$ Magnitude of force = $\left(\frac{pE}{r}\right)$ and directed along $\vec{p}$

Now, torque in each case

(i) $\tau = \vec{p} \times \vec{E} = pE = \frac{p\lambda}{2\pi \epsilon_0 r}$ [$\therefore$when dipole is placed perpendicular to length of thread]

(ii) $\tau = 0$

(iv) $\tau = \vec{p} \times \vec{E} = pE$

Tip : Use $\int \tau\, d\theta$ method only when the potential energy function is not available.

Example 38

An electric dipole of length 2a is placed at a distance 4a from an infinitely long thread with linear charge density λ. Find the work required to rotate the dipole by π, π/2, π/3.

Solution :

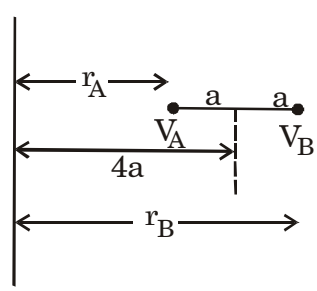
In the question, you can calculate the torque and then apply $\int \tau\, d\theta$ to find the work. But that would be lengthy, so use the potential energy method.

(i) $\Delta W_{ex} = \Delta U$ [work done in rotation by π]

$$= (U_f - U_i)$$

$$U_f = qV_A - qV_B$$

$$U_i = -qV_A + qV_B$$



$$\therefore \quad \Delta W_{ex} = \Delta U = U_f - U_i = 2q(V_A - V_B)$$

$$= 2q\,(V_A - V_B)$$

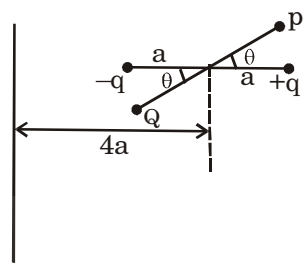
$$= 2q\,\frac{\lambda}{2\pi \epsilon_0}\ln\left(\frac{r_B}{r_A}\right)$$

$$\Rightarrow \frac{q\lambda}{\pi \epsilon_0}\ell n\left(\frac{5a}{3a}\right) = \frac{q\lambda}{\pi \epsilon_0}\ell n\left(\frac{5}{3}\right)$$

(ii) Rotation by $\pi/_2$

$$\Delta W_{ex} = \Delta U$$
$$= U_f - U_i = 0 - U_i$$
$$= [-qV_A + qV_B] = q\,[V_A - V_B]$$
$$= \frac{q\lambda}{2\pi\epsilon_0}\ln(5/3)$$

(iii) In case of rotation by an general angle θ (Here θ is the made by dipole with positive direction of $\vec{r}$)



$$\Delta W_{ex} = \Delta U$$
$$= (U_f - U_i)$$

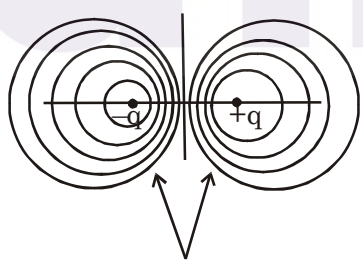
Now,

$$U_j = q\left(V_p - V_\theta\right) = q\frac{\lambda}{2\pi\epsilon_0}\left(\frac{4a - a\cos\theta}{4a + a\cos\theta}\right)$$

$$\therefore \quad \boxed{u_\theta = \frac{q\lambda}{2\pi\epsilon_0}\ln\left(\frac{4 - \cos\theta}{4 + \cos\theta}\right)}$$

Now, $\theta = \frac{\pi}{3} \quad \therefore \; U_f = \frac{q\lambda}{2\pi\epsilon_0}\ell n\left(\frac{4 - 1/2}{4 + 1/2}\right)$

11.5 Nature of Equipotential surfaces is case of dipole

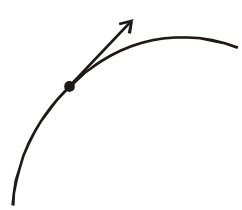


This area has higher field intensity due to overcrowding of curve

13. LINE OF FORCES

Line of forces are imaginary curves tangent to which at any point gives the direction of field intensity at that point.

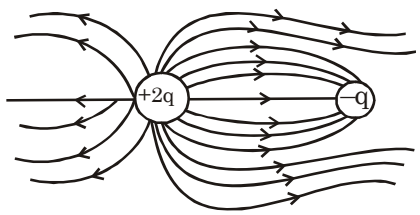
(The tangent direction must be along the direction of field intensity)



13.1 Properties of line of forces :

(1) The number of field lines passing per unit normal area at a point gives the field intensity at that point (in magnitude). Hence crowder the field lines, more strong in the field intensity.

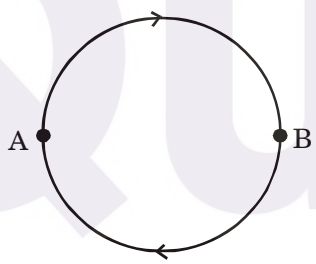
(2) Line of force are always emitted by a positive charge and terminates at a negative or extends to infinity.



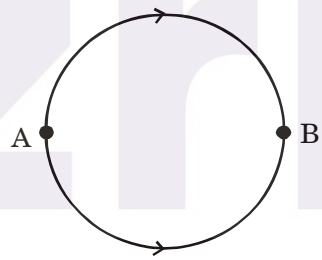
(3) Two lines of force never intersect each other, (otherwise there will be two directions of field intensity at a single point which is absurd).

(4) Electric field lines never form a closed loop (except the non-conservative induced electric field due to time varying magnetic fields).

$\int \vec{E}.d\vec{l} = 0$ **(Property of a conservative field** : work done in moving a positive charge round a loop is zero).



This is not possible (electric field line)



This line of force is possible in electric field

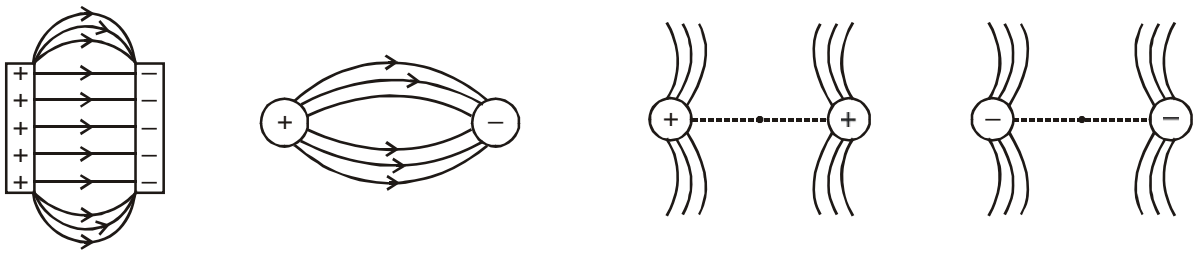
(5) The number of lines of force emitted (or terminating) at a charge is proportional to the magnitude of the charge.

$$n \propto q$$

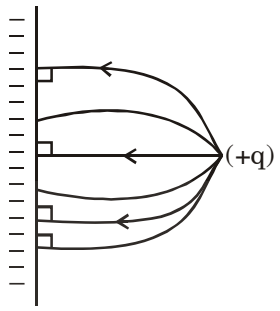
$$\therefore \quad n = \left(\frac{q}{\epsilon_0}\right)$$

(6) The line of forces have a tendency to contract longitudinally (just like a stretched elastic string) to produce force of attraction between unlike charges and have a tendency to repel laterally to produce the force of repulsion between like charges, and to produce **EDGE EFFECT.**

Bending of the line of forces near the edge of the conductor is termed as the edge effect.



(7) Line of forces are always terminating or emitting normally to the surface of a conductor.



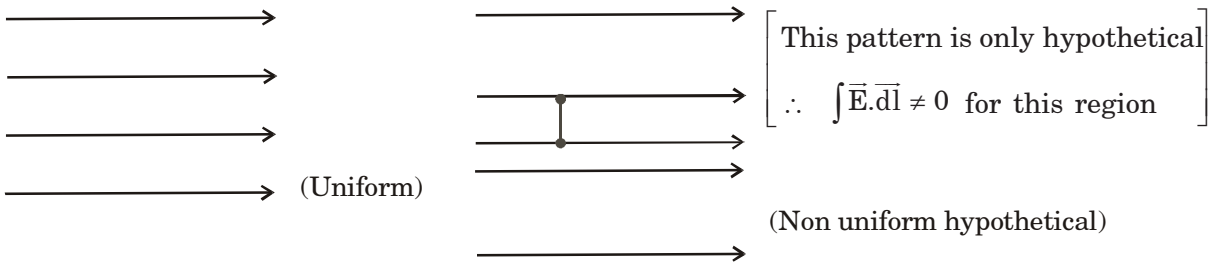
(8) Line of forces are absent when the field intensity is zero. That's why no line of forces exist inside the conductor as (E = 0).

Few important remarks :

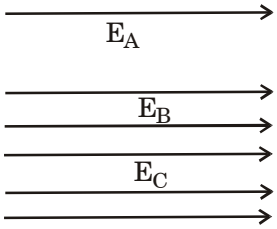
(i) If a point charge free to move, is placed on a line of force curve, then it will follow the curve if the curve is a straight line, otherwise it will not.

Reason : No normal acceleration component to change its direction.

(ii) Equidistant field lines show electric field is constant in both magnitude and direction.



(iii)



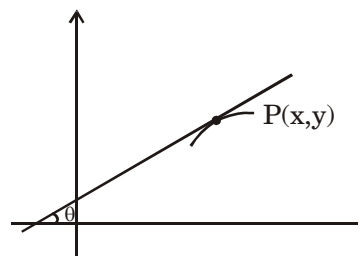
$(E_B > E_c > E_A)$

Example 39

Find the equation of the line of force in the region where

(i) $\vec{E}$ **varies as** $\vec{E} = a\left(-x\tilde{i} + y\tilde{j}\right)$, $a > 0$

Solution : $\left(\frac{dy}{dx}\right) = \left(\frac{y}{-x}\right)$



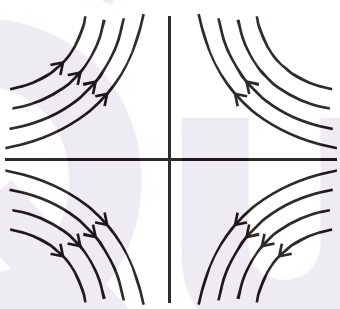
$\int\left(\frac{dy}{y}\right) = -\int\left(\frac{dx}{x}\right)$ [direction of tangent line is same as the direction of electric field]

$\ln|y| = -\ln|x| + \ln c$

$\Rightarrow$ xy = + c which represents a rectangular hyperbola

$\boxed{xy = \pm c}$

Now, plot of the given curve xy = ± c (which represents the line of force)



$\left|\vec{E}\right| = a\sqrt{x^2 + y^2}$ magnitude of the $\vec{E}$ increases as one moves away from the curve while means crowder lines.

Note that how you decide the direction of the arrow, just use the logic, $E_x = -ax$, $Ey = ay$

13.2 Flux of An Electric Field (ϕ_E) :

Let us consider a plane surface of area S placed in an electric field $\vec{E}$. Electric flux through an elementary area $\vec{dS}$ is defined as the scalar product of $\vec{ds}$ and $\vec{E}$ i.e., $d\phi_E = \vec{E}\cdot\vec{ds}$, where $\vec{ds}$ is the area vector, whose magnitude is the area ds of the element and whose direction is along the outward normal to the elementary area. Hence, the electric flux through the entire surface is given by

$$\phi_E = \int \vec{E}\cdot\vec{ds} \qquad ....(i)$$

or $$\phi_E = \int E\,ds\cos\theta$$

If the electric field is uniform, then $\phi_E = \int E\,ds\cos\theta = E\cos\theta\int ds = ES\cos\theta$

To show the integration over a closed surface, a circle is shouwn on the integration symbol :

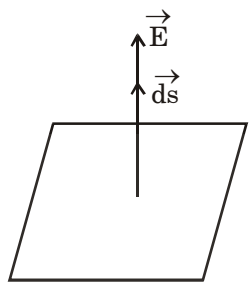
$\phi_E = \int E.ds$ .....(ii)

13.4 Important Points Regarding Electric Flux :

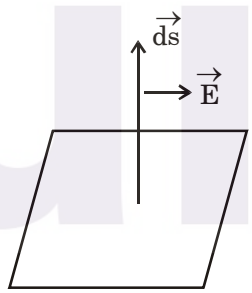
(i) It is a scalar physical quantity with units (volt m) or $N/m^2/c$

(ii) It will be maximum when cos θ = max = 1 i.e.

θ = 0 , i.e., electric field is normal to the surface with $(d\phi_E)_{max} = EdS$



(iii) for θ = 90 , cos θ = 0 mens ϕ_E is zero.



14. GAUSS'S LAW :

This law gives the relation between the electric flux through any closed hypothetical surface (called a Gaussian surface) and the charge enclosed by the surface. It states **"the electric flux (ϕ_E) through any closed surface is equal to $\frac{1}{\epsilon_0}$ times the 'net' charged enclosed".**

That is,

$$\phi_E = \oint \vec{E}.\vec{ds} = \frac{\sum q}{\epsilon_0}$$

where Σq denotes the algebraic sum of all the charges enclosed by the surface. If there are several charges $+ q_1, + q_2, + q_3, - q_4, - q_5$ inside the Gaussian surface then

$$\Sigma_q = q_1 + q_2 + q_3 - q_4 - q_5$$

Remarks :

(1) Gaussian surface is an imaginary closed surface.

(2) Gaussian surface cannot be passed through a discrete charges but can be passed through continuous charge distributions.

Note : The electric field is directed in radial direction for spherically symmetry charge distribution hence the use of Gauss's law simplifies the position.

14.1 Electric field intensity obtained using the Gauss's law

I. Uniformly charged non conducting sphere

(a) For an internal point

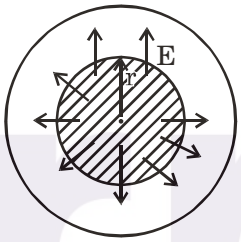
Consider a gaussion surface at a distance from the centre inside the surface

Note that the direction of the electric field will be directed radially outward

$$\Delta\phi = \int \vec{E}.d\vec{s} = E\int ds = E\left(4\pi r^2\right)$$

$$E\left(4\pi r^2\right) = \frac{\rho\left(\frac{4}{3}\pi r^3\right)}{\epsilon_0}$$

$$\therefore \quad \boxed{\mathbf{E} = \left(\frac{\rho r}{3\,\epsilon_0}\right)}$$

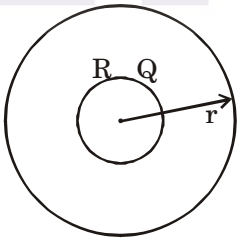


when r > R

(b) For an external point

$$E\left(4\pi r^2\right) = \left(\frac{Q}{\epsilon_0}\right)$$

$$\therefore \quad \boxed{\mathbf{E} = \frac{Q}{4\pi\epsilon_0 r^2}}$$



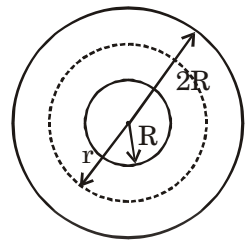
when r > R

Illustration

Find the field intensity inside a thick conducting shell of inner and outer radii, R and 2 R, whose volume charge density varies as, $\rho = \left(\frac{\rho_0 r}{R}\right)$

Solution : Using gauss's law by making a gaussian surface at a distance r from the centre

$$q_{inside} = \rho(4\pi r^2)dr = \frac{\rho_0 4\pi r^3 dr}{R}$$



Hence according to gauss's Law

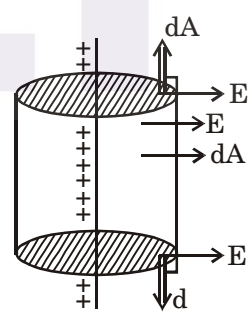
$$E\left(4\pi r^2\right) = \int_R^r \left(\frac{\rho_0 r\left(4\pi r^2\right)dr}{\frac{R}{\epsilon_0}}\right)$$

$$\therefore \quad \boxed{E = \frac{\rho_0\left(r^4 - R^4\right)}{Rr^2\left(4\epsilon_0\right)}}$$

II. UNIFORMLY CHARGED INFINITELY LONG THREAD

Assume a coaxial cylindrical gausian surface, at a distance r.

Charge enclosed by it (q) = (λl),



Hence, $E(2\pi rl) = \frac{(\lambda l)}{\epsilon_0}$

$$\therefore \quad \boxed{\mathbf{E} = \frac{\lambda}{2\pi\,\epsilon_0\, r}}$$

III. UNIFORMLY CHARGED INFINITE PLANE SHEET (NON-CONDUCTING)

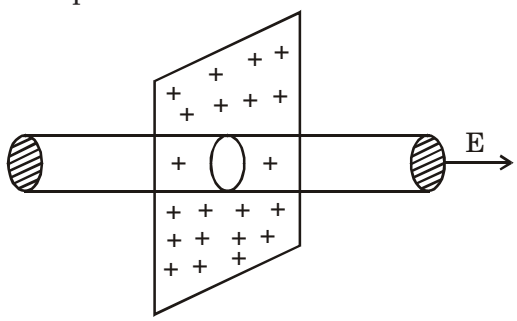
Let us consider an infinite non-conducting charged plane sheet having a surface charge density $\sigma\ \frac{C}{m^2}$

Now, assume a cylindrical gaussian surface through the plane sheet.

The area of each end is πr^2.

By symmetry, the electric field intensity at all points on either side near the sheet will be perpendicular to the sheet, directed outward (if the sheet is positively charged). Thus $\vec{E}$ is perpendicular to the plane ends of the cylinder and parallel to the curved surface.

Therefore, flux through the two plane ends is



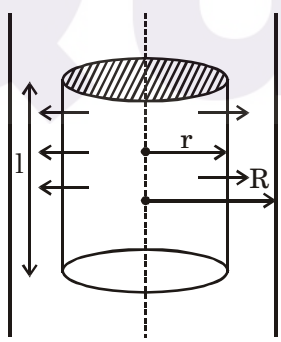
$$\phi_E = \int E.\ \pi r^2 + \int E.\pi r^2 = 2\int E.\pi r^2$$

$$\phi = \frac{q_{inside}}{\epsilon_0} = \frac{\sigma\left(\pi r^2\right)}{\epsilon_0} = 2E.\left(\pi r^2\right)$$

$$\Rightarrow \quad \boxed{E = \frac{\sigma}{2\epsilon_0}}$$

IV. INFINITELY UNIFORMLY CHARGED CYLINDER

(a) For an internal point



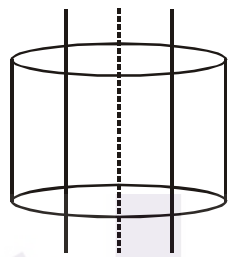
Since charge is uniformly distributed through out the volume of cylinder of radius R. For finding the electric field at a distance (r < R) from the centre, we choose a cylindrical Gaussian surface of radius r, co-axial with the charge distribution.

The magnitude E of the electric field has the same value at every point on the Gaussian surface and the direction of $\vec{E}$ is radial at every point on the lateral surface. Since, direction of electric field on the circular ends is taken to be zero. So, applying Gauss's law

$$\oint E.ds = E\left(2\pi rl\right) = \frac{\rho\left(\pi r^2 l\right)}{\epsilon_0}$$

$\therefore$ $\boxed{E = \frac{\rho r}{2\epsilon_0}}$ in vector $\boxed{\vec{E} = \frac{\rho \vec{r}}{2\epsilon_0}}$

(b) For an external point



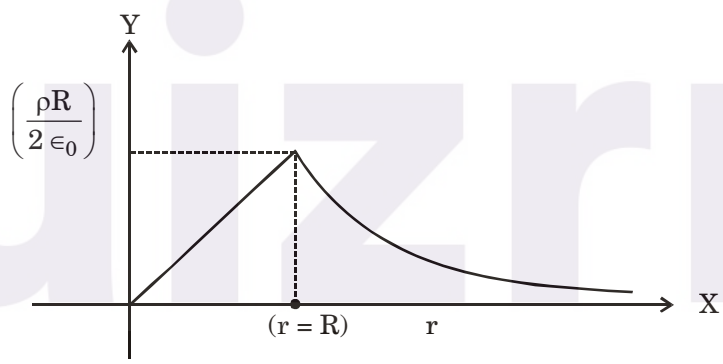
Similarly, as done above consider a gaussian surface co-axial with the uniformly charged cylinder at radius r (r > R)

Thus, $\vec{E}.\left(2\pi rl\right) = \frac{\rho\left(\pi R^2\right)l}{\epsilon_0}$

$\Rightarrow$ $E = \left(\frac{\rho R^2}{2\epsilon_0 r}\right)$

in vector form

$\vec{E} = \left(\frac{\rho R^2}{2\epsilon_0 r^2}\right)\vec{r}$



Plot of electric field v/s distance from the axis

Illustration

Find the field intensity inside long cylinder having inner and outer radii, R and 2R and having charge density given by $\rho = \rho_0\left(1 + \frac{r}{R}\right)$ for

(i) r < R (ii) 2R ≥ r > R (iii) r > 2R

Solution :

(i) for r < R

$q_{inside} = 0$ $\therefore$ $\Delta\phi = 0$ $\Rightarrow$ $\boxed{E = 0}$

Consider a cylindrical gaussian surface at a distance r

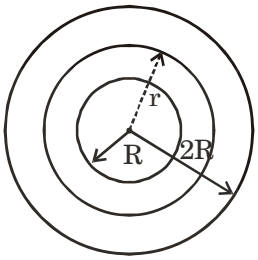
Now, $q_{inside} = \int_R^r \rho(2\pi rl)dr$

(ii) For $2R \geq r > R$

Hence, from gauss's law

$$\Delta\phi = \left(\frac{q_{inside}}{\epsilon_0}\right)$$

$$\Rightarrow \quad \int \vec{E}.d\vec{s} = \int_R^r \frac{\rho\,(2\pi rl)\,dr}{\epsilon_0}$$



$$\Rightarrow \quad E\,(2\pi rl) = \int_R^r \rho_0 \frac{\left(1+\frac{r}{R}\right)2\pi rl\,dr}{\epsilon_0}$$

$$\Rightarrow \quad E(r) = \left(\frac{\rho_0}{\epsilon_0}\right)\left|\frac{r^2}{2}+\frac{r^3}{3R}\right|_R^r$$

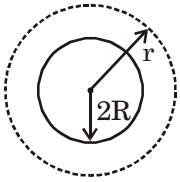
$$\therefore \quad E = \left(\frac{\rho_0}{r\,\epsilon_0}\right)\left[\frac{5r^2}{6}-\frac{5R^2}{6}\right]$$

$$= \; E = \left(\frac{\rho_0}{6\,\epsilon_0}\right)\left[\frac{r^2-R^2}{r}\right]$$

(iii) For $r > 2R$:

Again consider a cylindrical gaussian surfare.

$\therefore$ From gauss's law



$$q_{inside} = \rho_0 \int_R^{2R}\left(1+\frac{r}{R}\right)2\pi rl\;dr = \rho_0\,2\pi l\left\{\frac{r^2}{2}+\frac{r^3}{3R}\right\}_R^{2R}$$

$$= 2\pi\rho_0\,l\left\{2R^2+\frac{8R^2}{3}-\left(\frac{R^2}{2}+\frac{R^2}{3}\right)\right\} = \frac{\pi\rho_0 l(23R^2)}{3}$$

$$\therefore \quad E.\,2\pi r\,l = \frac{\pi\rho_0 l(23R^2)}{3\,\epsilon_0}$$

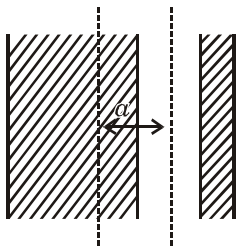
$$\Rightarrow \quad \boxed{E = \frac{23\rho_0 R^2}{6\,\epsilon_0\,r}}$$

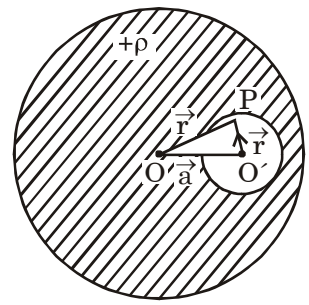
14.2 Cavity inside a cylindrical infinitely long volume :

Example 40

Find the field intensity inside a long cylindrical cavity which is uniformly charged and distance between the axis of cylinder and that of cavity is a**.**

Solution :





Fill the cavity with charge density + ρ and – ρ.

Now, find the field intensity at any point P inside the cavity.

$$E_{net} = \vec{E}_1 + \vec{E}_2$$

Where E_1 is the field intensity due to uniformly Charged cylinder of density + ρ and $\vec{E}_2$ is the field due to uniformly charge cylinder of density – ρ centred at 0′

$$= \left(\frac{\rho\vec{r}}{2\epsilon_0}\right) + \frac{\left(-\rho\vec{r}'\right)}{2\epsilon_0} = \frac{\rho}{2\epsilon_0}\left(\vec{r} - \vec{r}'\right)$$

Now, $\vec{r} = \vec{r}' + \vec{00'}$ or $\vec{r} - \vec{r}' = \vec{a}$

$$\therefore \qquad E_{net} = \frac{\rho}{2\epsilon_0}\left|\vec{00'}\right| = \left(\frac{\rho\vec{a}}{2\epsilon_0}\right)$$

Thus, just like is the case of spherical cavity inside a sphere, a cavity in a cylinder also has constant electric field inside it.

Example 41

A system consists of a uniformly charged sphere of Radius R and the surrounding has a volume charge density $\rho = \left(\frac{\alpha}{r}\right)$. Then find the charge on the sphere so that the field intensity at any point outside the sphere becomes independent of r **and also find its magnitude.**

Solution :

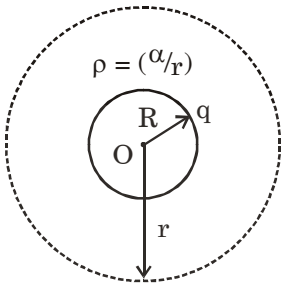
Consider a guassian spherical surface centred at O at a distance r.

Let Charge on sphere be q and the charge surrounding it inside the gaussian surface is

$$Q = \int_R^r \frac{\alpha}{r}\left(4\pi r^2\right)dr$$

Hence, according to Gauss Law

$$\Delta\phi = \left(\frac{q_{inside}}{\epsilon_0}\right)$$



$$\int \vec{E}.\vec{ds} = \frac{q}{\epsilon_0} + \int_R^r \frac{\left(\frac{\alpha}{r}\right)\left(4\pi r^2\right)dr}{\epsilon_0}$$

$$\Rightarrow \quad E\int ds = \frac{q + \frac{4\pi r}{2}\left[r^2 - R^2\right]\alpha}{\epsilon_0}$$

$$E\left(4\pi r^2\right) = \frac{q + 2\pi r\left(r^2 - R^2\right)\alpha}{\epsilon_0}$$

$$E = \frac{q + 2\pi r\left(r^2 - R^2\right)\alpha}{4\pi r^2\ \epsilon_0}$$

$$= \frac{\alpha}{2\,\epsilon_0} + \frac{1}{r^2}\left[\frac{q}{4\pi\,\epsilon_0} - \frac{\alpha R^2}{2\,\epsilon_0}\right]$$

For E to be independent of x,

$$\frac{q}{4\pi\,\epsilon_0} - \frac{\alpha R^2}{2\,\epsilon_0} = 0 \qquad \therefore$$

$$\boxed{q = 2\pi\alpha R^2}$$

and then, the constant electric field magnitude will be

$$\boxed{E = \frac{\alpha}{2\,\epsilon_0}}$$

Example 42

A sphere is charged with volume charge density $\rho = \rho_0 (1 - {}^{r}/_{R})$. Then find the field intensity at an internal and external point. Also find the maximum field intensity due to the given charge distribution.

Solution :

(a) At an internal point

Using the Gauss's Law

$$\int \vec{E}.d\vec{s} = \int_0^r \frac{\rho_0 \left(1 - \frac{r}{R}\right) 4\pi r^2 dr}{\epsilon_0}$$

$$E\left(4\pi r^2\right) = \left(\frac{\rho_0}{\epsilon_0}\right) 4\pi \left| \frac{r^3}{3} - \frac{r^4}{R(4)} \right|_0^r$$

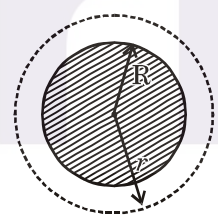
$$\therefore \quad E = \frac{\rho_0}{r^2 \epsilon_0} \left(\frac{4r^3R - 3r^4}{12R}\right)$$

$$= \frac{\rho_0}{12R\epsilon_0}\left(4rR - 3r^2\right)$$

(b) At an external point

Again by gauss's Law

$$\int \vec{E}.d\vec{s} = \int_0^R \frac{\rho_0 \left(1 - \frac{r}{R}\right)\left(4\pi r^2\right) dr}{\epsilon_0}$$



$$E\left(4\pi r^2\right) = \left(\frac{\rho_0}{\epsilon_0}\right) 4\pi \left(\frac{R^3}{12}\right)$$

$$\therefore \quad E = \left(\frac{\rho_0 R^3}{12 \epsilon_0 r^2}\right)$$

Maximum field exists at a point inside the given sphere; for an internal point we have,

$$E = \frac{\rho_0}{12R\epsilon_0}\left(4\pi R - 3r\right) \quad \therefore \text{ For } E_{max}, \ \frac{dE}{dr} = 0$$

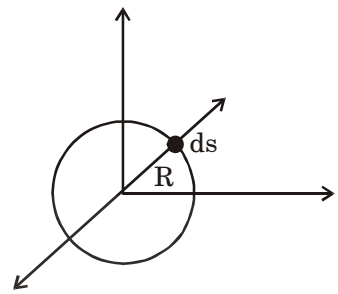
$$\Rightarrow \quad \left(\frac{dE}{dr}\right) = \left(4R - 6r\right) = 0 \qquad \therefore \quad \boxed{r = \left(\frac{2R}{3}\right)}$$

Example 43

The field intensity in a region is given by $\vec{E} = \dfrac{px\tilde{i} + qy^2\tilde{j} + z\tilde{k}}{px^3 + qy^3 + z^2}$. Then find the net charge enclosed by a sphere of radius R, centred at origin.

Solution :

To find the net charge enclosed by the sphere consider flux passing through the charged sphere. Now, for a sphere centred at origin, we can take the area vector as



$$\vec{ds} = ds\left\{\frac{x\tilde{i} + y\tilde{j} + z\tilde{k}}{R}\right\}$$

$$\therefore \quad \phi = \oint E.ds$$

$$= \oint \frac{\left(px^2\tilde{i} + qy^2\tilde{j} + z\tilde{k}\right)}{\left(px^3 + qy^3 + z^2y\right)} \cdot \frac{\left(x\tilde{i} + y\tilde{j} + z\tilde{k}\right)ds}{R}$$

$$= \oint \left(\frac{ds}{R}\right)\frac{\left(px^3 + qy^3 + z^2\right)}{\left(px^3 + qy^2 + z^2\right)}$$

$$= \left(\frac{4\pi R^2}{R}\right) = (4\pi R)$$

$\therefore$ Net charge enclosed can be obtained from Gauss's law.

$$\Delta\phi = (4\pi R) = \left(\frac{q_{inside}}{\epsilon_0}\right)$$

$$\boxed{\therefore \quad q_{net} = (4\pi\,\epsilon_0\,R)}$$

Example 44

The field intensity in a region is given by $\vec{E} = \left(2xy + z^2\right)\tilde{i}\left(2yz + x^2\right)\tilde{j} + \left(2x + y^2\right)\tilde{k}$.

Find the net charge enclosed by a sphere centred at origin.

Solution :

As done in the above example we take $\vec{ds} = ds\dfrac{\left(x\tilde{i} + y\tilde{j} + z\tilde{k}\right)}{R}$

Now, $\phi = \vec{E}.\vec{ds}$

$$= \left\{\left(2xy + z^2\right)\tilde{i} + \left(2y^2 + x^2\right)\tilde{j} + \left(2x + y^2\right)\tilde{k}\right\}.\left(\frac{x\tilde{i} + y\tilde{j} + z\tilde{k}}{R}\right)ds$$

$$= \frac{ds}{R}\left\{2\underline{x}^2y + \underline{z^2x} + 2\underline{y}^2z + x^2y + 2\underline{x}z^2 + \underline{y}^2z\right\}$$

$$= \frac{ds}{R}\left\{3\left(x^2y + y^2z + z^2x\right)\right\}$$

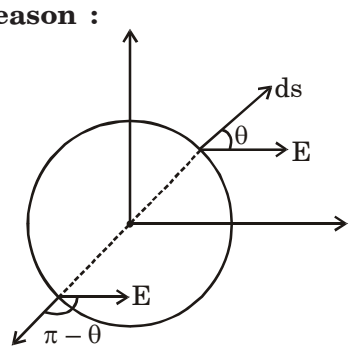
Now, eliminating x, y, z from the equation is not easy, hence we use another method (shortkut) to solve it.

Note : Net charge enclosed = 0, if the following 2 conditions are satisfied :

(i) always when the electric field is symmetrical i.e. if you replace $x \to -x$, $y \to -y$, $z \to -z$ and no change in the function, then field intensity is a symmetrical function.

(ii) Again the sphere must be centred at origin, for this to be applicable.

Reason :



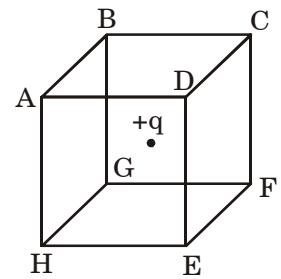
The dot product of $\vec{E}.\vec{ds}$ will cancel out for each oppositely located infinitesimal area elements.

14.3 Electric flux passing through different surfaces

Example 45

(1) Calculate the flux through each face of a cube when a point charge of + q is placed at the centre of the cube.

Solution :



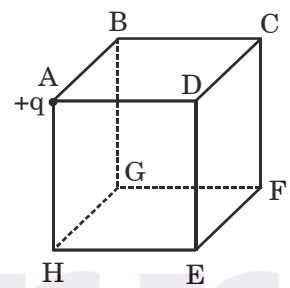
The flux through each face remains same, since the charge q is symmetrically located with respect to all the faces of the cube.

Hence, We have flux through the cube = $\frac{q}{\epsilon_0}$

$\therefore$ flux through each face = $\frac{q}{6\epsilon_0}$

(2) Calculate the flux through each face of a cube when a point charge q is placed at one of the vertex of the cube.

Solution :



Electric flux through the face ABCD, ABGH and ADEH are zero.

The reason the this statement lies in the fact that these faces contain the Charge q, Hence electric field strength in these faces lie perpendicular to the area vector of these faces. Thus $\vec{E}.\vec{ds} = \phi = 0$

Remember : The electric flux through a surface containing the charge is zero.

Now, the flux through each of the remaining faces will be same since, they are symmetrically located.

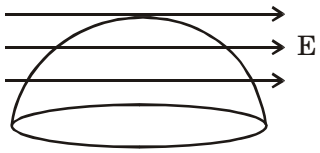
Concept :
Complete the symmetry by placing seven more cube around given cube so that the change q lies at the centre of the eight cubed volume.

Flux through each cube = $\frac{q}{8\epsilon_0}$

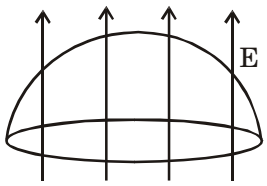
$\therefore$ Flux through each of the 3 faces of the 8 cubes = $\frac{q}{24\epsilon_0}$

(3) Calculate the flux in the given case through the curved hemispherical portion only, when placed is uniform electric field as shown :

(a)

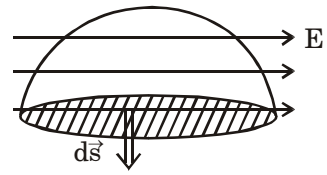


(b)



Solution :

(a) Complete the hemisphere by enclosing it from the bottom



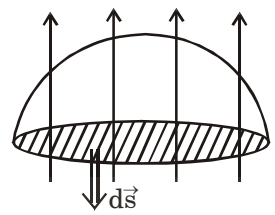
$\phi_{\text{curved}} + \phi_{\text{base (circular)}} = 0$

The above statement is stated because the charge enclosed by the hermisphere is zero hence total flux through it is zero.

$\therefore \quad \phi_{\text{curved}} = 0$

$[\because \phi_{\text{base}} = 0$, as $\vec{E}$ is perpendicular to $\vec{ds}]$

(b) Again complete the hemisphere



$\phi_{\text{curved}} + \phi_{\text{circular base}} = 0 \; [q_{\text{enclosed}}]$

$\phi_{\text{curved}} + E\,(-4\pi r^2) = 0$

Here, $\phi_{\text{circular base}} = \vec{E}.\vec{ds} = \vec{E}(-4\pi r^2)$,

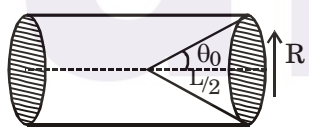
$\therefore \quad \phi_{\text{curved}} = E(4\pi r^2)$

Example 46

A point charge is placed at the centre of a cylinder of radius R and length l. Find the flux through the curved surface.

Solution :

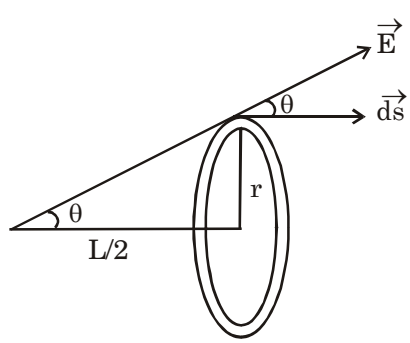
(1) First, close the cylinder's circular ends and apply gauss's law



$$\Delta\phi_{\text{net}} = \frac{q}{\epsilon_0}$$

$$\cos\theta_0 = \frac{L/2}{\sqrt{\frac{L^2}{4} + R^2}}$$

(2) The flux through each of the flat base $\Delta\phi_{\text{net}}$ circular ends $= 2\int_0^r \vec{E}.\vec{ds}$



$$\Delta\phi_{net}\ (\text{Circular ends}) = \int_0^r 2 \times \frac{q}{4\pi\epsilon_0} \frac{2\pi r\, dr}{\frac{\left(\frac{L}{2}\right)^2}{\cos^2\theta}} \cdot \cos\theta \qquad \left[\begin{array}{l} r = \frac{L}{2}\tan\theta \\ dr = \frac{L}{2}\sec^2\theta\, d\theta \end{array}\right]$$

$$= \int_0^{\theta_0} \frac{q}{\epsilon_0} \cos^2\theta \tan\theta . \sec^2\theta\, d\theta . \cos\theta$$

$$= \int_0^{\theta_0} \frac{q}{\epsilon_0} \sin\theta\, d\theta = \frac{q}{\epsilon_0}\left(1 - \cos\theta_0\right)$$

$$\Delta\phi_{(curved)} = \frac{q}{\epsilon_0} - \frac{q}{\epsilon_0}\left(1 - \cos\theta_0\right) = \frac{q}{\epsilon_0}\cos\theta_0$$

$$\Delta\phi = \frac{q}{\epsilon_0} \frac{L/2}{\sqrt{\frac{L^2}{4} + R^2}} = \frac{q}{\epsilon_0} \frac{L}{\sqrt{L^2 + 4R^2}}$$

14.4 Differential form of Gauss's Law

Derivation :

Consider the flux through an infinitesimal volume is space. Net flux through the cube

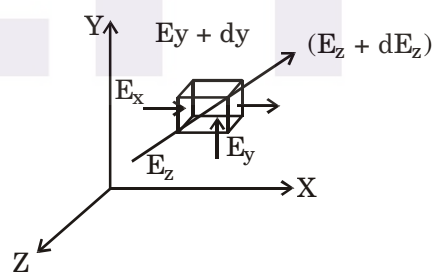
$= [-E_X\ (dy\ dz) + (E_X + dE_X)\ dydz]$

$+ [-E_y\ (\ dzdx) + (Ey + dEy)\ (dzdx]$

$+ [-Ez\ (dx\ dy) + (Ez + dEz)\ (dxdy)]$

$= (dEx\ dydz) + (dEy\ dzdx) + dEz\ (dxdy)$

Using Gauss law



$(dEx\ dy\ dz) + (dEy\ dx\ dz) + (dEz\ dxdy) =$

$$\left(\frac{dEx}{dx}\right) + \left(\frac{dEy}{dy}\right) + \left(\frac{dEz}{dz}\right) = \frac{\rho}{\epsilon_0}$$

$$\therefore \quad \boxed{\left(\frac{\partial E}{\partial x}\right) + \left(\frac{\partial E}{\partial y}\right) + \left(\frac{\partial E}{\partial z}\right) = \frac{\rho}{\epsilon_0}} \quad \text{(Differential form of Gauss law)}$$

In terms of potential :

$$-\left[\frac{\partial^2 V}{\partial x^2}+\frac{\partial^2 V}{\partial y^2}+\frac{\partial^2 V}{\partial z^2}\right]=\frac{\rho}{\epsilon_0}$$

In other forms

$$\vec{\nabla}=\left(\frac{\partial}{\partial x}\right)\tilde{i}+\left(\frac{\partial}{\partial y}\right)\tilde{j}+\left(\frac{\partial}{\partial z}\right)\tilde{k}$$

$\therefore$ Differential form of Gauss's law can also be stated as :

$$\boxed{\vec{\nabla}.\vec{E}=\frac{\rho}{\epsilon_0}} \quad \left(\text{Divergence of } \vec{E}=\frac{\rho}{\epsilon_0}\right)$$

Example 47

Electric potential in origin is given by

$V = (-\alpha x^3 + \beta)$

Then find the volume charge density in the region as function of space.

Solution :

$$-\left(\frac{\partial^2 V}{\partial x^2}\right)=\left(\frac{\rho}{\epsilon_0}\right) \quad \text{[Using differential form of Gauss law]}$$

$$\left(\frac{\partial}{\partial x}\right)3\alpha x^2=\left(\frac{\rho}{\epsilon_0}\right)$$

$$\therefore \quad (6\alpha x)=\frac{\rho}{\epsilon_0}$$

$$\boxed{\therefore \quad \rho = 6\epsilon_0\,\alpha\, x}$$

15.1 CONDUCTION

Flow of charge from one conductor to the other conductor is called conduction. (Charge on the conductor does not remain constant.]

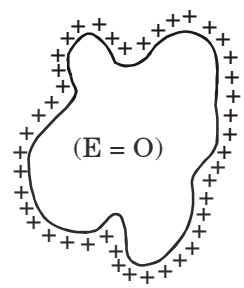
15.2 INDUCTION

Redistribution of charge over the surface of the conductor is called induction. In this case, the total charge of conductor remains constant.

15.3 PROPERTIES OF A CONDUCTOR

(1) When an external charge is given to the conductor, the entire charge appears over the surface of the conductor and the entire volume of the conductor is electrically neutral.

In the absence of an external field and charge around the conductor, the surface charge density of the charge distribution over the conductor surface is inversely proportional to the radius of curvature.



$\sigma \alpha \frac{1}{R}$ (flater portion has lesser charge density than curved part.)

Note : This is applicable only when external field and charges are absent

(2) The potential at any point inside (or on the surface) of the conductor is constant since the conductor is an equipotential body.

(3) Electric field is zero inside the conductor

Hence, the volume of the conductor is electrically neutral

(4) The field intensity near the surface of the conductor is given by : $E = \frac{\sigma}{\epsilon_0}$

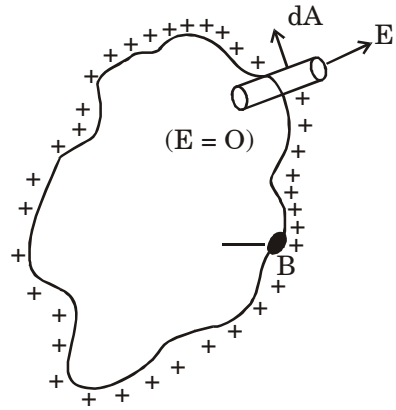
(normal to surface of the conductor)

Derivation :

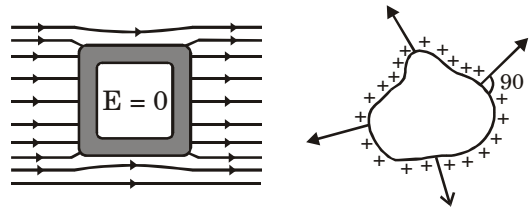
Consider a Gaussin surface near the surface of the conductor as shown

$E(ds) = \frac{[\sigma(ds)]}{C_0}$ [σ : surface density of the point]

$\therefore \quad E = \left(\frac{\sigma}{\epsilon_0}\right)$ ($\vec{E}$ is perpendicular to the surface)



15.4 Electric field and field lines are normal to the surface of a conductor



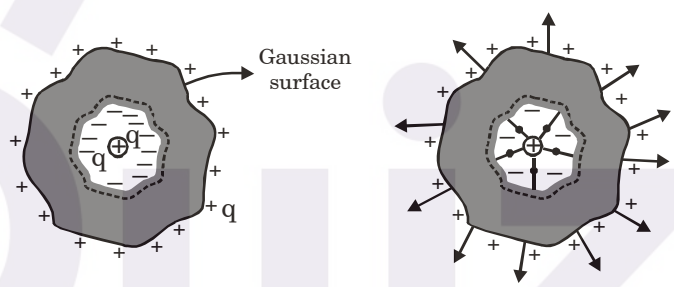
Net field inside a conductor is zero. It implies that no field lines enter a conductor. On the surface of a conductor, electric field and hence field lines are normal to the surface of the conductor.

If a conducting box is immersed in a uniform electric field, the field lines near the box are somewhat distorted. Similarly if a conductor is positively charged, the field lines originate from the surface and are normal at every point and if it is negatively charged the field lines terminate on the surface normally at every point.

15.5 Cavity inside a conductor

Consider a charge + q suspended in a cavity in a conductor. Consider a Gaussian surface just outside the cavity and inside the conductor $\vec{E} = 0$ on this Gaussian surface as it is inside the conductor.

Hence from Gauss's law, $\oint \vec{E}.d\vec{S} = \frac{q_{in}}{\epsilon_0}$ gives $q_{in} = 0$ as { E = 0 inside conductor}



This concludes that a charge of – q must reside on the metal surface of the cavity so that sum of this induced charge – q and the original charge + q within the Guassian surface is zero. In other words, a charge q suspended inside a cavity in a conductor induces an equal and opposite charge – q on the surface of the cavity. Further as the conductor is electrically neutral a charge + q is induced on the outer surface of the conductor. As field inside the conductor is zero the field lines coming from q cannot penetrate into the conductor. The field lines will be as shown in figure (b).

The same line of approach can be used to show that the field inside the cavity of a conductor is zero when no charge is suspended in it.

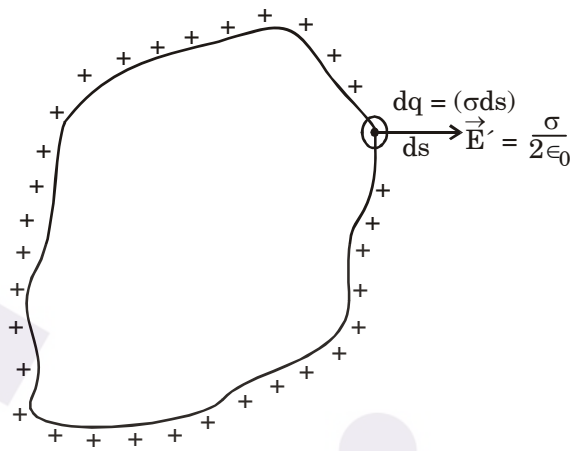
Electrostatic shielding

Suppose we have a very sensitive electronic instrument that we want to protect from external electric fields that might cause wrong measurements. We surround the instrument with a conducting box or we keep the instrument inside the cavity of a conductor. By doing this charge in the conductor is so distributed that the net electric field inside the cavity becomes zero and the instrument is protected from the external fields. This is called electrostatic shielding.

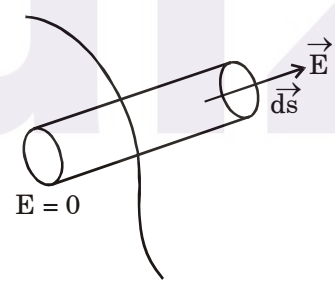
15.6 The potential of a charge conductor throughout its volume is same

In any region in which $\vec{E} = 0$ at all points, such as the region vary far from all charges or the interior of a charged conductor, the line integral of $\vec{E}$ is zero along any path. It means that the potential difference between any two points in the conductor are at the same potential or the interior of a charged conductor is an equipotential region.

16. ELECTROSTATIC PRESSURE



Let us consider a small elemental charge dq of area ds. Now let us calculate the net electric field $\vec{E}$ due to rest of the charges on this elemental area.



$$\phi = \vec{E}.\vec{ds} + \vec{E}\,\vec{ds} = \frac{q_{enclosed}}{\epsilon_0}$$

and $\quad q_{encl.} = dq = \sigma\, ds$

$$\therefore \qquad \boxed{\vec{E} = \frac{\sigma}{2\,\epsilon_0}}$$

Now, this electric fied is on the elemental charge dq, hence force experienced by it is,

$$F = dq\,\vec{E} = (\sigma\, ds)\left(\frac{\sigma}{2\,\epsilon_0}\right)$$

$$\Rightarrow \qquad F = \left(\frac{\sigma^2}{2\,\epsilon_0}\right) ds$$

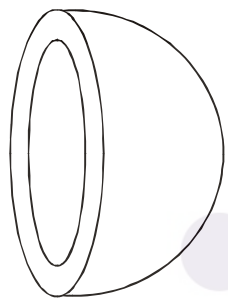
$$\Rightarrow \qquad \text{Electrostatic pressure} = P = \frac{\sigma^2}{2\,\epsilon_0}$$

Note : This eletrostatic pressure is bursting in nature i.e. directed outwards always.

Example 48

Find the maximum surface charge density that can be given to a conducting spherical shell of radius R and thickness ΔR ($\Delta R << R$). σ_0 is the bursting stress.

Solution :



By increasing the surface charge density, the electrostatic pressure also increses. Let us assume the maximum surface charge density be σ, then

$$\text{Pressure} = \left(\frac{\sigma^2}{2C_0}\right)$$

Now bursting stress $\geq$ pressure force {so that the shell do not burst}

When the shell bursts, the area available will be the area covered by shell in the middle i.e. $2\pi\Delta R$ and for electrostatic pressure area will be πR^2

$$\text{Hence, } \sigma_0(2\pi R\Delta R) \geq \frac{\sigma^2}{2\,\epsilon_0}\left(\pi R^2\right) \quad \Rightarrow \quad \boxed{\sigma \leq \sqrt{\frac{4\sigma_0\,\epsilon_0\,\Delta R}{R}}}$$

Example 49

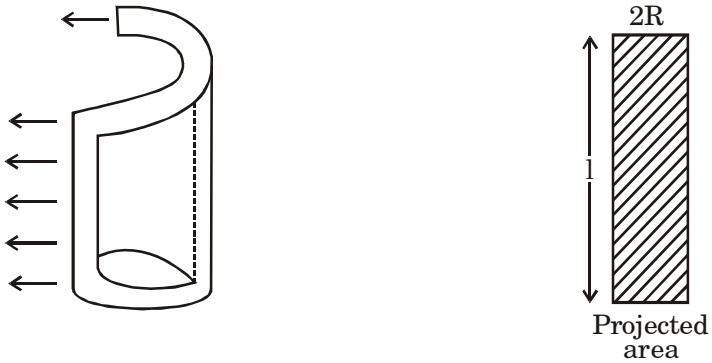
Find the maximum surface charge density that can be given to a thin conducting cylinder of thickness ΔR, and radius R, length l. σ_0 is the bursting stress.

Solution :

The question is similar to the previous example. However it is important for the students to know about the projected area concept used with the electrostatic pressure.

Here it is a thin conducting cylinder.

Now



Pressure force = (electrostatic pressure) (projected area)

$$= \left(\frac{\sigma^2}{2\epsilon_0}\right)\times(2Rl)$$

Bursting force = σ_0 $(2\ \Delta Rl)$

Now, bursting force $\geq$ pressure force

$$\Rightarrow \quad \sigma_0\ (2\Delta R\ l) \geq \frac{\sigma^2}{2\epsilon_0}(2Rl)$$

$$\therefore \quad \sigma \leq \sqrt{\frac{2\sigma_0\ \Delta Rl\epsilon_0}{R}}$$

V. CHARGED SPHERICAL SHELL

(A) Electric Field

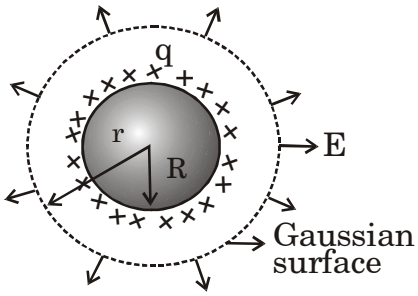
At all points inside the charged spherical conductor or hollow spherical shell, electric field $\vec{E} = O$, as there is no charge inside such a sphere. In an isolated charged spherical conductor any excess charge on it is distributed uniformly over its outer surface same as that of charged spherical shell or hollow sphere. The field at external points has the same symmetry as that of a point charge. We can construct a Gaussian surface (a sphere) of radius r > R. At all points of this sphere the magnitude of electric field is the same and its direction is perpendicular to the surface. Thus, we can apply

$$ES = \frac{q_{in}}{\epsilon_0}$$

or

$$E\left(4\pi r^2\right) = \frac{q}{\epsilon_0}$$

$$\therefore \quad E = \frac{1}{4\pi\epsilon_0}\cdot\frac{q}{r^2}$$



Hence, the electric field at any external point is the same as if the total charge is concentrated at centre.

At the surface of sphere r = R,

$$\therefore \qquad E = \frac{1}{4\pi \epsilon_0} \cdot \frac{q}{r^2}$$

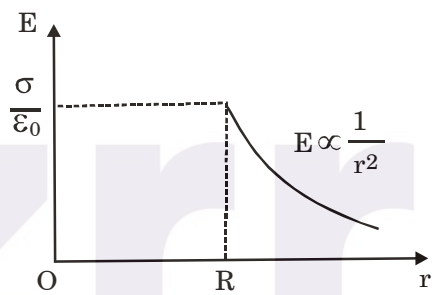
Thus, we can write,

$$\boxed{\begin{aligned} E_{inside} &= 0 \\ E_{surface} &= \frac{1}{4\pi \epsilon_0} \cdot \frac{q}{R^2} \\ E_{outside} &= \frac{1}{4\pi \epsilon_0} \cdot \frac{q}{r^2} \end{aligned}}$$

The variation of electric field (E) with the distance from the centre (r) is shown in Fig.

Note : (i) At the surface graph is discontinuous

(ii) $E_{surface} = \frac{1}{4\pi \epsilon_0} \cdot \frac{q}{R^2} = \frac{q/4\pi R^2}{\epsilon_0} = \frac{\sigma}{\epsilon_0}$



(B) POTENTIAL : As we have seen,

$$E_{outside} = \frac{1}{4\pi \epsilon_0} \cdot \frac{q}{r^2}$$

$$\therefore \qquad \left(\frac{-dV_{outside}}{dr}\right) = \frac{1}{4\pi \epsilon_0} \cdot \frac{q}{r^2} \qquad \left(E = -\frac{dV}{dr}\right)$$

$$\therefore \qquad \int_0^V dV_{outside} = \frac{-q}{4\pi \epsilon_0} \int_\infty^r \frac{dr}{r^2} \qquad (V_\infty = 0)$$

$$\therefore \qquad V = \frac{1}{4\pi \epsilon_0} \cdot \frac{q}{r} \quad \text{or} \quad V \propto \frac{1}{r}$$

Thus, at external points the potential at any point is the same when the whole charge is assumed to be concentrated at the centre. At the surface of the sphere, r = R

$$\therefore \qquad V = \frac{1}{4\pi \epsilon_0} \cdot \frac{q}{R}$$

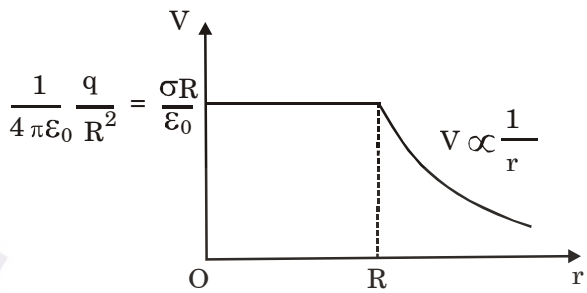
At some internal point electric field is zero everywhere, therefore, the potential is same at all points which is equal to the potential at surface. Thus, we can write,

$$V_{inside} = V_{surface} = \frac{1}{4\pi \epsilon_0} \cdot \frac{q}{R}$$

and

$$V_{outside} = \frac{1}{4\pi \epsilon_0} \cdot \frac{q}{r}$$

The potential (V) varies with the distance from the centre (r) as shown in Fig. below.



VI. A SOLID SPHERE

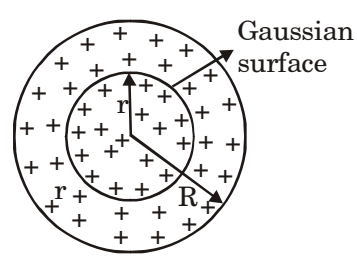
(A) Electric Field

Positive charge q is uniformly distributed throughout the volume of a solid sphere of radius R. For finding the electric field at a distance r (< R) from the centre let us choose as our Gaussian surface a sphere of radius r, concentric with the charge distribution. From symmetry the magnitude E of electric field has the same value at every point on the Gaussian surface, and the direction of $\vec{E}$ is radial at every point on the surface. So, applying Gauss's law,

$$ES = \frac{q_{in}}{\epsilon_0} \quad \text{... (i)}$$

Here, $S = 4\pi r^2$ and $q_{in} = (\rho)\left(\frac{4}{3}\pi r^3\right)$

here, ρ = charge per unit volume = $\frac{q}{\frac{4}{3}\pi R^3}$



Substituting these values in Eq. (i)

We have,

$$\mathbf{E = \frac{1}{4\pi \epsilon_0} \cdot \frac{q}{R^3} \cdot r}$$

or $E \propto r$

At the **centre** $r = 0$, so $\boxed{E = 0}$

At **surface** $r = R$, So $\boxed{E = \frac{1}{4\pi \epsilon_0} \cdot \frac{q}{R^2}}$

To find the electric field outside the charged sphere, we use a spherical Gaussian surface of radius r (> R). This surface encloses the entire charged sphere, so $q_{in} = q$, and Gauss's law gives

$$E\,(4\pi r^2) = \frac{q}{\epsilon_0} \quad \text{or} \quad \boxed{E = \frac{1}{4\pi \epsilon_0} \cdot \frac{q}{r^2}}$$

or $$E \propto \frac{1}{r^2}$$

Notice that if we set r = R in either of the two expressions for E (outside and inside the sphere), we get the same result,

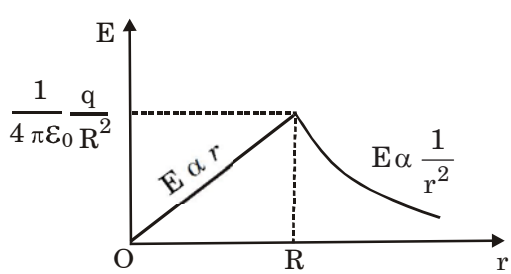
$$E = \frac{1}{4\pi \epsilon_0} \cdot \frac{1}{R^2}$$

this is because E is **continuous function** of r in this case. By contrast, for the charged conducting sphere the magnitude of electric field is discontinuous at r = R (it jumps from E = 0 to $E = \sigma/\epsilon_0$).

Thus, for a uniformly charged solid sphere we have the following formulae for magnitude of electric field.

$$\boxed{\begin{aligned} E_{inside} &= \frac{1}{4\pi \epsilon_0} \cdot \frac{q}{R^3} \cdot r \\ E_{surface} &= \frac{1}{4\pi \epsilon_0} \cdot \frac{q}{R^2} \\ E_{outside} &= \frac{1}{4\pi \epsilon_0} \cdot \frac{q}{r^2} \end{aligned}}$$

The variation of electric field (E) with the distance fromthe centre of the sphere (r) is shown in Fig. below.



(B) POTENTIAL : The field intensity outside the sphere is,

$$E_{outside} = \frac{1}{4\pi \epsilon_0} \cdot \frac{q}{r^2}$$

$$\frac{dV_{outside}}{dr} = -E_{outside}$$

$\therefore$ $$dV_{outside} = -E_{outside}\, dr$$

or $$\int_{\infty}^{V} dV_{outside} = -\int_{\infty}^{r} \frac{1}{4\pi \epsilon_0} \cdot \frac{q}{r^2}\, dr$$

$\therefore$ $$V = \frac{1}{4\pi \epsilon_0} \cdot \frac{q}{r} \quad \text{as } V_{\infty} = 0$$

or $$V \propto \frac{1}{r}$$

At $r = R$, $$V = \frac{1}{4\pi \epsilon_0} \cdot \frac{q}{R}$$

i.e., at the surface of the sphere potential is $V_S = \frac{1}{4\pi \epsilon_0} \cdot \frac{q}{R}$

The electric intensity inside the sphere,

$$E_{inside} = \frac{1}{4\pi \epsilon_0} \cdot \frac{q}{R^3} \cdot r$$

$$\frac{dV_{inside}}{dr} = -E_{inside}$$

$$\boxed{dV_{inside} = -E_{inside}\, dr}$$

$\therefore$ $$\int_{V_s}^{V} dV_{inside} = -\frac{1}{4\pi \epsilon_0} \cdot \frac{q}{R^3} \int_{R}^{r} r\, dr$$

$\therefore$ $$V - V_S = \frac{1}{4\pi \epsilon_0} \cdot \frac{q}{R^3} \left[\frac{r^2}{2}\right]_R^r$$

Substituting $V_S = \frac{1}{4\pi \epsilon_0} \cdot \frac{q}{R}$, we get $V = \frac{1}{4\pi \epsilon_0} \frac{q}{R} \left[\frac{3}{2} - \frac{1}{2} \frac{r^2}{R^2}\right]$

At the centre $r = 0$ and $V_c = \frac{3}{2}\left(\frac{1}{4\pi \epsilon_0} \cdot \frac{q}{R}\right) = \frac{3}{2} V_S$ i.e., **potential at the centre is 1.5 times the potential at surface.**

Thus, for a uniformly charged solid sphere we have the following formulae for potential.

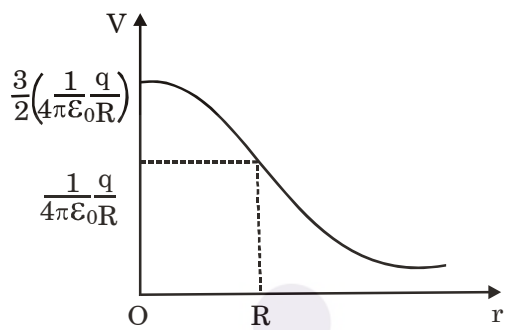
$$V_{outside} = \frac{1}{4\pi \epsilon_0} \cdot \frac{q}{r}$$

$$V_{surface} = \frac{1}{4\pi \epsilon_0} \cdot \frac{q}{R}$$

and

$$V_{inside} = \frac{1}{4\pi \epsilon_0} \cdot \frac{q}{R} \left[\frac{3}{2} - \frac{1}{2}\frac{r^2}{R^2}\right]$$

The variation of potential (V) with distance from the centre (r) is as shown in figure.



Points to Remember

1. To find the electric potential due to a conducting sphere (or shell) we should keep in mind the following two points :

(i) Electric potential on the surface and at any point inside the sphere is

$$V = \frac{1}{4\pi \epsilon_0} \cdot \frac{q}{R}$$

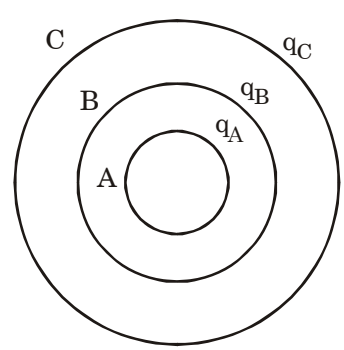
(R = radius of sphere)

(ii) Electric potential at any point outside the sphere is

$$V = \frac{1}{4\pi \epsilon_0} \cdot \frac{q}{r} \quad \text{(r = distance of the point from the centre)}$$

For example, in the figure shown, potential at A is

$$V_A = \frac{1}{4\pi \epsilon_0} \left[\frac{q_A}{r_A} + \frac{q_B}{r_B} + \frac{q_C}{r_C}\right]$$



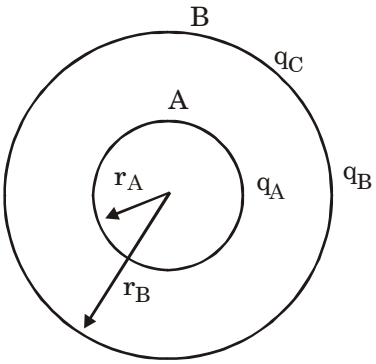
Similarly, potential at B is

$$V_B = \frac{1}{4\pi \epsilon_0} \left[\frac{q_A}{r_B} + \frac{q_B}{r_B} + \frac{q_C}{r_C}\right]$$

and potential C is,

$$V_C = \frac{1}{4\pi \epsilon_0} \left[\frac{q_A}{r_C} + \frac{q_B}{r_C} + \frac{q_C}{r_C}\right]$$

2. **Principle of a generator :** A generator is an instrument for producing high voltages in the million volt region. Its design is based on the principle that if a charged conductor (say A) is brought into contact with a hollow conductor (say B), all of its charge transfers to the hollow conductor no matter how high the potential of the latter may be. This can be shown as under.



In the figure,

$$V_A = \frac{1}{4\pi\epsilon_0}\left[\frac{q_A}{r_A} - \frac{q_B}{r_B}\right]$$

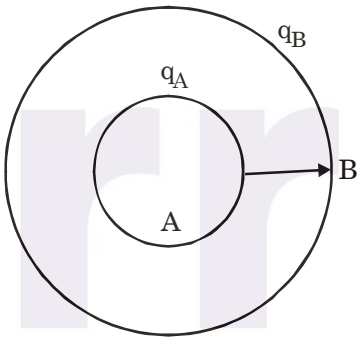
and

$$V_B = \frac{1}{4\pi\epsilon_0}\left[\frac{q_A}{r_B} - \frac{q_B}{r_B}\right]$$

$$\therefore \quad V_A - V_B = \frac{q_A}{4\pi\epsilon_0}\left[\frac{1}{r_A} - \frac{1}{r_B}\right]$$

From this expression following conclusions can be drawn :

(i) The potential difference (PD) depends on q_A only. It does not depend on q_B.



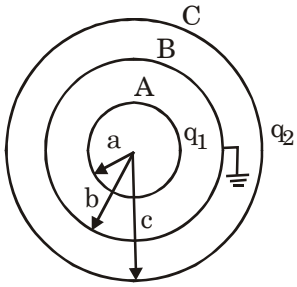
(ii) If q_A is positive, then $V_A - V_B$ is positive (as $r_A < r_B$), i.e., $V_A > V_B$. So if the two spheres are connected by a conducting wire charge flows from inner sphere to outer sphere (positive charge flows from higher potential to lower potential) till $V_A = V_B$ or $V_A - V_B = 0$. And potential difference will become zero only when $q_A = 0$, i.e., all charge q_A flows from inner sphere to outer sphere.

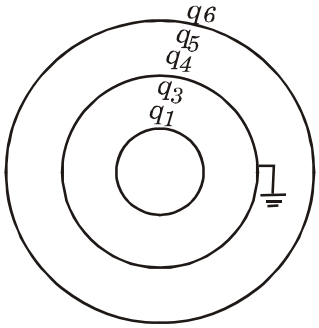
(iii) If q_A is negative, $V_A - V_B$ is negative, i.e. $V_A < V_B$. Hence, when the two spheres are connected by a thin wire all charge q_A will flow from inner sphere to the outer sphere. Because negative charge flows from lower potential to higher potential. Thus, we see that the whole charge q_A flows from inner sphere to outer sphere, no matter how high q_B is. Charge always flows from A to B, whether $q_A > q_B$ or $q_B > q_A$, $V_A > V_B$ or $V_B > V_A$.

3. **Earthing a conductor :** Potential of earth is often taken to be zero. If a conductor is connected to the earth, the potential of the conductor becomes equal to that of the earth, i.e., zero. If the conductor was at some other potential, charges will flow from it to the earthor from the earth to it to bring its potential to zero.

4. **Charges appearing on different surfacesof concentric spherical shells :** Figure shows three concentric thin spherical shells A, B and C of radii a, b and c. The shells A and C are given charges q_1 and q_2 and the shell B is earthed. We are interested in finding the charges on inner

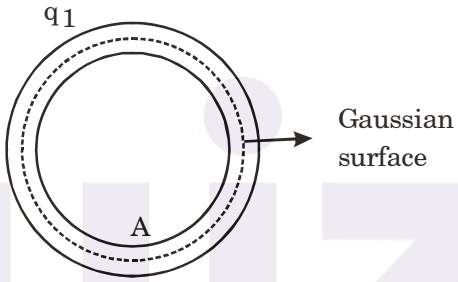
and outer surfaces of A, B and C. To solve such type of problems following points should keep in mind :





(i) The whole charge q_1 will come on the outer surface of A unless some charge is kept inside A.

To understand it let us consider a Guassian surface (a sphere) through the material of A. As the electric field in a conducting material is zero. The flux through this Gaussian surface is zero. Using Gauss's law, the total charge enclosed must be zero.



(ii) Similarly if we draw a Gaussian surface through the material of B we can see that

$$q_3 + q_1 = 0 \text{ or } q_3 = -q_1$$

and if we draw a Gaussian surface through the material of C, then

$$q_5 + q_4 + q_3 + q_1 = 0 \text{ or } q_5 = -q_4$$

(iii) $q_5 + q_6 = q_2$. As q_2 charge was given to shell C.

(iv) Potential of B should be zero, as it is earthed. Thus,

or
$$\frac{1}{4\pi\epsilon_0}\left[\frac{q_1}{b} + \frac{q_3 + q_4}{b} + \frac{q_5 + q_6}{c}\right] = 0$$

So, using the above conditions we can find charges on different surfaces.

We can summarise the above points as under

(1) Net charge inside a closed Gaussian surface drawn in any shell is zero. (provided the shell is conducting).

(2) Potential of the conductor which is earthed is zero.

(3) If two conductors are connected, they are at same potential.

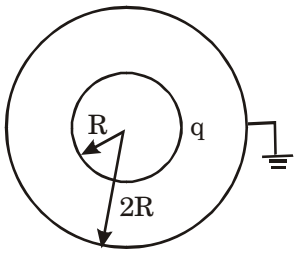
(4) Charge remains constant in all conductors except those which are earthed.

(5) Charge on the inner surface of the innermost shell is zero provided no charge is kept inside it. In all other shells charge resides on both the surfaces.

(6) Equal and opposite charges appear on opposite faces.

Example 50

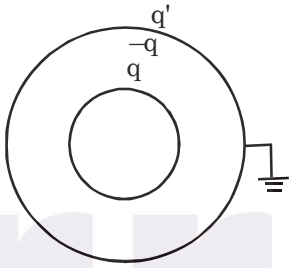
A charge q is distributed uniformly on the surface of a sphere of radius R. It is covered by a concentric hollow conducting sphere of radius 2R. Find the charges on inner and outer surfaces of hollow sphere if it is earthed.



Solution :

The charge on the inner surface should be – q, because if we draw a closed Gaussian surface through the material of the hollow sphere the total charge enclosed by this Gaussian surface should be zero. Let q´ be the charge on the outer surface of the hollow sphere.

Since, the outer sphere is earthed, its potential should be zero. The potential on it is due to the charges q, – q and q´, Hence,



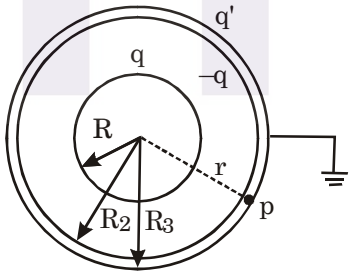
$$V = \frac{1}{4\pi \epsilon_0}\left[\frac{q}{2R} - \frac{q}{2R} + \frac{q'}{2R}\right] = 0$$

$\therefore$ $q' = 0$ **Ans.**

Therefore, there will be no charge on the outer surface of the hollow sphere.

Example 51

Solve the above problem if thickness of the hollow sphere is considerable.



Solution :

In this case, we can set V = 0 at any point on the hollow sphere. Let us select a point P a distance r from the centre. Where $R_2 < r < R_3$. So

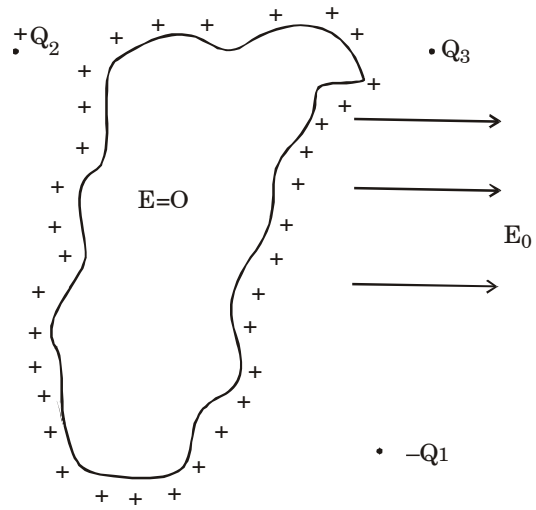
$$V_P = 0$$

$$\therefore \quad \frac{1}{4\pi \epsilon_0}\left[\frac{q}{r} - \frac{q}{r} + \frac{q'}{R_3}\right] = 0$$

$\therefore$ $q' = 0$ **Ans.**

i.e., in this case also there will be no charge on the outer surface of the hollow sphere.

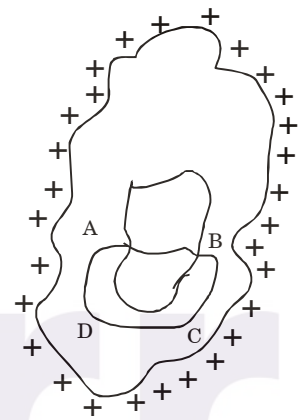
18. SHIELDING EFFECT



(1) If an external charge(s)/field are present in the surrounding of an electrical conductor then charge distribution on the surface of the conductor occurs in such a way that the net field intensity, due to the surrounding charge/field and the charge distributions (due to the surrounding charge) developed on the conductors surface; at any point inside the conductor becomes zero.

(2) If the external charge in the surrounding is arbitarily displaced/or the field charged; then re-distributing of charge occurs on the surface of the conductor at once, so that net field intensity at any point inside the conductor still remains zero.

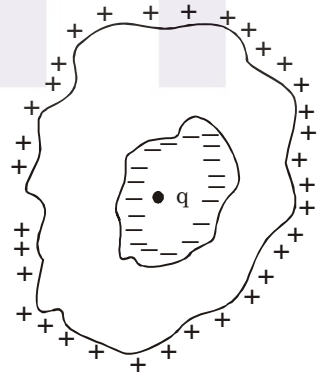
(3) If a cavity is created inside the electrical conductor, then electric field inside the cavity still remains zero (in the absence of external charge in the cavity).



Reason : To satisfy $\oint \vec{E}.\vec{dl} = 0$, $\vec{E}$ must zero inside the cavity.

(4) Charge placed in the cavity in a conductor :

When an external charge is placed inside the cavity of a conductor, the equal and opposite charge distributian is induced on the surface of the cavity, so that the field intensity due to both the external charge in the cavity and the induced charge at any point outside the cavity is zero.



$$\vec{E}_{\text{inside conductor}} = \vec{E}_{\text{inner}} + \vec{E}_{\text{outer}} = 0 + 0 = 0$$

(5) If the external charge in the cavity is arbitarily displaced, then, re-distribution of induced charge on the surface of cavity takes place so that field intensity outside the cavity still remains zero.

(6) Outer charge distribution remain unchanged if the external charge in the cavity is arbitarily displaced and the inner charge distribution remains unchanged if the outer charge distribution charges (i.e. vice-versa).

(7) Electric field at any point (inside the cavity) near its surface = $\left(\frac{\sigma}{\epsilon_0}\right)$. As a general idea, field is more strong at a point where, charge is closer to surface of the cavity.

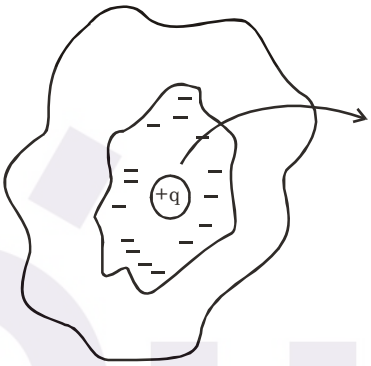
(8) The electric potential, due to the external charge and the induced charge on the surface of the cavity, at any point OUTSIDE THE CAVITY IS ZERO.

(Reason : No work done against field as E = due to both)

Remember : ELECTRIC POTENTIAL IS NOT ZERO due to the outer charge distribution inside the cavity.

(9) The net force experienced by any external charge inside the cavity is zero.

Proof :



Assume a Gaussian surface near the charge

$$\oint \vec{E}.\vec{ds} = \frac{(q_{inside})}{\epsilon_0}.$$

$\therefore$ $\vec{E}_{ext} = 0$ [$\because$ q = 0 excluding the charge itself]

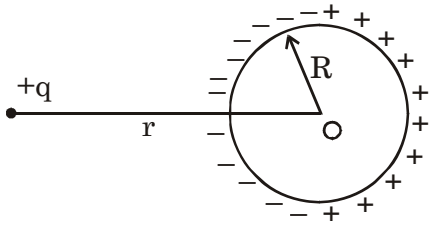
$\left(\vec{F} = 0\right)$

(10) Hence, a conducting shell divide the entire regions into two parts, inner and outer which are completely independent of each other, in terms of electrostatic field. This is called as the shielding effect in the electrical conductors.

Example 52

A charge q is placed at a distance r from the centre of an uncharged conductor sphere of radius R. Find the potential of the conducting sphere.

Solution :



Conductor has an equipotential space inside it.

$\therefore$ find the potential at the centre O for convenience

$$V = \left(\frac{Kq}{r}\right) + \int \frac{Kdq}{R} = \frac{Kq}{r} + \frac{K}{R}\int dq$$

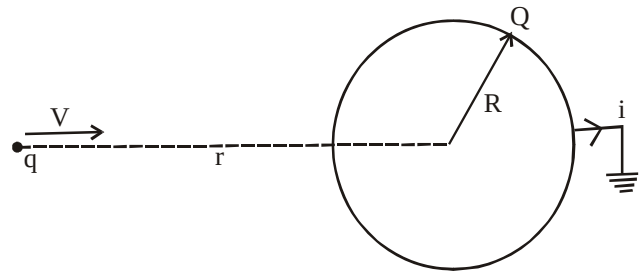
$$V = \frac{Kq}{r} + 0 = \left(\frac{Kq}{r}\right)$$

{Where dq is the elemental charge on the surface of conductor}

Example 53

In the previous question, the conductor is grounded and the charge q starts moving with constant velocity towards the sphere which is v. Then, find the current through the wire connecting the sphere to the earth.

Solution :



Let us assume that a charge Q is acquired by the conductor, so that its potential becomes zero.

$$V = \left(\frac{Kq}{r}\right) + \frac{KQ}{R} = 0 \quad [\because \text{ it is earthed}]$$

$$\therefore \quad Q = -\left(\frac{qR}{r}\right), \quad \frac{dQ}{dt} = +\frac{qR}{r^2}(-v)$$

$$\therefore \quad i = \left(\frac{qRv}{r^2}\right)$$

Now, since the charge on the conductor is decreasing, therefore the current flows from the conductor to the earth.

[However, the magnitude of the current is increasing]

Example 54

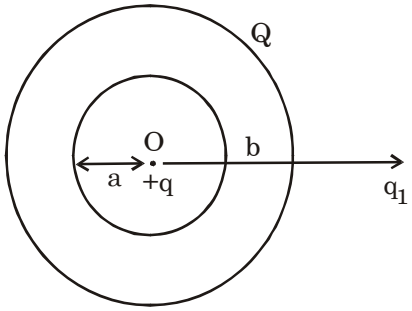
A point charge q is kept inside a charged conducting shell at a distance charged 'a' from its centre. The shell has an initial charge Q and external radii 'c'. Another charge 'q_1' is placed outside the shell at a distance 'b' from the centre.

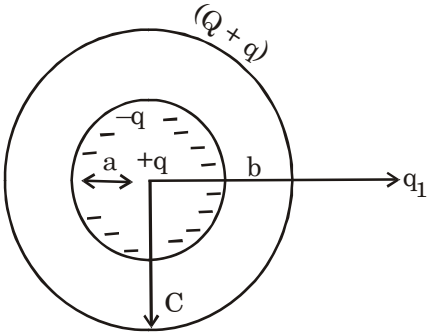
Find : (i) the net potential at O

(ii) the force experienced by q_1 due to the induced charge in the cavity of shell.

(iii) Potential at the position of q_1 due to the induced charge on the surface of cavity. (Assume shell is thin).

Solution :





(i) Potential at 0 :

$$V_0 = \left(\frac{Kq}{a}\right) + \left(\frac{KQ}{c}\right) + \left(\frac{Kq}{b}\right)$$

(Remember, when the shell is thin, write the potential together for the inner and the outer surface)

(ii) To calculate the force expression, use the concept that the force exerted by both, the induced charge as well as the external charge is zero. This can be easily seen by creating a gaussian surface centred at O and passing through the charge q_1,

So $q_{enclosed} = 0$, hence $\vec{E}_{net} = 0$

$\therefore$ $F = 0$ and $\vec{F}_{net} = \vec{F}_{cavity} + \vec{F}_q = 0$

$\therefore$ $F_{cavity} = -F_q$, $\vec{F}_q = \dfrac{K q q_1}{(a+b)^2}$ $\therefore$ $\boxed{\vec{F}_{cavity} = \dfrac{-K q q_1}{(a+b)^2}}$

(iii) Potential due to the charge on the surface of the cavity as done in the above part

$$V_{cavity} + V_{ext} = 0$$

$$\therefore \quad V_{cavity} = -V_{ext} = \frac{-Kq}{(a+b)}$$

19. PARALLEL PLATES

Two important concepts to be applied to all questions based on the parallel plates :

(1) The charges appearing on the faces opposite to each combination of pair of parallel plates are always equal is magnitudes and opposite in nature.

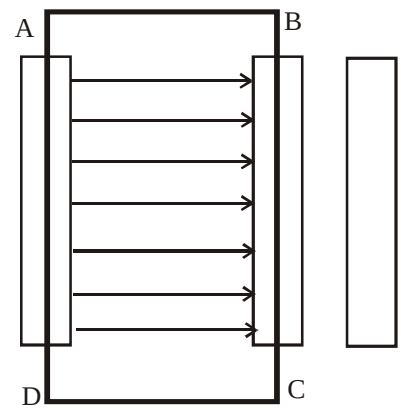
(2) The outer faces of the extreme plates have equal charges is magnitude and nature and equal to the (sum of charges on all plates divided by two) charge on the outer plates faces

$$\text{(outer)} = \frac{\sum Q_{all}}{2}$$

Derivation

(1) Use Gauss's law

Draw a Guasssian surface ABCD



Now $\oint \vec{E}.d\vec{s} = \left(\frac{q_{inside}}{C_0}\right)$

$$\oint_{ABCD} \vec{E}.d\vec{s} = \int_{AB} \vec{E}.d\vec{s} + \int_{BC} \vec{E}.d\vec{s} + \int_{CD} \vec{E}.ds + \int_{OA} \vec{E}.ds$$

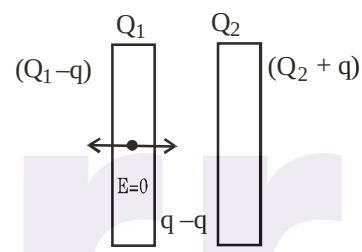
$$= \underset{[\because E \perp ds]}{0} + \underset{\left[\vec{E}=0\right]}{0} + \underset{\left[\vec{E} \perp ds\right]}{0} + \underset{\left[\vec{E}=0\right]}{0} = 0$$

$\therefore$ Net charge enclosed = 0

(2) To prove 2nd, use the concept that E = 0 inside the conductor which can happen only when the outer faces of the extreme plate have equal charges is magnitude as well as in direction.

$$\frac{(Q_1 - q)}{2 \in_0 A} = \frac{(Q_2 + q)}{2 \in_0 A}$$

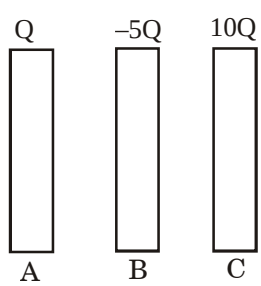
$$q = \frac{(Q_1 - Q_2)}{2}$$



$\therefore$ q_{outer} on each plate $= Q_1 - \frac{(Q_1 - Q_2)}{2} = \left(\frac{Q_1 + Q_2}{2}\right)$

Example 55

Show the charge distribution, on each surface when a charge of Q, – 5Q, 10 Q are given to each of the parallel plates.



Solution :

Charges on the outermost surfaces

$$= \frac{Q - 5Q + 10Q}{2} = 3Q$$

So, charge on the inner surface of plate A

$= Q - 3Q = -2Q$

Plate B

Charge on surface opposite to plate A

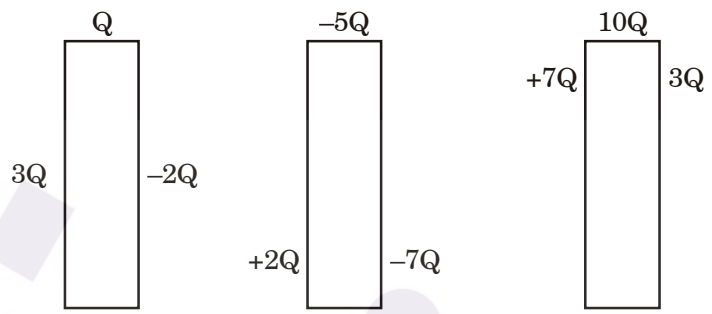
$= +2Q$

Now, charge on other surface

$= -5Q - (+2Q) = -7Q$

Plate C

Charge on surface opposite to plate C = + 7Q

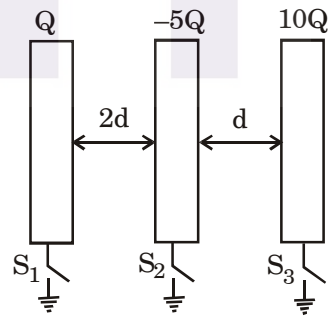


Here, we can check also that on plate C

7Q (as obtained) + 3Q = + 10Q (given)

Example 56

In the question, find the charge flown through the switch if :



(i) S_1 is grounded; S_2, S_3 open

(ii) S_3 is closed, S_1, S_2 open.

(iii) S_1, S_2 are simultaneously closed; S_3 open

(iv) S_2, S_3 simultaneously closed, S_1 open

(v) S_1, S_3 simultaneously closed, but S_2 remains open

(vi) S_2 closed and outer plates are connected by wire

(vii) If all switches are open and outer plates connected by wire, then find the charge distribution.

Solution :

(i) **Concept :** When any one of the outer plates is grounded then the charge on the outer faces of extreme plates becomes zero.

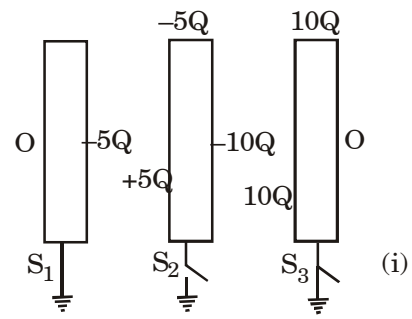
Reason : Its potential is now zero, hence, no work has to be done in bringing any charge from ∞ to the plate.

Start solving from the plate 3 which is not grounded.

Charge flown from earth to plate

$= Q_{final} - Q_{initial}$

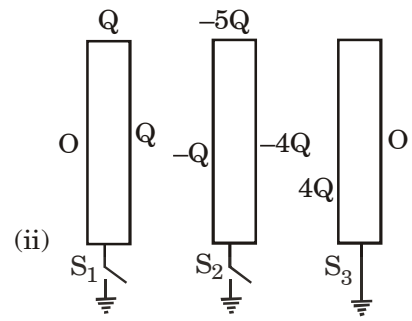
$= -5Q - Q = (-6Q)$



(ii) Charge flown to plate from the earth

$= Q_{final} - Q_{initial}$

$= 4Q - 10Q = (-6Q)$



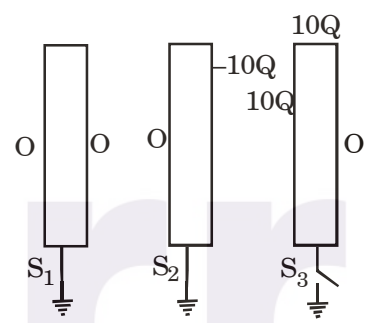
(iii) S_1, S_2 simultaneously closed; S_3 remains open.

Concept : In the question plate 1 and plate 2 which are adjacently grounded have no charge on their facing surface.

Reason : Their potential is same.

$\therefore$ Charge flown through $S_1 = Q - Q = -Q$

Charge flown through $S_2 = -10Q - (-5Q) = -5Q$



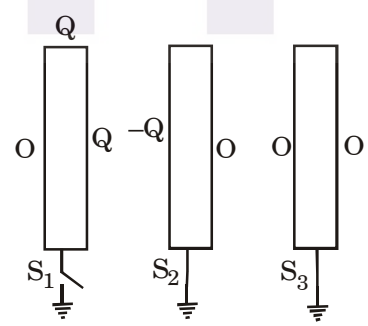
(iv) S_2, S_3 simultaneously closed; S_1 remains open

Charge flown through S_2

$= -Q - (-5Q) = 4Q$

Charge flown through S_3

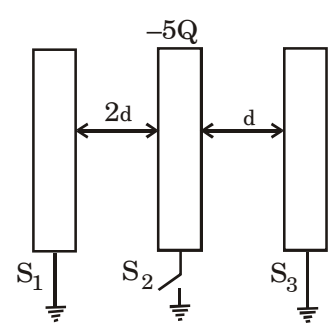
$= -O - (+10Q) = -10Q$



(v) S_1, S_3 are simultaneously closed but S_2 remaining open

In this case, the plates 1 and 3 are equipotential.

Now, these are 2 methods to solve this question.



Method 1 :

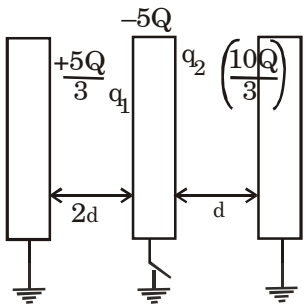
{This is to be read after going through capacitors)

The charge on surface of the plate 2 facing 1, and 3 are distributed in the ratio of their capacitance.

This is be cause potential drop acroos plates 1 and 2 and the drop across 2 and 3 is same, Hence, we can say that the capacitor formed by 1 and 2, and by 2 ad 3 are in paralle.

Where,

$$C_1 = \left(\frac{\epsilon_0 A}{2d}\right), C_2 = \left(\frac{\epsilon_0 A}{d}\right)$$



$$\therefore \quad \left(\frac{C_1}{C_2}\right) = \left(\frac{1}{2}\right)$$

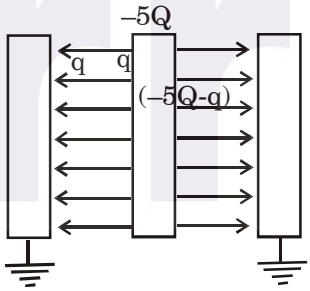
$$\therefore \quad q_1 = \frac{1}{3} \times (-5Q) = \left(\frac{-5Q}{3}\right), \quad q_2 = \frac{2}{3} \times 5Q = -\left(\frac{10Q}{3}\right)$$

Charge flow through $S_1 = \frac{5Q}{3} - Q = \left(\frac{2Q}{3}\right)$

Charge flow through $S_3 = \frac{10Q}{3} - Q = \left(\frac{7Q}{3}\right)$

Method 2 :

Potential drop across (plate 1 and plate 2) and (plate 2 and plate 3) must be equal and opp. so that plate 1 and plate 3 remain at same potential.

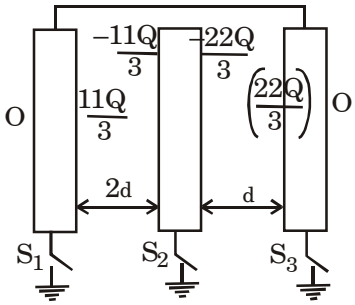


$$E_1 d_1 + E_2 d_2 = 0$$

$$\frac{q}{\epsilon_0 A}(2d) - \frac{(5Q - q)d}{\epsilon_0 A} = 0$$

$$2q + 5Q + q = 0 \quad \therefore \quad q = (-5Q/3)$$

(vi) Sum of the charges on the two outer plates remain constant = 11Q
But this charge will be distributed in the ratio of the capacitance

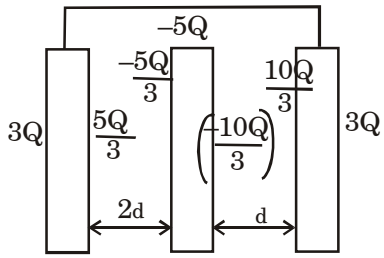


$\therefore$ Charge on plate 1 = $\frac{1}{3} \times 11Q$

Charge on plate 2 = $\frac{1}{3} \times 11Q = \frac{22Q}{3}$

Note : Here charges do not appear on the outer surface of the plate 1 and 3, since the entire charge is accomodated on their inner surfaces as the plate 2 is connected to a charge reservoir i.e. earth.

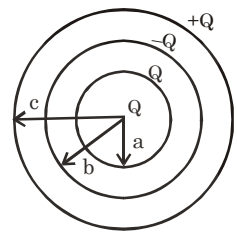
(vii) When only outer plates are connected :



Charges on the surface of the plate 2 facing 1 and 3 will be divided in the ratio of their capacitance.

Sum of the charges on the outer plates 1and 2 remain constant.

20. SPHERICAL CONDUCTING SHELLS



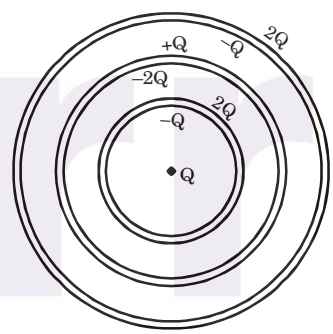
Example 57

Find the charge distribution on the inner and outer surfaces of the given conducting shells.

Solution :

Charge on the innermost surface can be found using the concept of shielding effect, from which we derived that equal and opposite charge was induced on the surface of the cavity.

Apply this conceptfor further surface.



Example 58

If the inner and outer shells are connected by wire and middle shell is grounded. Then, find the charge distribution on each shell? Initially A and C had Charge Q each.

Solution :

Concept :

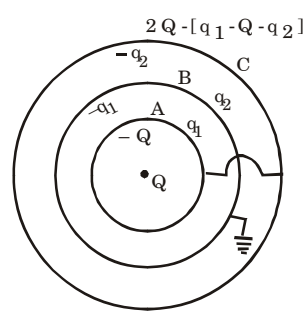
(i) $V_B = 0$ (grounded)

(ii) $V_A = V_C$ with $(\Sigma Q_{A+C})_{initial} = (\Sigma Q_{A+C})_{final}$

Assume the charge on the outer surface of A as q_1 and that on outer surface of B as q_2.

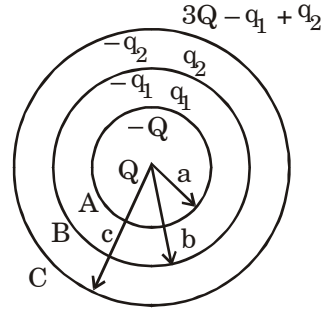
$\therefore$ charge on the outermost surface of C

$= 2Q - [q_1 - Q - q_2] = (3Q - q_1 + q_2)$



Now $V_B = 0$

$$\Rightarrow \quad 0 = \underbrace{\frac{K(Q)}{b}}_{\text{Charge at centre}} + \underbrace{\frac{K(q_1 - Q)}{b}}_{\text{Plate A}} + \underbrace{\frac{K(q_2 - q_1)}{b}}_{\text{Plate B}} + \underbrace{\frac{K(3Q - q_1 + q_2 - q_2)}{c}}_{\text{Plate C}}$$



$$\Rightarrow \quad \frac{Kq_2}{b} + \frac{K[3Q - q_1]}{c} = 0$$

$$\left(\frac{q_2}{b}\right) + \left(\frac{3Q - q_1}{c}\right) = 0 \qquad \text{... (1)}$$

also, $V_A = V_C$

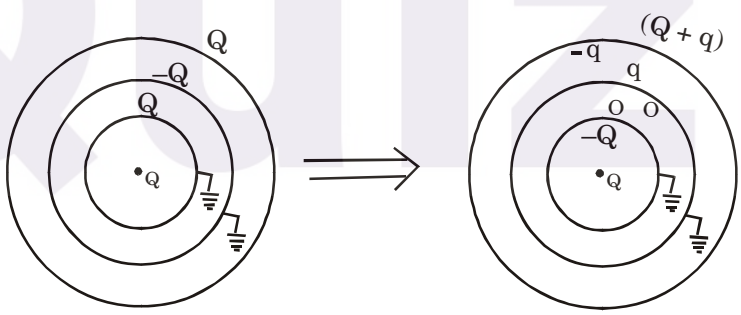
$$\Rightarrow \quad \frac{Kq_1}{a} + \frac{K(q_2 - q_1)}{b} + \frac{K(3Q - q_1)}{c} = K\frac{[3Q - q_1 + q_2]}{c} \qquad \text{... (2)}$$

Solve (1) and (2)

The rest solving is left as practice for the students

Example 59

In the previous question, find the charge distribution if the middle and inner shell are grounded.



Solution :

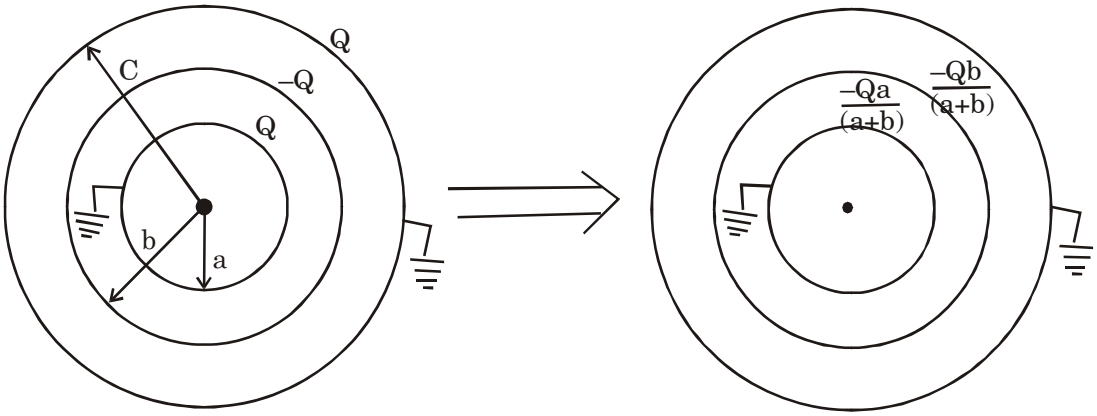
Concept to learn
There cannot be any charge on the surfaces which are equipotential and facing each other. [Otherwise, field lines would exist.]

Let q be the charge on outer surface of plate B.

Now, put $V_B = 0$

$$\therefore \quad \frac{Kq}{b} + \left(\frac{KQ}{c}\right) = 0 \Rightarrow q = -\left(\frac{Qb}{c}\right)$$

II. When the inner and outer spherical shells are ground.



Concept : The charge appearing on the inner and outer surfaces of the middle shell is divided in the ratio of the capacitances of the inner and outer shell which is of course the ratio of their radii.

21. CAPACITANCE

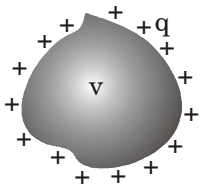
In practice we cannot handle free point charges or hold them fixed at desired positions. A practical way to handle a charge would be to put it on a conductor. Thus, one use of a conductor is to store electric charge (or electric potential energy). But every conductor has, its maximum limit of storing the electric charge or potential energy. Beyond that limitthe dielectric (or insulator) in which the conductor is placed becomes ionized. A capacitor is a device which can store more electric charge or potential energy compared to an isolated conductor.

Capacitors have a tremendous number of applications. In the flash light used by photographers the energy and charge stored in a capacitor are recovered quickly. In other applications the energy is released more slowly.

21.1 Capacitance of an Isolated Conductor

When a charge q is given a conductor it spreads over the outer surface of the conductor. The whole conductor comes to the same potential (say V). This potential V is directly proportional to the charge q, i.e.

$$V \propto q$$



when the proportionality sign is removed a constant of proportionality $\frac{1}{C}$ comes in picture. Hence,

$$V = \frac{q}{C}$$

or

$$\boxed{C = \frac{q}{V}}$$

Here, C is called the **capacitance** of the conductor. The SI unit of capacitance is called one **farad** (1 F). One farad is equal to coulomb per volt (1 C/V).

$\therefore$ 1 F = 1 farad = 1 C/V = 1 coulomb/volt

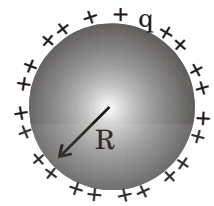
21.2 Method of Finding Capacitance of a Conductor

Give a charge q to the conductor. Find potential on it due to charge q. This potential V will obviously be a function of q and finally find q/V, which is the desired capacitance C.

21.3 Capacitance of a spherical conductor

When a charge q is given to a spherical conductor of radius R, the potential on it is,

$$V = \frac{1}{4\pi \in_0} \cdot \frac{q}{R}$$



From this expression we find that, $\frac{q}{V} = 4\pi \in_0 R = C$

Thus, capacitance of the spherical conductor is,

$$C = 4\pi\in_0 R$$

From this expression we can draw the following conclusions.

(i) $C \propto R$ or C depends on R only. Which we have already stated that C depends on the dimensions of the conductor. Moreover if two conductors have radii R_1 and R_2 then,

$$\frac{C_1}{C_2} = \frac{R_1}{R_2}$$

(ii) Earth is also a spherical conductor of radius R = 6.4 10^6 m. The capacity of earth is therefore

$$C = \left(\frac{1}{9 \times 10^9}\right)\left(6.4 \times 10^6\right) \approx 711 \times 10^{-6}\,F$$

or $$C = 711\ \mu F$$

From here, we can see that farad is a large unit. As capacity of such a huge conductor is only 711 μF.

21.4 Energy stored in a charged conductor

Work has to be done in charging a conductor against the force of repulsion by the already existing charges on it. The work is stored as a potential energy in the electric field of the conductor. Suppose a conductor of capacity C is charged to a potential V_0 and let q_0 be the charge on the conductor at this instant. The potential of the conductor when (during charging) the charge on it was q (< q_0) is,

$$V = \frac{q}{C}$$

Now work done in bringing a small charge dq at this potential is,

$$dW = Vdq = \left(\frac{q}{C}\right)dq$$

$\therefore$ total work done in charging it from 0 to q_0 is,

$$W = \int_0^{q_0} dW = \int_0^{q_0} \frac{q}{C}dq = \frac{1q_0^2}{2\,C}$$

This work is stored as the potential energy,

$$\therefore \qquad U = \frac{1}{2}\frac{q_0^2}{C}$$

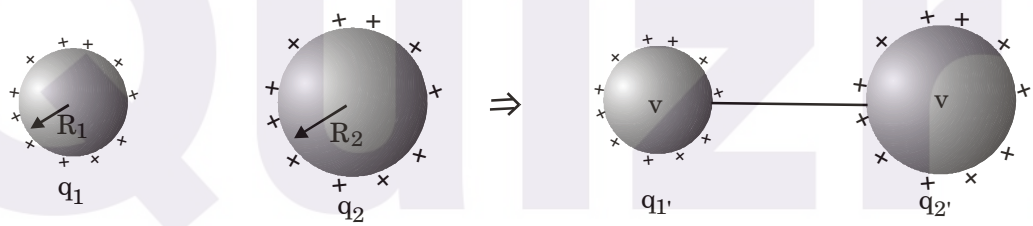
Further by using $q_0 = CV_0$ we can write this expression also as,

$$U = \frac{1}{2}CV_0^2 = \frac{1}{2}q_0\,V_0$$

In general if a conductor of capacity C is charged to a potential V by giving it a charge q, then

$$\boxed{U = \frac{1}{2}CV^2 = \frac{1}{2}\frac{q^2}{C} = \frac{1}{2}qV}$$

21.5 Redistribution of Charge



Suppose two conductors of capacities C_1 and C_2 have charges q_1 and q_2 respectively when they are joined together by a conducting wire, charge redistributes in these conductors in the ratio of their capacities. Charge redistributes till potential of both the conductors become equal. Thus, let q_1' and q_2' be the final charges on them, then

$$q' \propto C$$

or

$$\frac{q_1'}{q_2'} = \frac{C_1}{C_2}$$

and if they are spherical conductors, then $\frac{C_1}{C_2} = \frac{R_1}{R_2}$

$$\therefore \qquad \frac{q_1'}{q_2'} = \frac{C_1}{C_2} = \frac{R_1}{R_2}$$

Since, the total charge is $(q_1 + q_2)$. Therefore,

$$q_1' = \left(\frac{C_1}{C_1 + C_2}\right)(q_1 + q_2) = \left(\frac{R_1}{R_1 + R_2}\right)(q_1 + q_2)$$

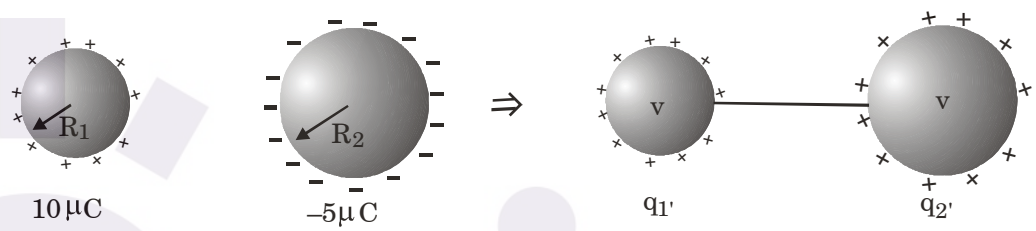
and

$$q_2' = \left(\frac{C_2}{C_1 + C_2}\right)(q_1 + q_2) = \left(\frac{R_2}{R_1 + R_2}\right)(q_1 + q_2)$$

Exampe 60

Two isolated spherical conductors have radii 10 cm and 20 cm respectively. They have charges of 10 μ and – 5 μC. Find the charges after they are connected by a conducting wire. Also find the common potential after redistribution.

Solution :



Net charge = (10 – 5) μC = 5 μC

charge is distributed in the ratio of their capacities (or radii in caseof spherical conductors), i.e.,

$$\frac{q_1'}{q_2'} = \frac{R_1}{R_2} = \frac{10}{20} = \frac{1}{2}$$

$\therefore$

$$q_1' = \left(\frac{1}{1+2}\right)(5) = \frac{5}{3}\mu C$$

and

$$q_2' = \left(\frac{2}{1+2}\right)(5) = \frac{10}{3}\mu C$$

common potential

$$V = \frac{q_1 + q_2}{C_1 + C_2} = \frac{\left(5 \times 10^{-6}\right)}{4\pi \epsilon_0 \left(R_1 + R_2\right)}$$

$$= \frac{\left(5 \times 10^{-6}\right)\left(9 \times 10^{9}\right)}{\left(15 \times 10^{-2}\right)}$$

$$= 30 \quad 10^4 \text{ volt}$$

Ans.

Exampe 61

An insulated conductor initially free from charge is charged by repeated contacts with a plate which after each contact is replenished to a charge Q. If q is the charge on the conductor after first operation prove that the maximum charge which can be given to the conducor in this way is $\frac{Qq}{Q-q}$

Solution :

Let C_1, be the capacity of plane and C_2 that of the conductor. After first contact charge on conductor is q. Therefore, charge on plate will remain Q – q. As the charge redistributes in the ratio of capacities.

$$\frac{Q-q}{q} = \frac{C_1}{C_2} \quad \text{... (i)}$$

Let q_m be the maximum charge which can be given to the conductor. Then flow of charge from the plate to the conductor will stop when,

$$V_{conductor} = V_{plate}$$

$$\therefore \quad \frac{q_m}{C_2} = \frac{Q}{C_1}$$

$$\therefore \quad q_m = \left(\frac{C_2}{C_1}\right)Q$$

Substituting $\frac{C_2}{C_1}$ from Eq. (i), we get

$$q_m = \frac{Qq}{Q-q}$$ **Proved**

Example 62

A conducting sphere S_1 of radius r is attached to an insulating handle. Another conducting sphere S_2 of radius R is mounted on an insulating stand S_2 is initially unchrged. S_1 is given a charge Q brought into contact with S_2 and removed. S_1 is recharged such that the charged on it is again Q and it is again brought into contact with S_2 and removed. This procedure is repeated n times :

(a) Find the electrostatic energy of S_2 after n such contacts with S_1.

(b) What is the limiting value of this energy as $n \to \infty$?

Solution :

Capacities of conducting sphere are in the ratio of their radii. Let C_1 and C_2 be the capacities of S_1 and S_2, then

$$\frac{C_2}{C_1} = \frac{R}{r}$$

(a) Charges are distributed in the ratio of their capacities. Let in the first contact, charge acquired by S_2 is q_1. Therefore, charge on S_1 will be $Q - q_1$.

$$\therefore \quad \frac{q_1}{Q - q_1} = \frac{C_2}{C_1} = \frac{R}{r}$$

It implies that Q charge is to be distributed in S_2 and S_1 in the ratio of R/r.

$$\therefore \quad q_1 = Q\left(\frac{R}{R + r}\right)$$

In the second contact, S_1 again acquires the same charge Q.

Therefore, total charge in S_1 and S_2 will be

$$Q + q_1 = Q\left(1 + \frac{R}{R + r}\right)$$

This charge is again distributed in the same ratio. Therefore, charge on S_2 in second contact,

$$q_2 = Q\left(1 + \frac{R}{R + r}\right)\left(\frac{R}{R + r}\right) = Q\left[\frac{R}{R + r} + \left(\frac{R}{R + r}\right)^2\right]$$

Similarly,
$$q_3 = Q\left[\frac{R}{R + r} + \left(\frac{R}{R + r}\right)^2 + \ldots \left(\frac{R}{R + r}\right)^3\right]$$

and
$$q_n = Q\left[\frac{R}{R + r} + \left(\frac{R}{R + r}\right)^2 + \ldots \left(\frac{R}{R + r}\right)^n\right]$$

or
$$q_n = Q\frac{R}{r}\left[1 - \left(\frac{R}{R + r}\right)^n\right] \quad \ldots \text{(i)} \qquad \left[\because S_n = \frac{a\left(1 - r^n\right)}{(1 - r)}\right]$$

Therefore, electrostatic energy of S_2 after n such contacts

$$U_n = \frac{q_n^2}{2C_2} = \frac{q_n^2}{2(4\pi \epsilon_0 R)}$$

or
$$U_n = \frac{q_n^2}{8\pi \epsilon_0 R}$$ **Ans.**

where q_n can be written from Eq. (i)

(b) $$q_n = \frac{QR}{R+r}\left[1 + \frac{R}{R+r} + \ldots + \left(\frac{R}{R+r}\right)^{n-1}\right] \qquad \text{as } n \to \infty$$

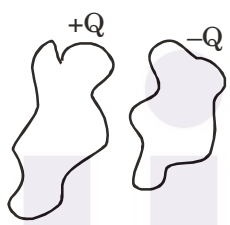
$$q_\infty = \frac{QR}{R+r}\left(\frac{1}{1-\frac{R}{R+r}}\right) = \frac{QR}{R+r}\cdot\left(\frac{R+r}{r}\right) = Q\frac{R}{r}; \qquad \left[S_\infty = \frac{a}{1-r}\right]$$

$\therefore$ $$U_\infty = \frac{q_\infty^2}{2C_2} = \frac{Q^2R^2/r^2}{8\pi\epsilon_0 R}$$

or $$U_\infty = \frac{Q^2R}{8\pi\epsilon_0 r^2}$$ **Ans.**

22. CAPACITOR

When two conductor are placed at a small separation (close to each other), then this combination is called capacitor.



$Q \alpha V$

$Q = CV$

(C = capacitance)

22.1 Factors on which capacitance of the arrangement depend

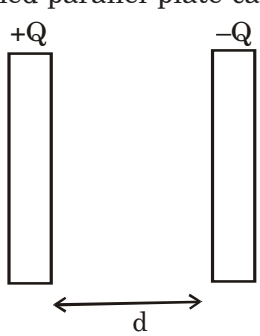
(i) Shape and size of the conductor

(ii) Relation orientation of the conductor

(iii) The medium in which they are placed

(1) PARALLEL PLATE CAPACITOR

When two conducting plates are placed parallel to each other at a small separation, then this combination is called parallel plate capacitor.



$$V = (V_+ - V_-)$$
$$= (Ed) \qquad [\because E \text{ is constant}]$$
$$= \left(\frac{\sigma d}{\epsilon_0}\right) = \left(\frac{Qd}{\epsilon_0 A}\right)$$
$$\boxed{C = \left(\frac{Q}{V}\right) = \frac{Q(\epsilon_0 A)}{Qd} = \left(\frac{\epsilon_0 A}{d}\right)}$$

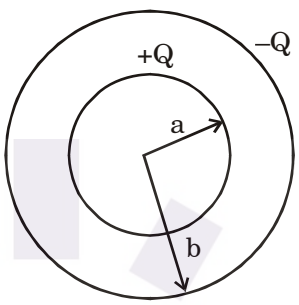
$\therefore$ $\boxed{\text{capacitance of a parallel plate capacitor} = \left(\frac{\epsilon_0 A}{d}\right)}$

22.2 General method to find the capacitance

Give charge Q, – Q to each surface and then find the potential difference V between the surface and use $C = \left(\frac{Q}{V}\right)$

(2) SPHERICAL CAPACITOR

When a hollow (or solid) spherical conductor is surrounded by another concentric hollow spherical conductor, placed at a separation, then this combination represents a **spherical capacitor.**



$$(V_a - V_b) = -\int_b^a \vec{E}.d\vec{r}$$

$$(V_a - V_b) = \int_a^b \left(\frac{KQ}{r^2}\right).dr$$

$$= KQ\left|\frac{-1}{r}\right|_a^b = KQ\left[\frac{1}{a} - \frac{1}{b}\right]$$

$$= \frac{Q(b-a)}{4\pi \epsilon_0 ab}$$

$$\boxed{\therefore\ C = \frac{Q}{(V)} = \frac{4\pi\epsilon_0 ab}{(b-a)}}$$

Cases :

(i) When $b \to \infty$ i.e. the inner conductor is an isolated sphere, then

$$C = (4\pi \epsilon_0 a)$$

(ii) When $d = (b - a) << a, b$

[i.e. the two conductors are nearly touching]

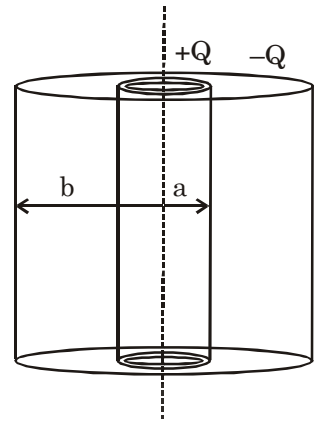
$$C = \left(\frac{4\pi \epsilon_0 a^2}{d}\right) = \frac{(4\pi a^2)\epsilon_0}{d} = \left(\frac{\epsilon_0 A}{d}\right)$$

then, this combination for $d = b - a << a, b$ behaves as a parallel plate capacitor.

(3) CYLINDRICAL CAPACITOR

When a hollow (or solid) cylindrical capacitor is surrounded by another cylindrical capacitor placed coaxially, then this combination is called cylindrical capacitor.

Here (l >> a, b)



$$(V_a - V_b) = -\int_b^a \vec{E}.d\vec{r}$$

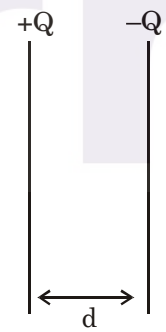
$$(V_a - V_b) = \int_a^b \frac{\lambda}{(2\pi \epsilon_0)r} dr$$

$$= \frac{\lambda}{2\pi \epsilon_0} \ln\left(\frac{b}{a}\right)$$

$$\therefore \quad C = \left(\frac{Q}{V_{ab}}\right) = \frac{\lambda L}{\left(\frac{\lambda}{2\pi \epsilon_0}\right)\ln\left(\frac{b}{a}\right)}$$

$$\therefore \quad \boxed{C = \frac{2\pi \epsilon_0 L}{\ln\left(\frac{b}{a}\right)}}$$

22.3 Force between the plates of a capacitor



Electric field due to a single plate = $\left(\frac{\sigma}{2 \epsilon_0}\right)$

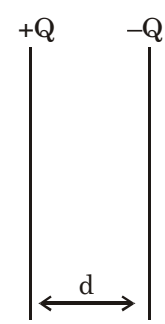
$\therefore$ Force on each plate = $\left(\frac{Q.Q}{2\epsilon_0 A}\right) = \frac{Q^2}{2 \epsilon_0 A}$

22.4 Potential Energy stored in the capacitor

Electrostatic potential energy is stored in the capacitor is the form of the electric field

$$\therefore \quad \Delta W_{ex} = F(d)$$

$$= \frac{Q^2}{(2 \epsilon_0 A)} \times d = \left(\frac{1}{2}\frac{Q^2}{C}\right)$$



$\therefore$ Potential energy stored in the capacitor = $\frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2 = \frac{1}{2}QV$

- When the capaitor is connected with a cell use $\frac{1}{2}CV^2$
- When the charge on the capacitor is constant, use $\left(\frac{1}{2}\frac{Q^2}{C}\right)$
- When distance between the plates is increased $\Delta W_{ex} = \frac{1}{2}\frac{Q^2}{\epsilon_0 A}(d_2 - d_1)$

22.5 Energy density between the capacitor plates

Energy density = potential energy stored per unit volume

$$= \left(\frac{U}{Ad}\right) = \frac{1}{2}\frac{Q^2 d}{(\epsilon_0 A)Ad} = \frac{1}{2}\frac{Q^2}{\epsilon_0 A^2}$$

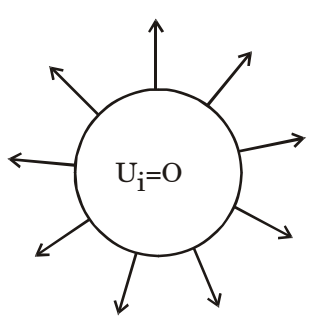
$$\therefore \quad \boxed{\text{Energy density} = \frac{1}{2}\epsilon_0 E^2}$$

Although, we have derived the expression for energy density in the capacitor, but is general this is valid for energy density is electric field, that is the energy stored per unit volume is the electric field.

$$\boxed{U = \frac{1}{2}\epsilon_0 E^2}$$

Using this we can calculate the self potential energy of a uniformly charged conducting sphere.

$$U_0 = \int_R^\infty \frac{1}{2}\epsilon_0 E^2 \, dV$$



$$U_0 = \int_R^\infty \frac{1}{2}\epsilon_0 \left(\frac{KQ}{r^2}\right)^2 (4\pi r^2)\, dr$$

$$= \frac{1}{2}K^2 \epsilon_0 Q^2 (2\pi)\int_R^\infty \frac{1}{r^2}\, dr$$

$$= \frac{1}{16\pi^2 \epsilon_0} Q^2 (2\pi)\left|\frac{-1}{r}\right|_R^\infty = \left(\frac{KQ^2}{2R}\right)$$

(ii) **For a non-conducting sphere**

$$U_i \neq 0,\ U_0 = \left(\frac{KQ^2}{2R}\right)$$

$$U_i = \int_0^R \frac{1}{2}\epsilon_0 \left(\frac{\rho\, r}{3\epsilon_0}\right)^2 \left(4\pi r^2\right) dr$$

$$= \frac{\rho^2\,(4\pi)}{9 \times 2\,\epsilon_0} \times \int_0^R r^4\, dr$$

and $$\rho = \left(\frac{Q}{\frac{4}{3}\pi R^3}\right)$$

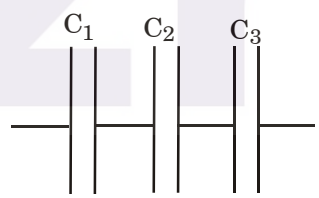
$$\Rightarrow \quad \frac{9Q^2(4\pi)}{(4\pi)^2 R^6 18\,\epsilon_0} \times \left(\frac{R^5}{5}\right) = \frac{KQ^2}{10R}$$

$$\therefore \quad U_T = U_0 + U_i = \left(\frac{KQ^2}{2R}\right) + \left(\frac{KQ^2}{10R}\right) = \left(\frac{3KQ^2}{5R}\right)$$

22.6 Series and parallel combination of capacitors

(A) Series Combination

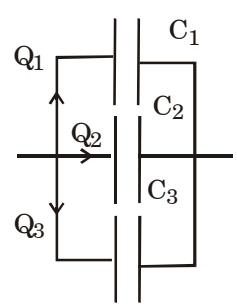
$V = (V_1 + V_2 + V_3)$



$$\frac{Q}{(C_{eq})} = \frac{Q}{C_1} + \frac{Q}{C_2} + \frac{Q}{C_3}$$

$$\therefore \quad \boxed{\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \ldots\ldots \frac{1}{C_n}} \quad \text{(Series combination)}$$

(In series combination, charge on each capacitor is constant).



(B) Parallel Combination

$Q = Q_1 + Q_2 + Q_3$

$C_{eq}\, V = C_1V + C_2V\ C_3V$

$$\therefore \quad \boxed{C_{eq} = C_1 + C_2 + C_3}$$

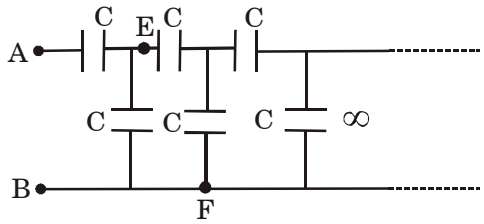
(In parallel, the voltage across each capacitor is constant).

Example 63

Find the equivalent capacitance of the following networks :

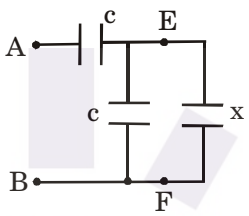
Solution :

(i)



Since the Chain repeats itself, let us assume after one link i.e. capacitance between E and F to be x, then

Equivalent circuit diagram,



Then, C_{eq} between A and B will also be x as, it is an infinite loop Hence

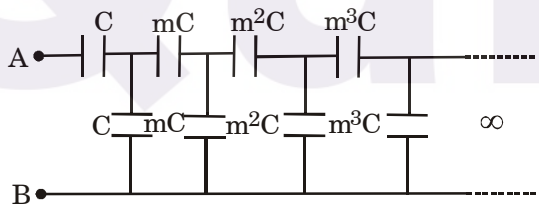
$$\boxed{x = \frac{C(C+x)}{(2C+x)}}$$

$\Rightarrow x\,(2C + x) = C\,(C + x)$

$\Rightarrow x^2 + Cx - C^2 = 0$

$$x = \frac{-C \pm \sqrt{C^2 + 4C^2}}{2} = C\left(\frac{\sqrt{5}-1}{2}\right)$$

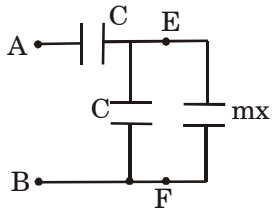
(ii)



Consider the equivalent circuit of the given infinite series

Assume the capacitance between E and F be mx, then

$$\boxed{x = \frac{C(mn + C)}{(2C + mn)}}$$



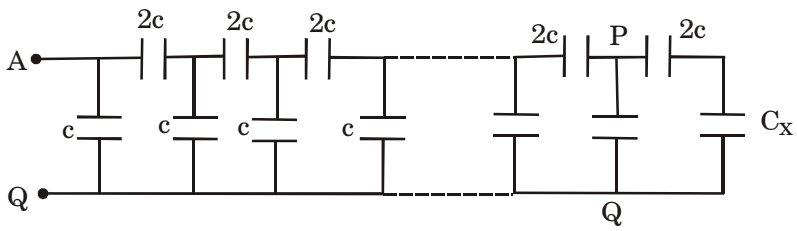
$x\,(2\,C + mx) = mCx + C^2$

$mx^2 + (2C - mc)x - C^2 = 0$

C_{eq} between E and F is written as mx because if we take m common from all the values of capacitance, the new circuit between E & F is same as the original circuit between A & B.

Example 64

Find the value of C_x so that equivalent capacitance between A and B becomes independent of the number of repeative links.



Solution :

For the equivalent capacitance to be independent of the number of links, the value of equivalent capacitance between P and Q should be same as C_x.

$$C_{eq}(PQ) = C_x$$

$$\Rightarrow \quad \frac{(2C.C_x)}{(2C + C_x)} + C = C_x$$

$$\Rightarrow \quad (2C.C_x) + C\,(2C + C_x) = C_x\,(2C + C_x)$$

$$\Rightarrow \quad C_x^2 - C_x\,C - 2C^2 = 0$$

$$\therefore \quad C_x = C\left[\frac{1+\sqrt{9}}{2}\right]$$

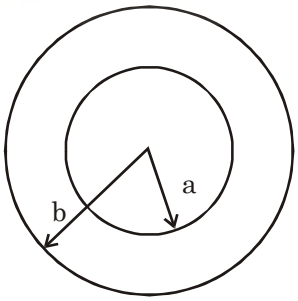
$$\boxed{C_x = (2C)}$$

Example 65

Find the equivalent capacitance of the given arrangement of spherical capacitors,

(i) When inner is given a charge Q and outer is earthed,

(ii) When inner is earthed and the outer is given a charge Q.

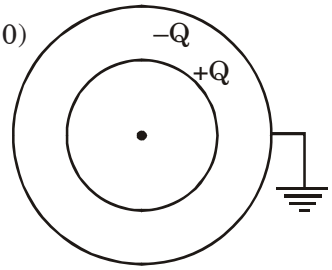


Solution :

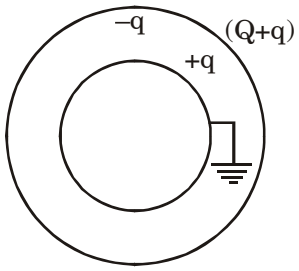
(i) The outer surface of the outer sphere will have no charge ($V_{outer} = 0$)

Now this combination behaves as a singe spherical capacitor

$$\boxed{C = \frac{4\pi \epsilon_0\, ab}{(b-a)}}$$



(ii) When inner is earthed



$$V_{inner} = 0$$

$$\left(\frac{Kq}{a}\right) + \left(\frac{KQ}{b}\right) = 0$$

$$q = -\left(\frac{Qa}{b}\right)$$

$\therefore$

$$V_{ab} = -\int_a^b \vec{E}.d\vec{r} = -\int_a^b K\left(\frac{-Qa}{b}\right)\left(\frac{1}{r}\right)dr$$

$$= +K\left(\frac{Qa}{b}\right)\left|\frac{1}{r}\right|_a^b = +\left(\frac{KQa}{b}\right)\left(\frac{b-a}{ab}\right)$$

$\therefore$

$$C = \left(\frac{Q}{V_{ab}}\right) = \frac{b^2(4\pi\in_0)}{(b-a)} = \frac{4\pi\in_0 b^2}{(b-a)}$$

2nd method :

In the above arrangement, we have two capacitors in parallel arrangement, the inner shell and inner surface of the outer capacitor and the outer surface of the outer sphere.

$\therefore$

$$C_{eq} = \frac{4\pi\in_0(ab)}{(b-a)} + 4\pi\in_0 b$$

$$= 4\pi\in_0\left[\frac{ab + b(b-a)}{(b-a)}\right]$$

$$= \frac{4\pi\in_0 b^2}{(b-a)}$$

3rd method :

Write the individual potential of each surface

$$V_{inner} = 0$$

$$V_{outer} = \left(\frac{Kq}{b}\right) + \left(\frac{KQ}{b}\right)$$

$$= \left(\frac{K}{b}\right)\left|-\frac{Qa}{b} + Q\right| \quad \left\{q = \frac{Qa}{b}\right\}$$

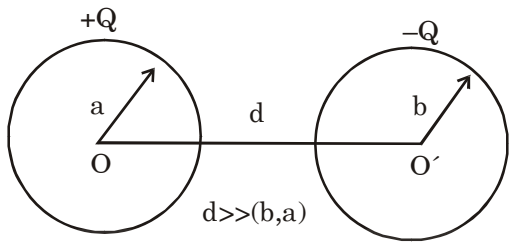
$$= \frac{KQ}{b^2}\left[(b-a)\right]$$

$$\therefore \qquad \Delta V = \frac{KQ}{b^2}(b-a)$$

$$\therefore \qquad C = \left(\frac{Q}{\Delta V}\right) = \frac{4\pi \in_0 b^2}{(b-a)}$$

Example 66

Find the equivalent capacitance of the given arrangement.



Solution :

Let V_+ be the potential of sphere centred at 0 and V_- be the potential of shere centred at 0´.

$$\therefore \qquad V_+ = \left(\frac{KQ}{a}\right) - \left(\frac{KQ}{d}\right)$$

$$V_- = \left(\frac{-KQ}{b}\right) + \left(\frac{KQ}{d}\right)$$

$$\therefore \qquad \Delta V = V_+ - V_-$$

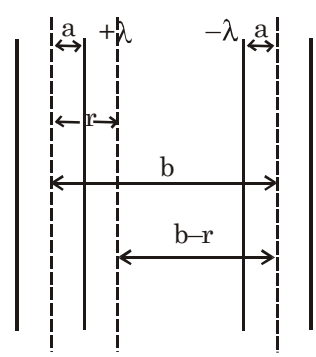
$$= \left(\frac{KQ}{a} + \frac{KQ}{b}\right) - \left(\frac{2KQ}{d}\right)$$

$$= KQ\left[\frac{1}{a} + \frac{1}{b} - \frac{2}{d}\right]$$

$$\therefore \qquad C = \left(\frac{Q}{\Delta V}\right) = \frac{4\pi C_0}{\left[\frac{1}{a} + \frac{1}{b} - \frac{2}{d}\right]}$$

Example 67

Two long cyindrical wires each of cross section radius a **are separated by distance** b **(>>** d**). Find the capacitance per unit length of the system.**



Solution :

Consider + λ and – λ charge density on each. Find the expression for electric field and then

Use $\Delta V = -\int \vec{E}.dr$

Let us Calculate electric field at a distance r from centre of cylindrical wire of charge density +λ. Then the net electric field is given by :

$$E = \frac{\lambda}{2\pi \in_0 r} + \frac{\lambda}{2\pi \in_0 (b-r)}$$

$$= \frac{\lambda}{2\pi \in_0}\left[\frac{1}{r} + \left(\frac{1}{b-r}\right)\right]$$

$$\therefore \quad \Delta V = \int_a^{(b-a)} \frac{\lambda}{2\pi \in_0}\left[\frac{1}{r} + \frac{1}{b-r}\right] dr$$

$$= \frac{\lambda}{2\pi \in_0}\left[\int_a^{(b-a)} \frac{1}{r} dr + \int_a^{(b-a)} \frac{1}{(b-r)}\right]$$

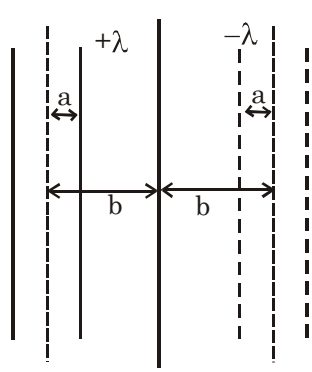
$$= \frac{\lambda}{2\pi \in_0}\left[\ln\left(\frac{b-a}{a}\right) - \ln\left(\frac{a}{ab-a}\right)\right]$$

$$= \frac{\lambda}{\pi \in_0}\left[\ln\left(\frac{b-a}{a}\right)\right]$$

$$\therefore \quad C = \left(\frac{\lambda l}{\Delta V}\right) = \frac{\pi \in_0 l}{\ln\left(\frac{b-a}{a}\right)} = \frac{\pi \in_0 l}{\ln\left(\frac{b}{a}\right)} \; (b >> a)$$

Example 68

A long cylindrical wire of radius a **is placed at a distance** b **from an infinite large plane sheet. Find the equivalent capacitance of the given arrangement.**



Soution :

Whenever we en counter an infinite large place sheet along with another conductor, we assume a similar distance **placed conductor on the opposite side (just as a mirror image about the sheet) with opposite charged. This is called image method.**

Using the image method we obtain the image of the long wire in infinite plane sheet.

$$E = \frac{\lambda}{2\pi \in_0 r} + \frac{\lambda}{2\pi \in_0 (2b-r)} \text{ [Note the distance is 2b]}$$

$$= \frac{\lambda}{2\pi \in_0}\left[\frac{1}{r} + \frac{1}{2br}\right]$$

$$\therefore \quad \Delta V = \int_a^b \frac{\lambda}{2\pi \in_0}\left[\frac{1}{r} + \frac{1}{2b-r}\right].dr$$ [Note again the limits put as the real potential to be calculated is upto the Large plane sheet.]

$$= \frac{\lambda}{2\pi \in_0}\left[\ln\left(\frac{b}{a}\right) - \ln\left(\frac{b}{2b-a}\right)\right]$$

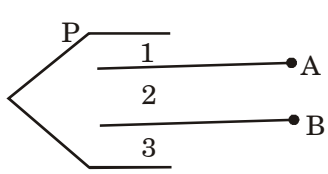
$$= \frac{\lambda}{2\pi \in_0}\left[\ln\left(\frac{b}{a} \times \frac{2b-a}{b}\right)\right]$$

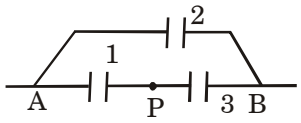
$$\therefore \quad \boxed{C = \frac{2\pi \in_0}{\ln\left(\frac{2b-a}{a}\right)}}$$

Example 69

Find the equivalent capacitance in the given arrangement.

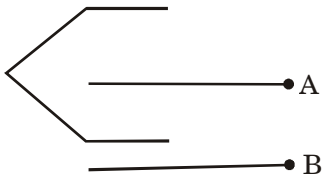
(i)

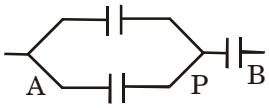




$$C_{eq} = C + \left(\frac{C}{2}\right) = \left(\frac{3C}{2}\right)$$

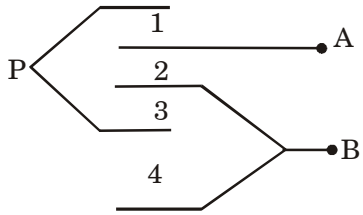
(ii)

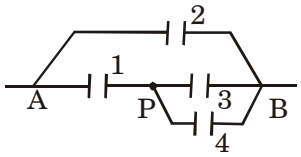




$$C_{eq} = \left(\frac{2C^2}{3C}\right) = \left(\frac{2C}{3}\right)$$

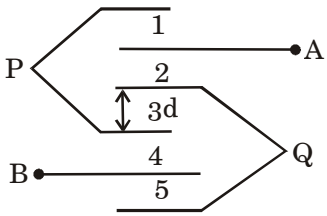
(iii)

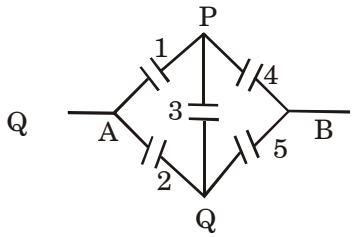




$$C_{eq} = \left(\frac{2C}{3}\right) + C = \left(\frac{5C}{3}\right)$$

(iv)





$$C_{eq} = \left(\frac{C}{2} + \frac{C}{2}\right) = C$$

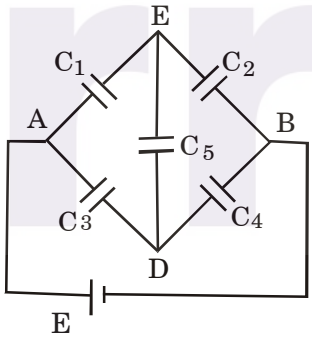
{Please refer to the video's for more explanation}

22.7 Wheatstone's bridge circuits

If $\frac{C_1}{C_2} = \frac{C_3}{C_4}$, bridge is said to be balanced and in that case

$$V_E = V_D \text{ or } V_E - V_D \text{ or } V_{ED} = 0$$

i.e., no charge is stored in C_5. Hence, it can be removed from the circuit.

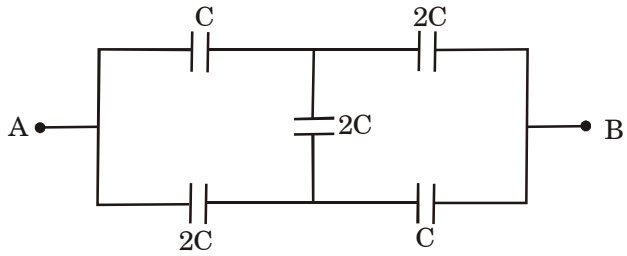


By distributing charge

Sometimes the above methods is not applicable. So this method can be applied everywhere. Of course this method is a little bit lengthy but is applicable everywhere, under all conditions. In this method we assume a charge q. Distribute it in different capacitors as q_1, q_2, ... etc. Using Kirchhoff's laws we find through any path and equate it with q/C_{net}. By doing so we can calculate C_{net}. Following example is in support of the thoery.

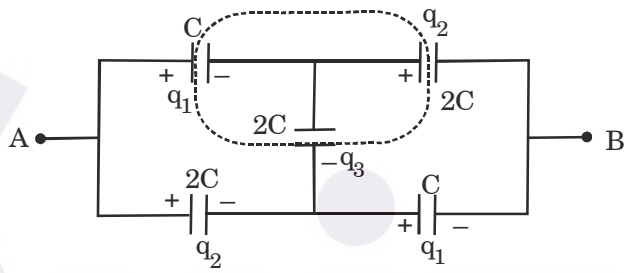
Example 70

Find the equivalent capacitance between A and B.



Solution :

The given circuit forms a Wheatstone bridge. But the bridge is not balanced. Let us suppose point A is connected to the positive terminal of a battery and B to the negative terminal of the same battery; so that a total charge q is stored in the capacitors. Just by seeing input and output symmetry we can say that charges will be distributed as shown below.



Applying kirchoff's second law, we have

$$q_1 + q_2 = q \qquad \text{... (i)}$$

$$-\frac{q_1}{C} - \frac{q_3}{2C} + \frac{q_2}{2C} = 0$$

or $$q_2 - q_3 - 2q_1 = 0 \qquad \text{... (ii)}$$

Plates inside the dotted line form an isolated system. Hence,

$$q_2 + q_3 - q_1 = 0 \qquad \text{... (iii)}$$

Solving these three equations, we have $q_1 = \frac{2}{5}q$, $q_2 = \frac{3}{5}q$ and $q_3 = -\frac{q}{5}$

Now, let C_{eq} be the equivalent capacitance between A and B. Then,

$$V_A - V_B = \frac{q}{C_{eq}} = \frac{q_1}{C} + \frac{q_2}{2C}$$

$$\therefore \quad \frac{q}{C_{eq}} = \frac{2q}{5C} + \frac{3q}{10C} = \frac{7q}{10C}$$

$$\therefore \quad C_{eq} = \frac{10}{7}C$$

23. EFFECT OF DIELECTRIC

Michel Faraday found that when certain non-conducting materials such as glass, paper or plastic are introduced between the plates of a capacitor, its capacity increases. These materials are called **dielectrics** and the ratio of capacity of a capacitor when completely filled with dielectric C to that without dielectric C_0 is called dielectric constant K, relative permitivity $\varepsilon_r = \frac{C}{C_0}$

Note : Regarding **dielectrics,** it is worth nothing that :

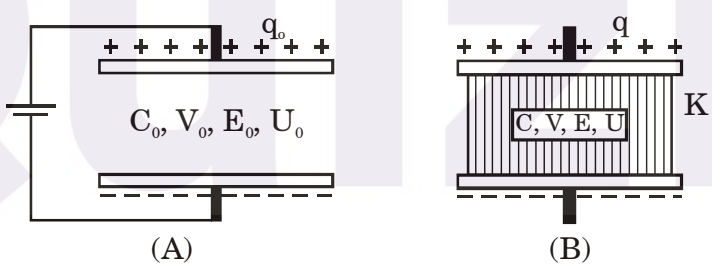
(i) These are non-conductors upto a certain value of field depending on its nature. If the field exceeds this limiting value called **dielectric-strength**, dielectric lose its insulating property and begins to conduct.

(ii) These have either permanent dipole moment (**polar-dielectrics,** e.g., water) or acquire induced dipole moment **(non-polar dielectrics)** when placed in an electric field.

(iii) The dielectric constant of polar dielectric depends on its temperature and due to thermal agitation with rise in temperature decreases.

The effect of dielectric on other physical quantities such as charge, PD, field and energy associated with a capacitor depends on the fact that whether the charged capacitor is isolated (i.e., charge is held constant).

So if q_0, C_0, V_0, E_0 and U_0 represent the charge, capacity, PD, electric field and energy associated with charged air capacitor respectively, with introduction of a dielectric slab of dielectric constant K between the plates.



(a) If charge is held constant, i.e., battery disconnected

(1) Charge remains unchanged, i.e., $q = q_0$, as in isolated system charge is conserved.

(2) Capacity increases, i.e., $C = KC_0$, as by presence of a dielectric capacity becomes K times.

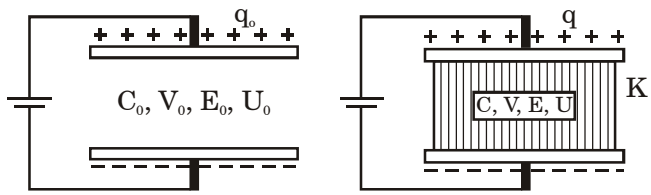
(3) PD between the plates decreases, i.e., $V = (V_0/K)$ as $V = \frac{q}{C} = \frac{q_0}{KC_0}$ [as $q = q_0$ and $C = KC_0$]

(4) Field between the plates decreases, i.e. $E = (E_0/K)$, as

$$E = \frac{V}{d} = \frac{V_0}{Kd} = \frac{E_0}{K} \qquad \left[\text{as } V = \frac{V_0}{K} \text{ and } E_0 = \frac{V_0}{d}\right]$$

(5) Energy stored in the capacitor decreases, i.e., $U = (U_0/K)$, as

$$U = \frac{q^2}{2C} = \frac{q_0^2}{2KC_0} = \frac{U_0}{K} \quad [\text{as } q = q_0 \text{ and } C = KC_0]$$



(A) (B)

(b) If potential is held constant, i.e. battery remains attached

(1) PD remains constant, i.e., $V = V_0$, as battery is a source of constant potential difference.

(2) Capacity increases, i.e., $C = KC_0$, as by presence of a dielectric capacity becomes K times.

(3) Charge on capacitor increases, i.e. $q = Kq_0$ as, $q = CV = (KC_0)V = Kq_0$ $\{\text{as } q_0 = CV_0\}$

(4) Electric field remains unchanged, i.e., $E = E_0$ as,

$$E = \left[\frac{V}{d}\right] = \frac{V_0}{d} = E_0 \quad \left[\text{as } V = V_0 \text{ and } \frac{V_0}{d} = E_0\right]$$

(5) Energy stored in the capacitor increases, i.e., $U = KU_0$ as, $U = \frac{1}{2}CV^2 = \frac{1}{2}(KC_0)(V_0)^2$

$$= \frac{1}{2}KU_0 \quad [\text{as } C = KC_0 \text{ and } U_0 = \frac{1}{2}C_0V_0^2]$$

Note : While solving problems of this type proceed in the following steps :

(i) If battery is disconnected $q' = q$ and if battery remains attached $V' = V$

(ii) For a capacitor as $C = (\varepsilon_0 KA/d)$, find whether C' is greater or lesser than C in accordance with given problem (as C depends on K, A and d only)

(iii) Now using $q' = C'V'$ and points (1) and (2), find q' if V' is given and V' if q' is given.

(iv) To calculate E' use the formula :

$$E' = \frac{V'}{d'} \text{ or } \frac{q'}{AK\varepsilon_0} \quad [\text{as } E' = \frac{\sigma'}{\varepsilon}] \quad \text{as the case may be}$$

(v) To calculate U' use the formula : $U' = \frac{1}{2}q'V' = \frac{1}{2}C'(V')^2 = (q')^2/2C'$ as the case may be

(vi) To calculate W apply conservation of energy i.e., $W_{IF} = (U_F - U_I)$

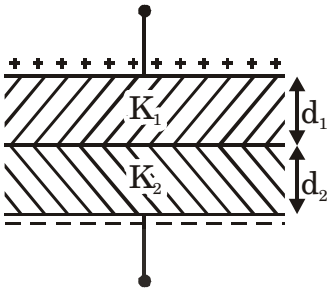
23.1 Capacitors with more than one Dielectric slabs :

(I) A parallel plate capacitor contains two dielectric slabs of thickness d_1, d_2 and dielectric constants k_1 & k_2 respectively. The area of the capacitor plates and slabs is equal to A.

Considering the capacitor as a combination of two capacitors in series, the equivalent capacitance C is given by :

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$\frac{1}{C} = \frac{d_1}{k_1 \epsilon_0 A} + \frac{d_2}{k_2 \epsilon_0 A}$$



$$C = \frac{\epsilon_0 A}{\frac{d_1}{k_1} + \frac{d_2}{k_2}}$$

In general for more than one dielectric slab :

$$C = \frac{\epsilon_0 A}{\sum \frac{d_i}{k_i}}$$

If V is the potential difference across the plates, the electric fields in the dielectrics are given as :

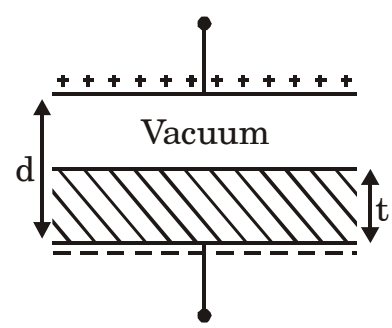
$$E_1 = \frac{V_1}{d_1} = \frac{Q}{C_1 d_1} = \frac{CV}{C_1 d_1} \qquad \left(\text{use } C_1 = \frac{k_1 \epsilon_0 A}{d_1} \text{ and } C = \frac{A \epsilon_0}{\frac{d_1}{K_1} + \frac{d_2}{K_2}} \right)$$

$$E_1 = \frac{1}{\frac{d_1}{k_1} + \frac{d_2}{k_2}} \left(\frac{V}{k_1} \right) \qquad E_2 = \frac{1}{\frac{d_1}{k_1} + \frac{d_2}{k_2}} \left(\frac{V}{k_2} \right)$$

Note : $k_1 E_1 = k_2 E_2$ and $E_1 d_1 + E_2 d_2 = V$

(II) If there exists a dielectric slab of thickness t inside a capacitor whose plates are separated by distance d, the equivalent capacitor is given as :

$$C = \frac{\epsilon_0 A}{\frac{t}{K} + \frac{d-t}{1}} \qquad (K = 1 \text{ for vacuum})$$



$$C = \frac{\epsilon_0 A}{\frac{t}{K} + d - t}$$

The equivalent capacitance is not affected by changing the distance of slab from the parallel Plates.

If the slab is of metal the equivalent capacitance is :

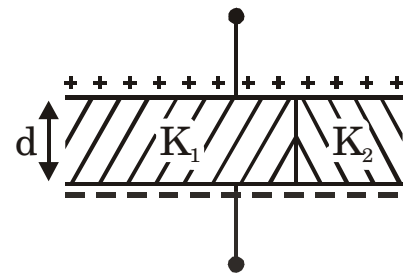
$$C = \frac{\epsilon_0 A}{d\text{-}t} \quad (\text{for a metal, } K = \infty)$$

(III) Consider a capacitor with two dielectric slabs of same thickness d placed inside it as shown. The slabs have dielectric constant k_1 & k_2 and areas A_1 & A_2 respectively. Treating the combination as two capacitors in parallel,

$$C = C_1 + C_2$$

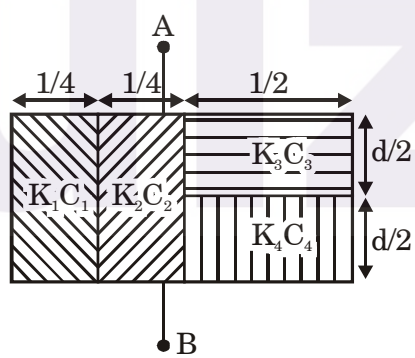
$$C = \frac{k_1 \epsilon_0 A_1}{d} + \frac{k_2 \epsilon_0 A_2}{d}$$

$$C = \frac{\epsilon_0}{d}[k_1 A_1 + k_2 A_2]$$



Example 71

The plates of the capacitor formed by inserting four-dielectric slabs (as shown) have an area equal to S. Find the equivalent capacitance between A and B if $K_1 = 2K_2 = K_3 = K_4 = 5$.



Solution :

Consider the capacitor as a parallel combination of C_1 and C_2 and series combination of C_3 and C_4 Equivalent Capacitance = $C_1 + C_2 + C_3C_4/C_3 + C_4$

$$\therefore \quad \boxed{C = \frac{\varepsilon_0}{d}\left(k_1 \frac{S}{4} + k_2 \frac{S}{4}\right) + \frac{\varepsilon_0 S/2}{\frac{d/2}{k_3} + \frac{d/2}{k_4}}}$$

Example 72

Two metal plates form a parallel plate capacitor. The distance between the plates is d. A metal plate of thickness b (= d/2) and same area is inserted completely between the plates. What is the ratio of the capacitance in the two cases ?

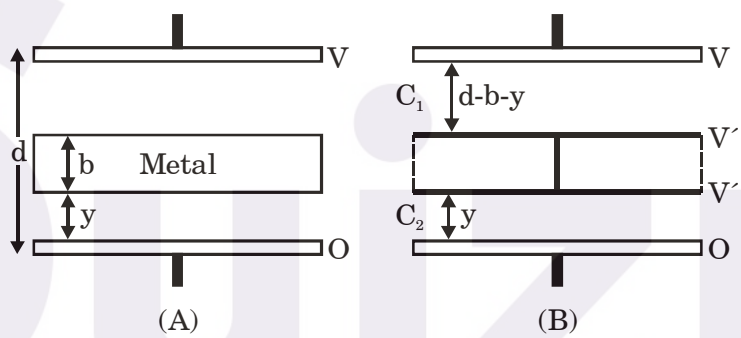
Solution :

Before the introduction of the slab the capacity of the system

$$C_0 = \frac{q}{V} = \frac{q}{(Ed)} \qquad [\text{as } E = (V/d)]$$

or, $$C_0 = \frac{\sigma A}{(\sigma/\varepsilon_0)d} = \frac{\varepsilon_0 A}{d} \quad \left[\text{as } q = \sigma A \text{ and } E = \frac{\sigma}{\varepsilon_0}\right] \qquad ...(1)$$

When the metal slab of thickness b is introduced between the plates, the system becomes equal to two capacitors C_1 and C_2 in series (as metal slab is an equipotential surface) as shown in (B).



So the capacity C of the system now becomes

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}, \qquad \text{i.e.,} \quad \frac{1}{C} = \frac{(d-b-y)}{\varepsilon_0 A} + \frac{y}{\varepsilon_0 A} \left[\text{as } C = \frac{\varepsilon_0 A}{d}\right]$$

or, $$\frac{1}{C} = \frac{(d-b)}{\varepsilon_0 A}, \qquad \text{i.e.,} \quad C = \frac{\varepsilon_0 A}{(d-b)} \qquad ...(2)$$

But as here b = (d/2)

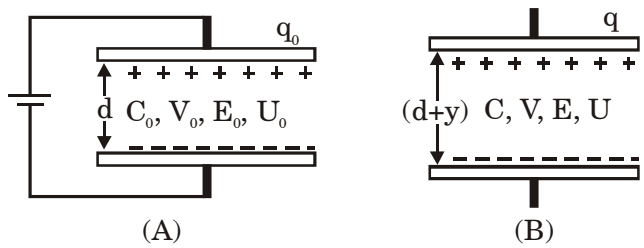
$$C = \frac{\varepsilon_0 A}{(d/2)} = 2\frac{\varepsilon_0 A}{d} = 2C_0. \qquad \text{i.e.,} \quad \frac{C}{C_0} = 2$$ **Ans.**

Example 73

How does the charge, capacity, potential, field and energy associated with a capacitor change qualitatively when the separation between its plates is increased (a) with battery disconnected (b) with battery attached.

Solution :

(a) When battery is disconnected.



In this situation :

(1) Charge remains unchanged, i.e., $q = q_0$ as charge in an isolated system is conserved.

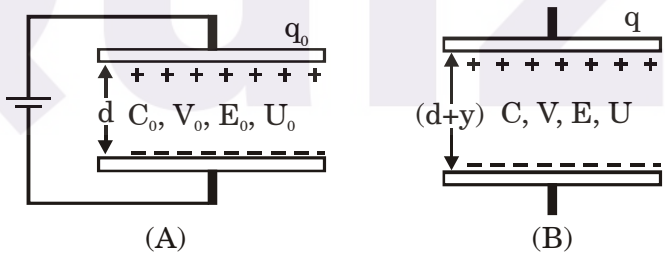
(2) Capacity decreases, i.e., $C < C_0$ as $C \propto (1/d)$.

(3) PD increases, i.e. $V > V_0$ as $q = CV$ and for constant q, $V \propto (1/C)$.

(4) Electric field remains constant, i.e., $E = E_0$ as $E = \sigma/\varepsilon = \sigma_0/\varepsilon$ = constant.

(5) Energy stored increases, i.e., $U > U_0$ as $U = 1/2qV$, i.e., $U \propto V$ for constant q.

(b) When battery is attached.



(1) PD across the capacitor remains constant, i.e. $V = V_0$ as battery is a source of constant emf.

(2) Capacity decreases, i.e., $C < C_0$ as $C \propto (1/d)$.

(3) Charge on the capacitor decreases, i.e., $q < q_0$ as $q = CV$, i.e., $q \propto C$ (for constant V.

(4) Electric field between the plates decreases, $E < E_0$ as $E = (V/d)$, i.e., $E \propto (1/d)$ for constant V.

(5) Energy stored in the capacitor decreases, i.e., $U < U_0$ as $U = \frac{1}{2}qV$, i.e., $U \propto q$ for constant V.

MISCELLANEOUS EXAMPLES

Example 1

A point charge q is placed on the top of a cone of semi vertex angle θ. Show that the electric flux through the base of the cone is $\dfrac{q(1-\cos\theta)}{2\epsilon_0}$

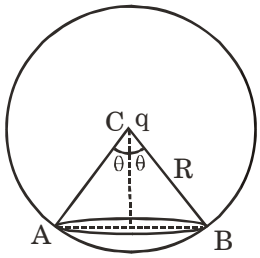
How to Proceed : This problem can be solved by the method of symmetry. Consider a Gaussian surface, a sphere with its centre at the top and radius the slant length of the cone. The flux through the whole sphere is q/ϵ_0. Therefore, the flux through the base of the cone can be calculated by using the following formula,

$$\phi_e = \left(\frac{S}{S_0}\right).\frac{q}{\epsilon_0}$$

Here, S_0 = area of whole sphere

and S = area of sphere below the base of the cone

Solution : Let R = slant length of cone = radius of Gaussian sphere



$\therefore$ S_0 = area of whole sphere = $(4\pi R^2)$

S = area of sphere below the base of the cone

$= 2\pi R^2 (1 - \cos\theta)$

$\therefore$ The desired flux is, $\phi_e = \left(\dfrac{S}{S_0}\right).\dfrac{q}{\epsilon_0}$

$$= \frac{(2\pi R^2)(1-\cos\theta)}{(4\pi R^2)}.\frac{q}{\epsilon_0} = \frac{q(1-\cos\theta)}{2\epsilon_0}$$

Proved

Note : $S = 2\pi R^2 (1 - \cos\theta)$ can be calculated by integration.

At $\theta = 0$, $S = 2\pi R^2 (1 - \cos 0) = 0$

$\theta = 90$, $S = 2\pi R^2 (1 - \cos 90) = 2\pi R^2$

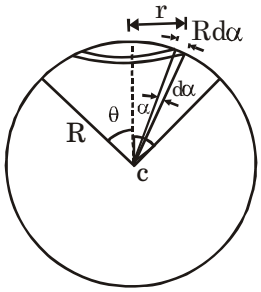
and $\theta = 180$, $S = 2\pi R^2 (1 - \cos 180) = 4\pi R^2$

Proof :

$dS = (2\pi r)\, R d\alpha$

$= (2\pi R \sin\alpha)\, R d\alpha$ as $r = R \sin\alpha$

$= (2\pi R^2) \sin\alpha\, d\alpha$



$\therefore$ $$S = \int_0^{\theta} (2\pi R^2) \sin\alpha\, d\alpha$$

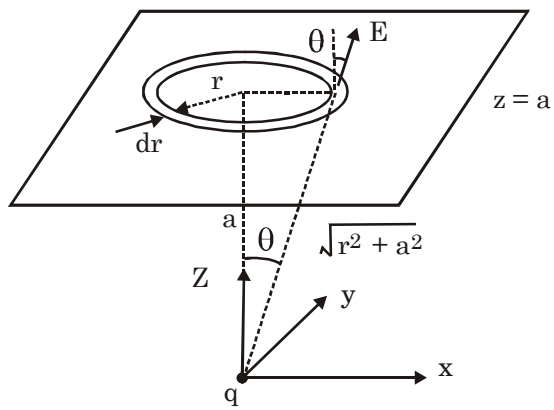
$\therefore$ $$\boxed{S = 2\pi R^2 (1 - \cos\theta)}$$

Example 2

A point charge q is fixed at the origin. Calculate the electric flux through the infinite plane z = a.

Solution :

Consider a ring of radius r on the plane z = a. The thickness of ring is dr. Electric field on the ring due to charge q is,



$$E = \frac{1}{4\pi \epsilon_0} \cdot \frac{1}{\left(r^2 + a^2\right)}$$

Area of the ring $\quad dS = (2\pi r).dr$

$$\cos\theta = \frac{a}{\sqrt{r^2 + a^2}}$$

$\therefore$ Flux passing through the ring,

$$d\phi_e = E\, dS \cos\theta = \frac{1}{4\pi \epsilon_0} \cdot \frac{q}{r^2 + a^2} \cdot (2\pi r\, dr) \cdot \frac{a}{\sqrt{r^2 + a^2}}$$

$$= \frac{q}{2\epsilon_0} \frac{ar\, dr}{\left(r^2 + a^2\right)^{3/2}}$$

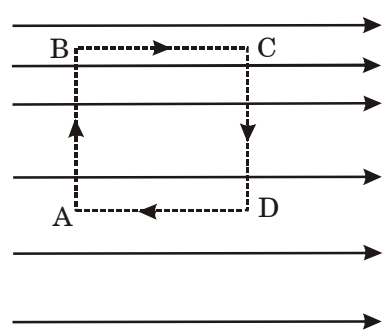
$\therefore$ Total flux passing through the plane,

$$\phi_e = \int_0^\infty d\phi_e = \frac{qa}{2\epsilon_0} \int_0^\infty \frac{r dr}{\left(r^2 + a^2\right)^{3/2}}$$

or $\quad \boxed{\phi_e = \frac{q}{2\epsilon_0}}$

Example 3

Is the configuration of an electrostatic field shown in figure is possible.



Solution :

No, it is not possible. To prove it, suppose we take a closed path ABCDA in the electric field, the line integral over this path should be zero.

$$\oint \vec{E}.\vec{dl} = 0$$

Because this quantity represents $V_P - V_P$, where P is any point on the path and $V_P - V_P$ is obviously zero.

Along the path AB and CD the line integral of electric field is zero. Since, $\vec{E} \perp \vec{dl}$ and

$$\int_A^B \vec{E}.\vec{dl} = \int_C^D E.\vec{dl} = 0$$

Thus, for $\int \vec{E}.\vec{dl} = 0$, $\int_B^C \vec{E}.\vec{dl}$ should be equal to $-\int_D^A \vec{E}.\vec{dl}$

But since, lines are closer along BC, i.e., $\left|\vec{E}_{BC}\right| > \left|\vec{E}_{DA}\right|$

$$\therefore \quad \int_B^C \vec{E}.\vec{dl} \neq -\int_D^A \vec{E}.\vec{dl}$$

or $$\int \vec{E}.\vec{dl} \neq 0$$

Thus, such a configuration of an electrostatic field shown in figure can't be realised in practice.

Example 4

A solid conducting sphere A with a radius R_1 = 6 cm has an initial electrostatic potential V_0 = 30 kV relative to ground. Sphere A is then symmetrically surrounded by two conducting and initially neutral hemispheres B_1 and B_2. In this way a sphere B with inner radius R_2 = 10 cm and outer radius R_3 = 12 cm is formed.

(a) What is the net charge on sphere A ?

(b) What are the charges induced on the inner and outer surface of sphere B ?

(c) What are the electric potentials of spheres A and B ?

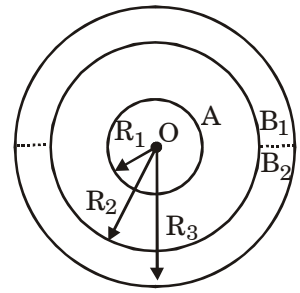
(d) What are the electric fields at the surface of sphere A and at the inner and outer surfaces of sphere B ?

(e) Draw separate graphs of the electric potential and the electric fields as a function of distance x from the centre of the system.

(f) How would the charge be distributed if the spheres A and B were connected by a conducting wire ?

Solution : (a)

$$V_0 = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1}{R_1}$$



$$\therefore \quad q_1 = \frac{V_0 R_1}{\left(\frac{1}{4\pi\epsilon_0}\right)}$$

$$= \frac{\left(30 \times 10^3\right)\left(6 \times 10^{-2}\right)}{\left(9.0 \times 10^9\right)}$$

$$= 2.0 \quad 10^{-7} \text{ C} = 0.2 \ \mu\text{C}$$ **Ans.**

(b) Charges induced on inner side of B = $-q_1 = -0.2\ \mu$C and on outer side of B = $q_1 = 0.2\ \mu$C **Ans.**

(c)

$$V_A = \frac{1}{4\pi\epsilon_0}\left[\frac{q_1}{R_1} + \frac{q_2}{R_2} + \frac{q_3}{R_3}\right]$$

$$= \frac{q_1}{4\pi\epsilon_0}\left[\frac{1}{R_1} - \frac{1}{R_2} + \frac{1}{R_3}\right]$$

Substituting the values,

$$V_A = \left(0.2 \times 10^{-6}\right)\left(9.0 \times 10^9\right)\left(10^2\right)\left[\frac{1}{6} - \frac{1}{10} + \frac{1}{12}\right]$$

$$= 2.7 \quad 10^4 \text{ volt}$$

$$= 27 \text{ kV}$$ **Ans.**

$$V_B = \frac{1}{4\pi\epsilon_0}\left(\frac{q_1}{R_3}\right) = \frac{\left(9.0 \times 10^9\right)\left(0.2 \times 10^{-6}\right)}{12 \times 10^{-2}} = 15\,\text{kV}$$ **Ans.**

(d) Electric field at the surface of sphere A,

$$E_1 = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1}{R_1^2} = \frac{\left(9.0 \times 10^9\right)\left(0.2 \times 10^{-6}\right)}{\left(6 \times 10^{-2}\right)^2} = 5 \times 10^5 \text{ V/m}$$

$$= 0.5 \text{ MV/m}$$ **Ans.**

Electric field at the inner side of sphere B,

$$E_2 = \left(\frac{1}{4\pi\epsilon_0}\right)\frac{q_1}{R_2^2} = E_1\left(\frac{R_1}{R_2}\right)^2 = (0.5)\left(\frac{6}{10}\right)^2$$

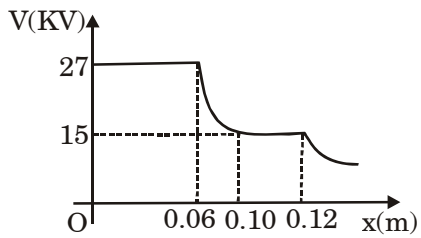
$$= 0.18 \text{ MV/m}$$ **Ans.**

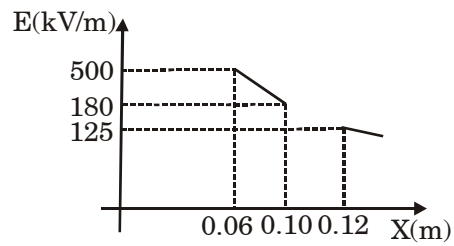
Electric field at the outer side of sphere B,

$$E_3 = \left(\frac{1}{4\pi\epsilon_0}\right)\frac{q_1}{R_3^2} = E_1\left(\frac{R_1}{R_3}\right)^2 = (0.5)\left(\frac{6}{12}\right)^2$$

$$= 0.125 \text{ MV/m}$$ **Ans.**

(e)





(f) When the two spheres are brought in contact, then all of the charge q_1 transfers to the outer sphere B.

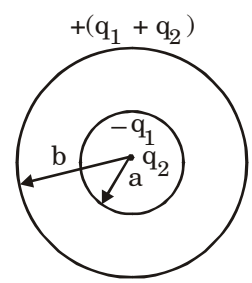
Example 5

A thick metallic shell of inner radius a and outer radius b has a charge q_1 on it. A point charge q_2 is fixed at the centre of the shell. Calculate the charge on each surface of the shell and also potential and field everywhere.

Solution :

The charges on inner and outer surfaces of the shell are as shown in figure. Field lines starting from q_2 terminates on the inner surface of the shell. The field at external points ($r \geq b$) is the same as that due to a point charge ($q_1 + q_2$) at the centre, i.e.

$$E(r \geq b) = \frac{(q_1 + q_2)}{4\pi\epsilon_0 r^2}$$



Within the thickness of the shell the field is zero.

$$E\ (a < r < b) = 0$$

The field at points $r < a$ is due to charge q_2 only. Hence,

$$E(r < a) = \frac{q_2}{4\pi\epsilon_0 r^2}$$

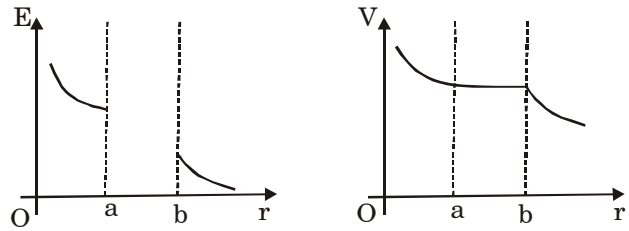
Similarly,

$$V(r \geq b) = \frac{q_1 + q_2}{4\pi\epsilon_0 r}$$

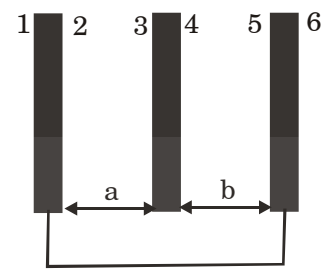
$$V\ (r < a) = \frac{1}{4\pi\epsilon_0}\left\{\frac{q_1 + q_2}{b} - \frac{q_2}{a} + \frac{q_2}{r}\right\}$$

From r = b to r = a the potential is uniform and has the value, $\frac{1}{4\pi\epsilon_0}\frac{(q_1+q_2)}{b}$

Figure shows the variation of V (r) and E (r)

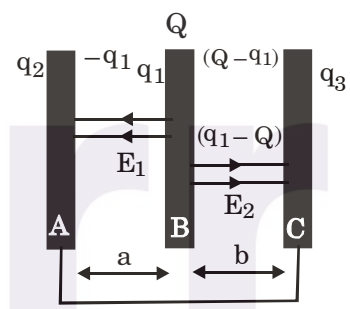


Example 6



Three identical metallic plates are kept parallel to one another at a separation of a **and** b**. The outer plates are connected by a thin conducting wire and a charge Q is placed on the central plate. Find final charges on all the six plate's surfaces.**

Solution :



Let the charge distribution in all the six faces be as shown in figure. While distributing the charge on different faces, we have used the fact that two opposite faces have equal and opposite charges on them.

Net charge on plates A and C is zero. Hence,

$$q_2 - q_1 + q_3 + q_1 - Q = 0$$

or $$q_2 + q_3 = Q \quad \text{..... (i)}$$

Further A and C are at same potentials.

Hence,

$$V_B - V_A = V_B - V_C \quad \text{or} \quad E_1 a = E_2 b$$

$$\therefore \quad \frac{q_1}{A\epsilon_0}.a = \frac{Q-q_1}{A\epsilon_0}.b \qquad \text{(A = Area of plates)}$$

$$\therefore \quad q_1 a = (Q - q_1)\, b \quad \therefore \quad q_1 = \frac{Qb}{a+b} \qquad \text{... (ii)}$$

Electric field inside any conducting plate (say inside C) is zero. Therefore,

$$\frac{q_2}{2A\epsilon_0} - \frac{q_1}{2A\epsilon_0} + \frac{q_1}{2A\epsilon_0} + \frac{Q-q_1}{2A\epsilon_0} + \frac{q_1-Q}{2A\epsilon_0} - \frac{q_3}{2A\epsilon_0} = 0$$

$$\therefore \quad q_2 - q_3 = 0 \qquad \text{... (iii)}$$

Solving these three equations, we get $q_1 = \frac{Qb}{a+b}$, $q_2 = q_3 = \frac{Q}{2}$

Hence, charge on different faces are as follows.

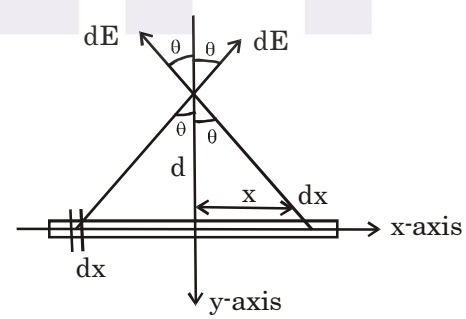
Face	Charge
1	$q_2 = \frac{Q}{2}$
2	$-q_1 = -\frac{Qb}{a+b}$
3	$q_1 = \frac{Qb}{a+b}$
4	$Q - q_1 = \frac{Qa}{a+b}$
5	$q_1 - Q = -\frac{Qa}{a+b}$
6	$q_3 = \frac{Q}{2}$

Example 7

Find the field intensity and potential due to a uniformly charged rod with charge per unit length λ and length l at a distance d on the perpendicular bisector and along its length.

Solution :

(a) At a distance d on the perpendicular bisector from the centre horizontal component of E will cancel each other



$$dE_y = dE \cos\theta$$

$$= \frac{K\lambda\, dx}{r}\cos\theta$$

$$\therefore \quad E = \int_{-l/2}^{+l/2} dE\cos\theta$$

$$= \int_{-l/2}^{+l/2} \frac{K\lambda\, dx}{\left(x^2+d^2\right)} \times \frac{d}{\sqrt{x^2+d^2}}$$

$$= K\lambda d \int_{-l/2}^{l/2} \frac{dx}{\left(x^2+d^2\right)^{3/2}}$$

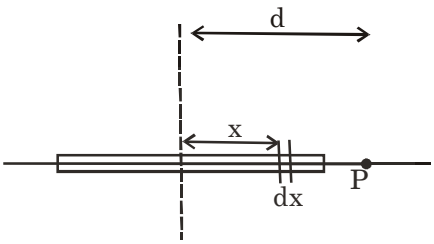
$$= \frac{K\lambda d}{d^2}\left[\frac{x}{\sqrt{x^2+d^2}}\right]_{-l/2}^{l/2} = \frac{K\lambda}{d}\left|\frac{2l}{\sqrt{l^2+4d^2}}\right|$$

$$= \frac{\lambda}{4\pi\epsilon_0\, d}\frac{2l}{\sqrt{l^2+4d^2}}$$

$$\boxed{E = \frac{\lambda}{2\pi\epsilon_0\, d}\frac{l}{\sqrt{l^2+4d^2}}}$$

(b) At an axial point

$$dE = \frac{K\lambda\, dx}{(d-x)^2}$$



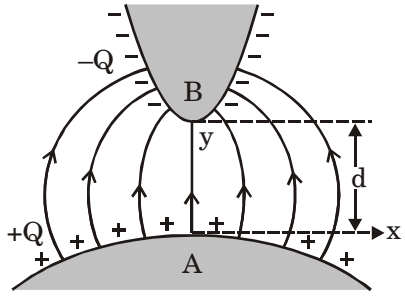
$$\therefore \quad E = K\lambda\int_{-l/2}^{l/2}\frac{dx}{(d-x)^2}$$

$$= K\lambda\left|\frac{1}{(d-x)}\right|_{-l/2}^{l/2} = K\lambda\left|\frac{(d+l/2)-(d-l/2)}{\left(d-\frac{l}{2}\right)\left(d+\frac{l}{2}\right)}\right|$$

$$= \frac{4K\lambda l}{(4d^2-l^2)} = \frac{4K\lambda l}{(4d^2-l^2)} = \left(\frac{\lambda}{\pi\epsilon_0}\right)\frac{l}{(4d^2-l^2)}$$

Example 8

Two conductors carrying equal and opposite charges create a non-uniform field as shown in fig. What is the capacity of this capacitor if the field along y-asix varies as ?



$$\mathbf{E = \frac{Q}{\epsilon_0 A}[1+By^2]} \text{ with } \mathbf{B} = \text{constant.}$$

Solution : As by definition $E = -(dV/dr)$

$$-\frac{dV}{dy} = \frac{Q}{\varepsilon_0 A}[1 + By^2]$$

or, $$-\int_{V_A}^{V_B} dV = \frac{Q}{\varepsilon_0 A}\int_0^d (1 + By^2)\, dy$$

or, $$V_A - V_B = \frac{Q}{\varepsilon_0 A}\left[d + \frac{1}{3}Bd^3\right]$$

or, $$V = \frac{Qd}{\varepsilon_0 A}\left[1 + \frac{1}{3}Bd^2\right]$$

So, $$C = \frac{Q}{V} = \frac{\varepsilon_0 A}{d\left[1 + \frac{1}{3}Bd^2\right]}$$

Note that this capacity is lesser than that of a parallel plate capacitor [i.e., $C = (\varepsilon_0 A/d)$] which has uniform field between the plates.

Example 9

A parallel plate capacitor has the space between the plates filled with a medium whose dielectric constant increases uniformly with distance. If d is the distance between the plates and K_1 and K_2 are the dielectric constant of the medium at the two plates (square each of area A) respectively show that the capacity of the capacitor.

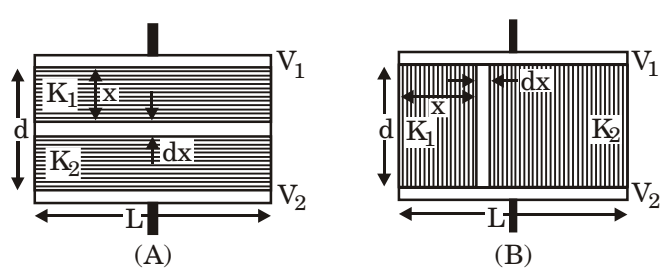
$$\mathbf{C = \frac{\varepsilon_0 A(K_2 - K_1)}{d \log_e (K_2 / K_1)}}$$

Solution : As dielectric constant varies uniformly with distance :

$$\frac{dK}{dx} = b = \text{const}, \qquad \text{i.e.,} \qquad K = a + bx$$

But as for $x = 0$, $K = K_1$ and for $x = d$, $K = K_2$, i.e., $a = K_1$ and $b = (K_2 - K_1)/d$

So, $K = K_1 + bx$ with $b = (K_2 - K_1)/d$



And hence, $$E = \frac{\sigma}{\varepsilon} = \frac{\sigma}{\varepsilon_0 K} = \frac{\sigma}{\varepsilon_0 (K_1 + bx)}$$

or, $$-\frac{dV}{dx} = \frac{\sigma}{\varepsilon_0 (K_1 + bx)} \quad \left[\text{as } E = -\frac{dV}{dx}\right]$$

or, $$-\int_{V_1}^{V_2} dV = \frac{\sigma}{\varepsilon_0}\int_0^d \frac{dx}{(K_1 + bx)}$$

i.e., $$V_1 - V_2 = \frac{\sigma}{\varepsilon_0 b}[\log_e (K_1 + bx)]_0^d$$

or, $$V = \frac{Q}{\varepsilon_0 Ab}[\log_e (K_1 + bd) - \log_e K_1] \quad \left[\text{as } \sigma = \frac{Q}{A}\right]$$

or, $$V = \frac{Qd}{\varepsilon_0 A(K_2 - K_1)}\left[\log_e \left(\frac{K_2}{K_1}\right)\right] \quad \left[\text{as } b = \frac{(K_2 - K_1)}{d}\right]$$

or, $$C = \frac{Q}{V} = \frac{\varepsilon_0 A(K_2 - K_1)}{d \log_e (K_2 / K_1)}$$ **Ans.**

Note : In this problem if the dielectric constant increases at a uniform rate from one edge to the other (so that the value of K at the two edges are K_1 and K_2 respectively [fig.(b)], the capacitor may be supposed to be constituted of large number of elemental capacitors connected in parallel. In this situation if we consider an elemental capacitor of width dx at distance x from the edge having dielectric constant K_1 :

$$dC = \frac{\varepsilon_0 (K_1 + bx)(L' dx)}{d} \quad \left[\text{as } C = \frac{\varepsilon_0 KA}{d}\right]$$

So, on integrating

$$C = \frac{\varepsilon_0 L'}{d}\int_0^L (K_1 + bx) = \frac{\varepsilon_0 LL'}{d}[K_1 + \frac{1}{2}bL]$$

or, $$C = \frac{\varepsilon_0 A}{d}\left[\frac{K_1 + K_2}{2}\right] \quad \left[\text{as } LL' = A \text{ and } b = \frac{K_2 + K_1}{L}\right]$$